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Boundary Layer Flashback of Non-Swirling Premixed Flames: Mechanisms, Fundamental Research, and Recent Advances

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Abstract

Boundary layer flashback in premixed jet flames has been the subject of detailed experimental and numerical investigation since the 1940's. The traditional approach for characterizing flashback has involved the critical velocity gradient concept, with higher values indicating a higher flashback propensity for a given situation. Recent studies in confined configurations have illustrated that a key assumption underlying the critical velocity gradient concept, namely a lack of interaction between the flame and the approaching flow, is fundamentally incorrect. However, for unconfined configurations, where this interaction is much less important, the critical velocity gradient concept is able to partially capture flashback characteristics. Historically, the critical velocity gradient concept predicts trends of flashback behavior in laminar configurations for a wide range of temperatures, pressures, and fuel compositions more consistently than in turbulent configurations. This is due in part to the fact that many laminar studies establish well behaved velocity conditions in the tube conveying the premixed reactants to the reaction zone. Yet many important practical systems are in the turbulent regime and cannot be approximated by a simplified analysis. Studies to date in either regime, while numerous, generally do not provide a comprehensive methodology for accounting for all parameters. Recent work has attempted to capture the effect of a large number of these parameters in the turbulent regime, with some emphasis on providing design tools that can be used to estimate flashback propensity in more general terms. These approaches have demonstrated reasonable performance for the limited data available at elevated temperature and pressure which are representative of important practical system such as lean premixed combustors for gas turbines. While progress has been made in the last few years relative to predicting flashback for practical systems with high Reynolds numbers, only limited data are available for developing and validating correlations. Open questions remain in terms of using detailed numerical simulations and complex reaction chemistry to predict flashback for unconfined flames. In addition, flame-wall interaction in terms of heat transfer, sensitivity to turbulence levels, the role of general velocity gradients (vs idealized fully developed flow), and the role of high pressure must be further evaluated.

Keywords: Premixed flames; Boundary layer flashback; Laminar; Turbulent; Burning velocity, Quenching distance

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1. Introduction

The combustion of fossil fuels is expected to play a key role as a power generation source in the foreseeable future. Petroleum and natural gas are an essential part of total power generation in many countries worldwide. Combustion driven engines are utilized to produce the required energy consumption by residential, commercial and manufacturing sectors. Combustion engines, including gas turbines, not only have high reliability, but also offer an excellent load following ability [1,2]. Given the inherent advantages of gas turbines, they have been consistently evolving to improve efficiency while also using an increasingly variable fuel composition [1,3]. As manufacturers develop more robust gas turbines in terms of performance and stability, pollutant emission remains a key concern. Environmental awareness has prompted development of combustion systems with remarkably low pollutant emission levels [4,5]. To achieve this goal, premixed combustion is implemented along with a precise control of the temperature inside the combustor [1,3,6]. The pollutant emission of a gas turbine is significantly controlled by decreasing temperature spatial variation and operating within the lean combustion limit. However, operability issues including lean blow-off, auto-ignition, flashback, and dynamic stability are challenging due to the intrinsic characteristics of premixed flames [1,3,6]. Current gas turbines operate within a narrow range of fuel/air ratio to avoid these aforementioned problems. As a result, the operating range of an engine must be determined for various fuels prior to its design and development. Experimental data, theories, and numerical modeling can effectively contribute to evaluate the combustion characteristics under various operating conditions. Operability issues in an engine can cause severe damage to the engine and the stability of the combustion system has been extensively studied in past few decades to address this problem [6,7].

In recent years, the ability to extract natural gas using advanced hydraulic fracturing technology has led to an abundance of this fuel [8,9]. Natural gas is appealing due to its inherently low emissions compared to liquid and solid fossil fuels. However, the combustion of all fossil derived hydrocarbons

generates pollutants including CO, unburned hydrocarbons (UHC), particular matter (C), and especially CO₂, a greenhouse gas produced by complete combustion of fuel. As a result, while natural gas is currently plentiful, near term strategies for using fuel derived from renewable sources (e.g., alternative fuels) must still be pursued if a reduction in carbon signature is to be achieved [10]. Using alternative fuels is an important step providing a realistic solution toward resolving a long term problem. For example, the combustion of hydrogen eliminates the direct generation of carbon species. Steam reforming of natural gas is a common method of hydrogen production [11,12] in which hydrogen-rich content fuels can be derived and combusted in a gas turbine while allowing management of the associated carbon [13]. In terms of coal as a fuel, integrated gasification combined cycle (IGCC) represents a technology to manage CO₂ emission for clean coal applications [14,15]. The synthesis gas produced through the IGCC process contains mainly hydrogen and carbon monoxide, the latter of which can be further shifted to H₂ and CO₂ through reactions with water [13]. The captured CO₂ can be transported and stored in suitable places, playing an important role in reducing greenhouse emission and mitigating climate change [16,17]. Hydrogen can also be produced renewably via electrolysis of water using electricity generated by wind or photovoltaic [18]. This hydrogen can be added to the existing natural gas infrastructure or combusted directly.

In light of these developments with increased consideration for carbon management, it is apparent that fuel flexibility remains a key consideration for gas turbine systems. It also underscores the importance of high hydrogen content fuels [19] for which the fuel/air mixture preparation and subsequent combustion behavior needs to be well understood. It is clear that syngas composition changes the flame behavior [6]. Of particular relevance to lean premixed gas turbines combustion systems is determination of how fuel composition impacts flashback propensity.

Flashback has been the subject of many studies in the past few decades. This paper summarizes the major findings from research dating to the 1940's, when a systematic experiment was developed to

investigate boundary layer flashback. Furthermore, recent progress, technical challenges, and recommendations for future studies are discussed. Despite the fact that prediction of flashback propensity is crucial to overcome the operability issues associated with gas turbines, the lack of a comprehensive study to summarize all observations, conclusions, and theories developed in the past is still apparent. This paper summarizes the main contributions of related research and is organized such that the reader can locate a specific topic and obtain an overview of previous studies.

2. Flashback Mechanisms

Flashback occurs when the flame propagates from the combustion zone into the premixing section of the combustor. Flame flashback significantly increases the temperature inside the premixer, which is not originally designed for this purpose. Furthermore, if the flame anchors within the wake of existing geometric features inside the premixer passage ways, it can cause serious damage to the engine. Flashback in gas turbine premixers can occur as a result of at least four mechanisms [6,20,21]: (1) core flow flashback, (2) combustion instability induced flashback, (3) combustion induced vortex breakdown (CIVB), and (4) boundary layer flashback. Each is briefly introduced for context.

2.1 Core Flow Flashback

Flame propagation in the core flow occurs when the turbulent burning velocity exceeds the local flow velocity in the core flow [6,20]. The turbulent burning velocity magnitude plays the main role in triggering the core flow flashback and is a function of turbulent-flame interaction as well as chemical kinetics [22]. As a result, the fuel composition and turbulence structure are essential parameters for determining the core flow flashback limits [23,24]. A typical combustor is designed to avoid core flashback by simply increasing the axial velocity. However, common flame stabilization methods used in practical devices may lead to core flashback. For instance, swirl significantly decreases the local axial velocity component. In addition, flame stretching caused by the vortex-flame interaction increases the turbulent flame speed

[25–27]. Consequently, these two effects, higher turbulent flame speed and lower flow axial velocity, can lead to core flow flashback.

The key characteristics of core flashback is the turbulent burning velocity which has been extensively studied in the literature and is often presented as a function of laminar flame speed and turbulence characteristics [28–31]. Fuels with higher flame speed, such as hydrogen, increase the potential risk of flashback compared to natural gas. Several correlations for turbulent burning velocity have shown promise as a prediction tool for core flow flashback [22,32]. However, these correlations cannot be generally extended to various fuels since the turbulent burning velocity is strongly dependent upon fuel compositions [23,24]. In addition, different concepts associated with the definition of turbulent burning velocity result in different values, which also contributes to inconsistency among the existing data in the literature and challenges in interpretation [22].

2.2 Combustion Instability Induced Flashback

Flashback due to combustion instability occurs as a result of large amplitude fluctuations of the flow field [6,21]. Instabilities can be generated by the interaction of the acoustic modes, unsteady heat release, and flow structure. Pressure and velocity oscillations associated with the instability can induce the flame movement and generate large vortices leading to the flow reversal [33], which can potentially result in flashback. Figure 1 indicates the instability of the flame stabilized by backward-facing step [33]. The pulsation cycle of the flow structure leads to the lifting of the flame from the edge of step and eventually triggers flashback. A large eddy simulation (LES) performed by Thibaut et al. [21] investigated the flashback induced by the flame instability. The interactions between acoustic, turbulence, and combustion were effectively captured by imposing an external velocity modulation at the inlet of the computational domain, illustrated in Figure 2. As the flow reversal is enhanced by the instability process, the flame propagates upstream of the step.

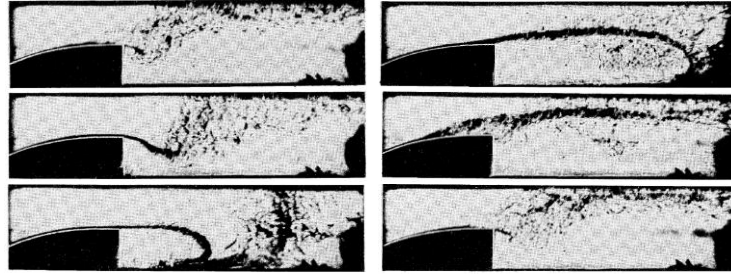


Figure 1: Combustion instability of a backward-facing step and cyclic process of the flame-flow interaction investigated by Keller et al. [33].

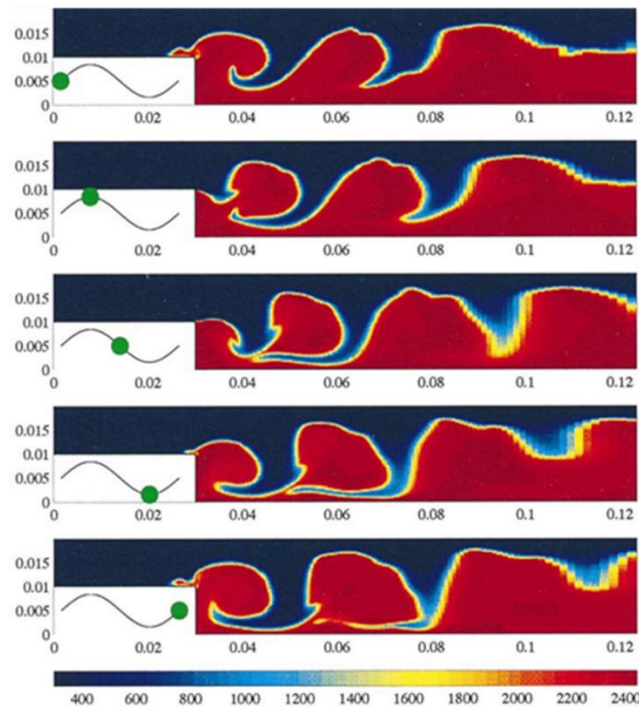


Figure 2: Temperature distribution of premixed flames stabilized by a backward-facing step obtained from two-dimensional LES [21].

2.3 Combustion Induced Vortex Breakdown

CIVB flashback is a prevalent mechanism in swirl stabilized gas turbine combustors. Vortex breakdown is a unique characteristic of swirling flows and depends on the swirl number, which describes the relative amount of swirl strength. If the swirl number is above a critical value, it leads to generation of vortex breakdown [34,35]; otherwise reverse flow would not occur. An increase in azimuthal velocity relative to axial velocity causes vortex breakdown and creation of a reverse flow region. Vortex break down is an

abrupt change in the vortex structure leading to the formation of a stagnation point and recirculation region downstream of it. In a study by Konle et al. [36], particle image velocimetry (PIV)-laser induced fluorescence (LIF) measurements with high temporal resolution identified the formation of a closed bubble at the tip of the internal recirculation zone, which is induced by small changes of the flow field as indicated in Figure 3. Positive vorticity generated by the volumetric expansion tends to move the bubble downstream and stabilizes the flame; however, the baroclinic torque produces negative vorticity, enhancing the negative axial velocity and causing the upstream propagation of the bubble [37–40].

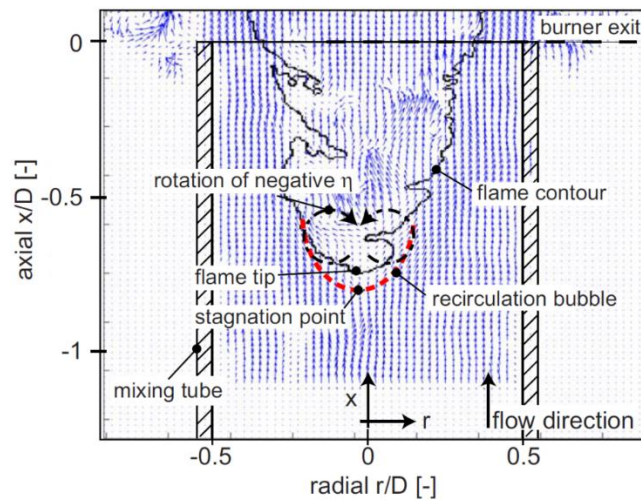


Figure 3: Instantaneous image showing the location of the flame front relative to the recirculation bubble studied by Konle et al. [36,41].

A competition between the volume expansion and baroclinic torques dictates the movement of the bubble and determines whether fast flame propagation occurs. The position of flame front relative to the recirculation bubble was observed to be a critical parameter in triggering CIVB flashback [36]. A flame front located downstream of the stagnation point provides a favorable condition for the generation of reverse flow caused by baroclinic torque. However, if the flame anchors upstream of the stagnation point, the volume expansion term compensates for an increase of baroclinic torque. Indeed, the flame tip location needs to be downstream of the stagnation point in order to propagate upstream; otherwise, the

flame remains stable. For the former case, the bubble propagates upstream under special circumstances and moves the flame front into the premixing zone, which can then lead to flashback.

A number of experimental studies investigated the effects of syngas compositions on CIVB flashback. Various fuel compositions, representing actual syngas mixtures [42] have been considered, and the results indicate that fuels with higher hydrogen concentration tends to have higher propensity for flashback via the CIVB mechanism. In general, for swirl stabilized systems, it is noted that (1) flashback characteristics depends on the swirl number and (2) multiple flashback mechanisms may result depending on swirl strength which creates some challenges in interpreting flashback for these systems. Some work has also been directed at this issue by varying the swirl number and indicated that the upstream flame propagation speed was significantly different based on flashback mechanisms [43].

2.4 Boundary Layer Flashback

Boundary layer flashback is considered the primary mechanism of flashback in jet/Bunsen flames and has been studied extensively in the literature. The free stream velocity of a gas turbine pre-mixer during regular operating conditions exceeds the turbulent flame speed. As a result, the upstream flame propagation cannot occur in the core flow region. But the flow velocity decreases close to the wall due to no-slip boundary condition and this can be a potential route for the onset of flashback. The burning velocity also decreases near the so called “quenching distance” where the reaction cannot sustain itself due to heat loss or flame stretch [44–46]. Hence, flashback occurs at a location where the local burning velocity is large enough to overcome the local flow velocity near the wall. Boundary layer flashback has been characterized by the “critical velocity gradient” concept in the literature [47]. The critical velocity gradient corresponds to the bulk flow velocity at the condition of flashback and thus is an indicator of the flashback propensity of a mixture for a given condition. Boundary layer flashback has been studied under laminar and turbulent flow conditions for different parameters including fuel/oxidizer composition, preheat temperature, operating pressure, burner material, and geometrical burner configuration. It is noted that

the most recent research has focused on high hydrogen content fuels as interest in the use of low carbon fuels increases.

With a brief overview of known flashback mechanisms now provided, emphasis is now focused on the boundary layer mechanism which is the primary mechanism considered in this review.

3. Boundary Layer Flashback Mechanism

Figure 4 summarizes studies from 1943 to 2017 that address boundary layer flashback in non-swirling premixed flames, including published papers and technical reports.

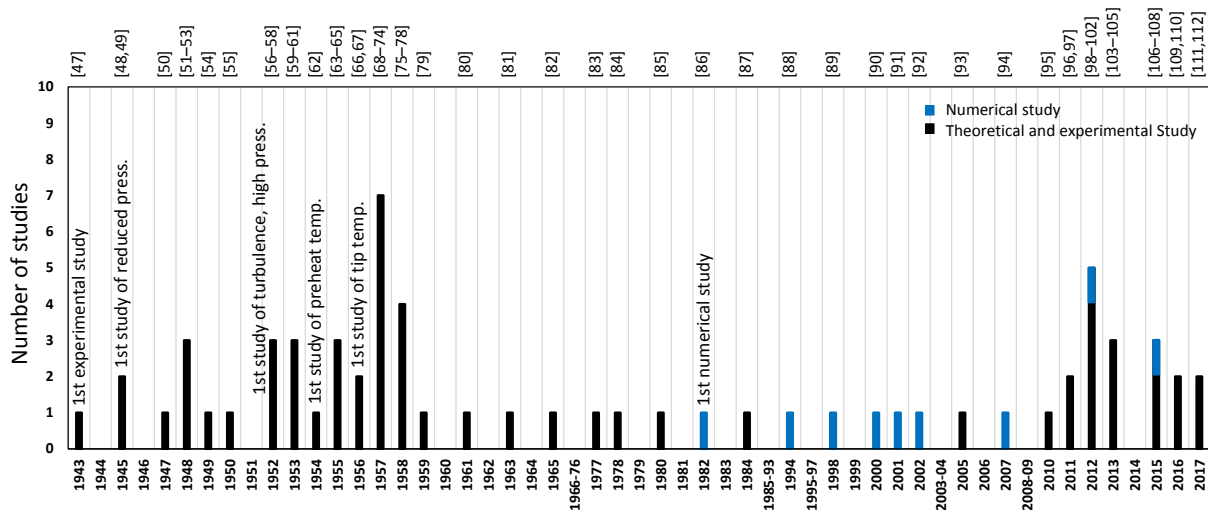


Figure 4: Number of studies per year addressing boundary layer flashback.

Lewis and Von Elbe first conducted a systematic experiment and developed a prediction model for boundary layer flashback [47]. Garside et al. carried out the first flashback experiment at sub-atmospheric pressure [49], while Edse first studied the turbulent boundary layer flashback at high pressure [56]. The effect of preheat temperature and tip temperature were first investigated by Grumer and Harris [62] and Bollinger and Edse [66], respectively. The first numerical study was conducted by Lee and T'ien [86] for propagation of a premixed laminar flame through a tube.

3.1 Simple Model for Boundary Layer Flashback Prediction

The critical velocity gradient model neglects the flame-flow interaction and assumes that the flow velocity remains undisturbed. Figure 5 indicates the flame front stabilized inside a burner close to the wall. The velocity profile of the unburned mixture $u(y)$ represents a laminar flow approaching the flame front. The flame speed denoted by $S_f(y)$ decreases close to the wall due to the heat loss and quenching of radicals. The flame is quenched at a certain distance from the wall and part of the flame close to the wall trails behind the central position as indicated in Figure 5 [47,48,63]. The quenching distance in Figure 5 is denoted by δ_q . The velocity gradient g at the wall in Eq. 1 is used to describe the flashback, in which τ_w is the shear stress and μ_u is the viscosity of the unburned mixture.

$$g = \left| \frac{\partial u}{\partial y} \right| = \frac{|\tau_w|}{\mu_u} \quad \text{Eq. 1}$$

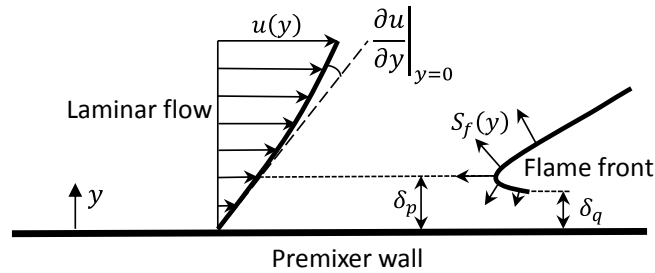


Figure 5: Schematic of the critical velocity gradient prediction model.

Based on this theory, flashback occurs when the flow velocity drops below the burning velocity somewhere in the flow field close to the wall. The location at which the burning velocity is equal to the flow velocity is defined as the penetration distance δ_p [113], indicated in Figure 5. The magnitude of the flow bulk velocity at the onset of flashback corresponds to the critical velocity gradient at the wall, g_c . Hence, the flame does not propagate as long as the velocity gradient is higher than this critical value (i.e. g_c).

To explain this concept in more detail, a tube burner is considered in which the flame stabilizes downstream of the burner rim. Figure 6 represents several flow conditions relative to the propagation of

the flame into the tube [47]. The burning velocity is zero within the quenching distance and it gradually increases to its constant normal value corresponding to the one-dimensional laminar flame speed. Both quenching and penetration distances are indicated in Figure 6. Line (2) indicates the critical velocity gradient condition at which flashback is possible. As the flow velocity is decreased (line (1)), the flame flashes back and propagates upstream. However, if the flow velocity is further increased than the critical value (line (3)), the flame is swept out of the tube.

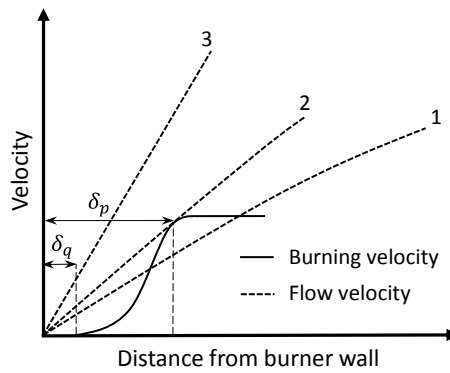


Figure 6: Critical velocity gradient model showing the flow velocity and burning velocity profiles at flashback condition, adapted from [47].

The velocity gradient at the wall is approximated by the ratio of the flow velocity to the penetration distance as follows in Eq. 2. This is based on the assumption that the flow velocity remains linear close to the wall.

$$g_c = \frac{u|_{y=\delta_p}}{\delta_p} \quad \text{Eq. 2}$$

The flow velocity at the penetration distance is equal to the burning velocity at the critical condition. Furthermore, the burning velocity magnitude at the penetration distance is assumed to be close to its normal value and can thus be approximated by the laminar flame speed S_L [47]. The laminar flame speed is calculated based on the one-dimensional steady flow assumption. With these simplifications, Eq. 2 reduces to:

$$g_c = \frac{S_L}{\delta_p} \quad \text{Eq. 3}$$

For fully developed laminar flow in a pipe with the radius of R , the velocity gradient can be analytically determined from Eq. 4.

$$u(r) = 2\bar{U} \left(1 - \left(\frac{r}{R} \right)^2 \right) \quad \text{Eq. 4}$$

in which r is radial distance and $\bar{U}$ is the bulk flow velocity. Combining Eq. 3 with the derivative of Eq. 4 at the tube wall ($r = R$), Eq. 5 is obtained at flashback condition (see, also [53]):

$$g_c = \frac{S_L}{\delta_p} = \frac{4\bar{U}}{R} \quad \text{Eq. 5}$$

The critical velocity gradient concept has been frequently used in the literature to characterize boundary layer flashback. It was used as a means to evaluate flashback propensity of a given fuel at different operating conditions. Regarding fuel composition impacts, it is evident that the influence of fuel composition on laminar flame speed and penetration distance can capture the effects.

While Eq. 3 was originally proposed to predict the flashback propensity of laminar flames [47], the concept was extended to turbulent conditions. For fully developed turbulent flow in a pipe upstream of location which the flame stabilizes, the velocity gradient is estimated from the Blasius correlations for smooth pipes with the diameter of D at a given Reynolds number Re [114]:

$$g = \frac{1}{8} \frac{f \bar{U}^2}{\nu} \quad \text{Eq. 6}$$

$$f = \frac{0.3164}{Re^{0.25}}, \quad 4 \times 10^3 < Re < 10^5 \quad \text{Eq. 7}$$

$$g = 0.03955 Re^{0.75} \frac{\bar{U}}{D} \quad \text{Eq. 8}$$

in which f and ν are friction factor and kinematic viscosity, respectively. Eq. 3 and Eq. 8 are used together to determine the flashback propensity of turbulent flames [63].

The only parameter not fully described in this simple analysis is the penetration distance for a specific fuel or distance from the wall at which the flow velocity is equal to the burning velocity. Indeed,

the penetration distance in Eq. 3 needs to be determined in order to predict flashback propensity of different fuel compositions. The overall variation of penetration distance is considered to be proportional to the quenching distance, as frequently reported in the literature [48,51,61,69,76,82,85]. Flame quenching near the wall can be categorized into three situations: (1) quenching between two parallel plates or inside a tube, (2) head-on quenching, and (3) sidewall quenching. The first quenching type is defined as a minimum distance between two parallel plates (or minimum tube diameter) at which the flame can propagate through the channel (or tube). Many experimental studies have been conducted to investigate the relation between the passage characteristic dimension as a function of fuel variation, equivalence ratio, pressure, wall temperature, and material. It is difficult to isolate the effect of one variable since all parameters are coupled and cannot be changed independently. Nevertheless, different studies have tried to address this problem by investigating various parameters [44,115–117] and finding empirical relations [118–121].

The measurement of quenching distance for a single wall is more complicated. Head-on quenching is defined as the flame propagates perpendicular to the wall (studied in [122–125]) as indicated in Figure 7, while for sidewall quenching, the flame propagates parallel to the cold wall [45,46,126]. In boundary layer flashback, the flame propagates upstream along the wall in which the sidewall quenching plays an important role. However, it is noted that the quenching distance between two parallel plates (or inside a tube) has been used in the literature to characterize boundary layer flashback [48,51,69,76,85].

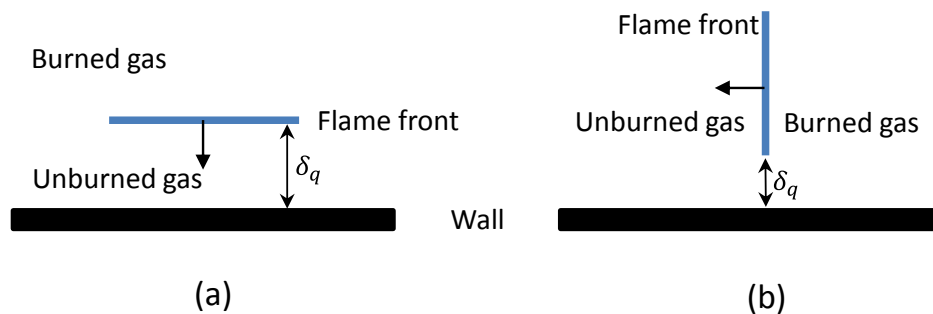


Figure 7: Flame-wall interaction: (a) head-on quenching and (b) sidewall quenching.

The critical velocity gradient model originally developed neglects flame-flow interaction which is a significant simplification. As suggested in Figure 6, the flow velocity profile at the onset of flashback was assumed to not be influenced by the flame. In addition, the burning velocity profile is assumed to only be affected by the wall and consideration of quenching distance. Yet this approach has been widely implemented in the literature and has proven capable of correlating the experimental data for laminar and certain types of turbulent flames, as will be shown in Sections 4 and 5, respectively.

3.2 Underlying Physics of Boundary Layer Flashback

More recent experimental and numerical simulations reveal that the flame-flow interaction near the wall can have a significant effect on flow velocity distribution for both laminar and turbulent flows at the onset of flashback [99,100,106,108]. In practical devices, the premixed fuel/air mixture is injected into the combustor by the burner premixer. At normal operating conditions, the flame stabilizes past the burner rim due to the sudden expansion as the flow is injected from the burner into the combustion chamber. The schematic of the stable flame front is shown in Figure 8(a). When the boundary layer flashback limit is reached, the flame propagates upstream into premixed reactants, indicated in Figure 8(b).

As a result, it is apparent that two regimes must be considered for physical interpretation and modeling. The first is the regime prior to transition of the stabilized flame at the burner rim into the injector. This regime is labeled as an “unconfined flame” situation. Once the flame enters the premixer/injector and propagates upstream along the wall, the underlying physics of the problem are different. This second regime is labeled as a “confined flame” throughout the paper.

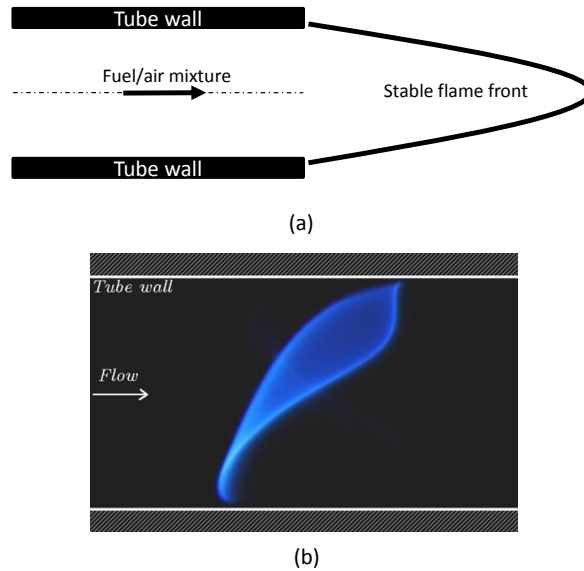


Figure 8: Flame configuration: (a) unconfined flame stabilized downstream of the premixer tube and (b) confined flame stabilized in a tube [94].

Until the transition of the unconfined flame from the burner rim into the premixer tube, the flame does not strongly affect the velocity distribution of the incoming flow and the flow velocity profile upstream of the flame front remains approximately undisturbed as experimentally observed and discussed in [106]. This is the main reason why the critical velocity gradient concept originally conceived was able to characterize boundary layer flashback propensity, especially under laminar conditions.

However, once the flame propagates into the burner (confined flames), the basic assumption of the critical velocity gradient concept is not valid [99,100]. The flow-flame interaction near the wall-region for confined flames cannot be neglected. The flame is strongly curved towards the unburned mixture as flame propagates upstream leading to a divergence of streamlines and adverse pressure gradient. When the gas expansion is large enough relative to the local flow velocity near the wall, it can cause the boundary layer separation as discussed in more detail in Section 5.4.

3.3 Parameters Impacting Boundary Layer Flashback

Many parameters impact flashback propensity; as a result, isolating the effect of a specific variable is difficult. In general, all parameters can be categorized into the following groups:

- Flow and combustion characteristics
- Operating conditions
- Flame configuration and boundary layer heating

How each impacts flashback propensity is discussed in the sections below.

3.3.1 Flow and Combustion Characteristics

The flow bulk velocity directly impacts boundary layer flashback by changing the streamwise momentum near the wall-region. As the mean flow velocity increases, the local flame speed must increase in order to propagate upstream. Hence, flashback propensity occurs at fuel/air ratios nearer to those associated with the highest flame speed as the bulk flow velocity increases. Furthermore, the velocity profile of the incoming flow, where the flame stabilizes, affects the flashback limits. For instance, it was noted experimentally that when the incoming flow is not fully developed, the flashback propensity decreases due to the higher wall shear stress, which is a function of entrance length [83].

The boundary layer flashback limit is quite different under laminar and turbulent boundary conditions. The flame-wall interaction is more complex under turbulent conditions [127,128]. Depending on the turbulent Reynolds number, the turbulent combustion regime dictates the flame structure and defines how the small scale eddies interact with the flame [129,130]. Hence, the turbulent flame speed is also a function of turbulence intensity and consequently higher turbulence intensity results in higher flashback propensity [22].

Combustion characteristics of premixed flames are dependent upon the fuel composition. The mixture burning velocity plays a key role in triggering the flashback and is largely a function of chemical kinetic mechanisms. Furthermore, the thermal and radical quenching by the wall varies significantly with different fuel compositions which, in turn, leads to different flashback limits [48,57,65,69]. The high flashback propensity of hydrogen mixtures is a result of high burning velocity and small quenching

distance. The flame propagation is enhanced for low Lewis number premixed flames as verified for various fuel compositions by experimental and numerical studies [131–136].

3.3.2 Operating Conditions

Pressure and preheat temperature are two important factors in boundary layer flashback. An increase in preheat temperature increases the burning velocity significantly [137]. Consequently, boundary layer flashback propensity increases for higher preheat temperatures, which will be discussed in detail in Section 4.3 and 5.3.

Boundary layer flashback experiments at high pressure are especially germane to gas turbine applications. Conclusions and results drawn from atmospheric pressure tests must be further verified at high pressure prior to any general conclusion for systems operated at elevated pressure. It is noted that the effect of sub-atmospheric pressures on flashback has been studied [67,70,76], but these results cannot be extended to elevated pressures as will be discussed in Section 5.3.

3.3.3 Flame Configuration and Boundary Layer Heating

Experimental results have indicated that the flame-flow interaction near the wall-region cannot be neglected for flames propagating through a channel (confined flames) [99,100]. As will be discussed in Section 5.4.1, confined flames are more prone to flashback and the flame-wall interaction for confined flames has received considerable attention in recent years in both experimental and numerical modeling studies. When the flame propagates upstream in the confined configuration, the flame leading edge is convex towards the reactants, enhancing the flame stretch effect and causing the approach flow to decelerate.

Burner heads constructed from different materials show varying flashback propensity [66,98,103,104]. The effect of thermal coupling between the burner head and flame on flashback propensity has been found to be an important parameter. A qualitative relationship has been observed between flashback and the variables involved in the construction and operation of the burners

[66,98,103,104]. The flame temperature and product species also vary with fuel composition, in turn influencing the heat transfer between the burner rim and flame front (see Figure 8(a)). This can cause the burner tip temperature to change depending on the fuel compositions for a given equivalence ratio. The heat is conducted upstream along the burner, causing the heat addition from the wall into the incoming reactants. The boundary layer heating leads to a point of inflection in the velocity profile, which can induce flow separation. As a result, flashback propensity is more pronounced for uncooled burners. Unsteady transitions in operating conditions (e.g. a sudden drop in reactant flow rates) can also trigger boundary layer flashback [138]. It is also noted that the flame, which does not initially flash back at a given condition, can eventually flash back after long periods of operation due to small perturbations which may lead to an increase in the premixer wall temperature.

Using an enclosure about an unconfined flame can also change the flashback propensity limits due to resulting recirculation regions in the sudden expansion region and change in the local heat transfer characteristics which will be explained in Section 5.2 [98,104].

4. Laminar Flames

Determination of limits for boundary layer flashback of laminar flames received much attention in the early literature. Different basic models and correlations, including Lewis and von Elbe's critical velocity concept, were developed in order to characterize the flashback propensity. These prediction models were used to compare the experimental results with various fuel compositions, preheat temperatures, and operating pressures. In addition, the sensitivity of flashback limit to the burner diameter, ranging from quenching limit to large values, was evaluated for different fuel/oxidizer mixtures. Numerical simulations of laminar flame provided more detail regarding the flame-flow interaction and have been used to reproduce the flashback limits obtained from experimental results. The detailed discussion is presented in following sections.

4.1 Prediction Models

As mentioned previously, Lewis and Von Elbe [47] developed the critical velocity gradient model to predict the flashback propensity of jet flames. The penetration distance of the flame near the wall calculated from Eq. 3 was found to be about the half of quenching distance of a flame through plane parallel plates and predicted to be one-third of quenching distance for cylindrical tubes. This concept was further developed by using non-dimensional parameters. Putnam et al. used Peclet numbers to present the critical velocity gradient model in non-dimensional form [52]. Peclet number represents the balance between the characteristic time scales of the flow and chemical reaction [52,139,140]. Based on this approach, flow and flame Peclet numbers are defined for a tube diameter of D in Eq. 9 and Eq. 10:

$$Pe_{flow} = \frac{D\bar{U}}{\alpha} \quad \text{Eq. 9}$$

$$Pe_{flame} = \frac{DS_L}{\alpha} \quad \text{Eq. 10}$$

where α is thermal diffusivity and $\bar{U}$ is the flow bulk velocity. In this model, the penetration distance (δ_p) is considered to be a multiple of the laminar flame thickness calculated from one-dimensional steady analysis as follows [141]:

$$\delta_p = K \frac{\alpha}{S_L} \quad \text{Eq. 11}$$

in which K is an arbitrary constant. Substituting Eq. 11 in Eq. 5 and using D for the burner diameter, this equation is written as:

$$\frac{8\bar{U}}{D} = \frac{S_L^2}{K\alpha} \quad \text{Eq. 12}$$

Multiplying both side of Eq. 12 by D^2 and dividing by α , this equation can be simplified to the non-dimensional form [52]:

$$Pe_{flow} = \frac{1}{8K} (Pe_{flame})^2 \quad \text{Eq. 13}$$

This approach assumes a linear velocity profile near the wall. Furthermore, the burning velocity is assumed to be equal to the one-dimensional laminar flame speed, which is not affected by the quenching in the vicinity of the tube wall. These assumptions were relaxed to improve the prediction model developed in [52]. Indeed, the penetration distance δ_p at which the flow velocity equals the flame speed is an unknown parameter and must be determined. Figure 9 illustrates the flow and flame velocity profiles at the onset of flashback. Parameter e denotes the distance from the wall at which the burning velocity starts falling as a parabolic curve to zero value at the distance, d . Therefore, the flame velocity profile close to the wall can be written as [52]:

$$S_f(r) = S_L \left(1 - \frac{(r + e - R)^2}{d^2} \right) \quad \text{Eq. 14}$$

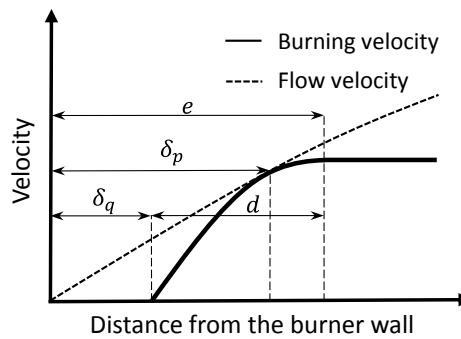


Figure 9. Schematic of the flame and flow velocity profiles for boundary layer flashback, adapted from [52].

in which r is the radial distance from the center of the tube with a radius of R . At the penetration distance, the flow velocity and flame speed are equal [52]:

$$u(r|_{\delta_p}) = 2\bar{U} \left(1 - \left(\frac{r}{R} \right)^2 \right) = S_f(r|_{\delta_p}) = S_L \left(1 - \frac{(r + e - R)^2}{d^2} \right) \quad \text{Eq. 15}$$

In addition, the flame and velocity profiles are tangent at the penetration distance as indicated in Figure 9 [52]:

$$\left. \frac{du}{dr} \right|_{\delta_p} = -4 \frac{\bar{U}}{R^2} r = \left. \frac{dS_f}{dr} \right|_{\delta_p} = -2S_L \left(\frac{r + e - R}{d^2} \right) \quad \text{Eq. 16}$$

The critical velocity at which flashback occurs is obtained by eliminating the r coordinate between Eq. 15 and Eq. 16, resulting in the following equation:

$$\left(\frac{2\bar{U}d}{S_L R}\right)^2 - \left(\frac{2\bar{U}d}{S_L R}\right)\left(\frac{d^2 - e^2 + 2eR}{Rd}\right) + 1 = 0 \quad \text{Eq. 17}$$

If the ratio of d/e is assumed to be β and the distance e is approximated by the planar flame thickness ($K\alpha/S_L$), Eq. 17 is simplified to the non-dimensional form as:

$$Pe_{flow} = \frac{Pe_{flame}^2}{4K\beta^2} \left(1 - \frac{K}{Pe_{flame}}(1 - \beta^2) - \sqrt{\left(1 - \frac{K}{Pe_{flame}}(1 - \beta^2) \right)^2 - \beta^2} \right) \quad \text{Eq. 18}$$

In this equation, it is assumed that $e \leq R$ or $Pe_{flame}/K \geq 2$ and the flame velocity in the vicinity of the wall (e) is affected by the quenching effect. For the special case where β equals zero (meaning that the burning velocity abruptly drops to zero from the normal value), Eq. 18 is simplified to:

$$Pe_{flow} = \frac{Pe_{flame}^2}{8K} \left(\frac{1}{1 - \frac{K}{Pe_{flame}}} \right) \quad \text{Eq. 19}$$

A similar equation was developed in [52] for $e > R$ or $Pe_{flame}/K < 2$ in which the burning velocity at the centerline of the tube burner is influenced by the wall quenching.

Figure 10 shows the performance of the Peclet number analysis, Eq. 13, for ethylene/air [50], acetylene/oxygen [49], and natural gas [48]. The solid line in this figure represents $Pe_{flow} = 0.0125(Pe_{flame})^2$ variation in logarithmic scales, where the slope of the trend line is equal to 2. As shown in Figure 10, good agreement between the model and data obtained using various fuels is apparent. However, for the simplified case (Eq. 19), the slope of log plot is always smaller than 2, depending on Pe_{flame}/K (see Eq. 20).

$$\frac{d(\log(Pe_{flow}))}{d(\log(Pe_{flame}))} = 2 - \frac{1}{Pe_{flame}/K - 1} \quad \text{Eq. 20}$$

The Peclet number analysis can predict the limited data for the laminar flame stabilized beyond a tube burner exit. However, significant deviation is observed at small Peclet numbers in Figure 10. The authors mention several effects as the main reasons for the observed deviation [52]. Namely that the equations derived do not take into account the difference between one-dimensional and two-dimensional heat transfer from the flame, the change in the flame front, and the change of the flow direction at the flame front. Furthermore, the quenching distance is dependent on the tube size and increases as the tube size decreases. A group of data at high Peclet numbers also deviates from the trend line in Figure 10 and this was associated with turbulent effects previously identified by von Elbe and Menster [48].

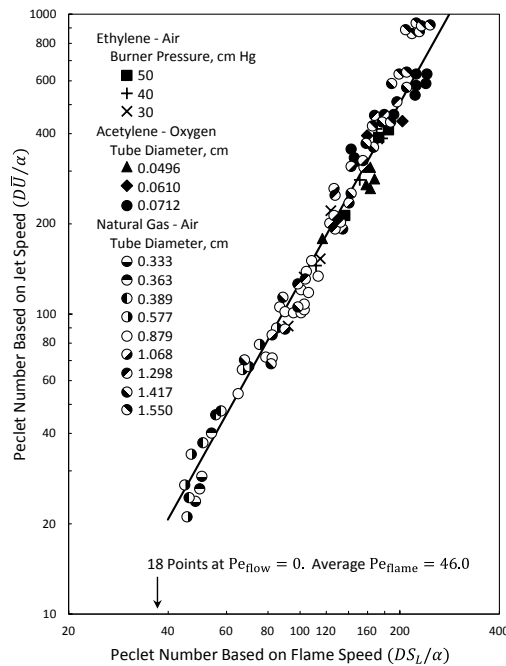


Figure 10: Peclet number non-dimensional analysis for boundary layer flashback, reproduced from [52].

Based on the conclusion made by Wohl et al. [61] who state that the penetration distance is proportional to flame thickness and quenching distance, Berlad et al. presented the critical velocity gradient model as a function of the quenching distance as follows [69]:

$$g_c = C \left(\frac{S_L}{\delta_q} \right) \quad \text{Eq. 21}$$

The penetration distance used by Von Elbe and Mentser [48] is about the half of the quenching distance for flame quenching between parallel plates. Assuming the quenching distance of circular tube is about 1.5 times that of parallel plates, the constant C should be 3 from Von Elbe and Mentser [48]. The application of Eq. 21 was further examined using a wide variety of published experimental data for various fuels [44,47,48,60,61,63,142]. Figure 11 presents the variation of critical velocity gradient versus the ratio of laminar flame speed to the quenching distance for various fuels at different pressures, preheat temperatures, and equivalence ratios. Based on this study, the value of C is not constant and increases from 2.5 to 5 as g_c increases from 10 to 10^5 1/s. Hence, the constant C can be represented as function of g_c itself, resulting in a power law variation suggested in [51] (Eq. 22). This correlation is indicated by the solid line in Figure 11.

$$g_c = 14.125 \left(\frac{S_L}{\delta_q} \right)^{1.168} \quad \text{Eq. 22}$$

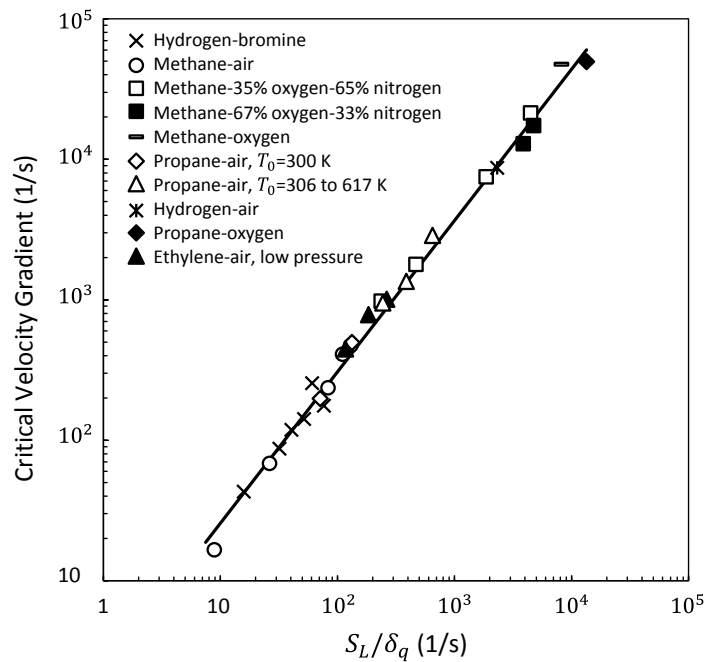


Figure 11: Critical velocity gradient variation using Eq. 22, reproduced from [69].

The effect of sub-atmospheric pressure on flashback was studied by Wohl [61]. The dependency of critical velocity gradient on pressure was determined from the original model developed in [47,48]. The

penetration distance is lower than the quenching distance and is approximated by the magnitude of the planar flame thickness δ_f as previously presented in Eq. 11.

$$g_c = \frac{S_L}{\delta_f} \quad \text{Eq. 23}$$

The variation of the laminar flame speed and the flame thickness with pressure are described in Eq. 24 and Eq. 25 [61] using the similar approach developed by Spalding [143], Mallard and Le Chatelier [144]:

$$S_L \propto P^{(n-2)/2} \quad \text{Eq. 24}$$

$$\delta_f \propto P^{-n/2} \quad \text{Eq. 25}$$

in which n is the global reaction order. As a result, the variation of critical velocity gradient with pressure according to Eq. 23 would be [61]:

$$g_c \propto P^{n-1} \quad \text{Eq. 26}$$

Assuming the reaction order of $1 < n < 2$, increasing pressure causes a decrease in burning velocity and flame thickness, while increases the critical velocity gradient.

The Damköhler number, defined as the flow time scale to the combustion time scale (Eq. 27), was also used as a non-dimensional parameter in numerical modeling of the laminar boundary layer flashback [90,94]. This non-dimensional group has been used in the literature to characterize flashback propensity of laminar as well as turbulent flames [90,94,110,138].

$$Da = \frac{S_L^2}{\alpha g_c} \quad \text{Eq. 27}$$

An increase in the laminar flame speed increases the Damköhler number which corresponds to higher flashback propensity. For laminar flows, the critical velocity gradient is small compared to turbulent flows.

Section 4.1 has summarized the key basic models developed for boundary layer flashback of laminar premixed flames. The main underlying assumption of predictive models is that the flame and flow do not interact. As a result, the laminar velocity profile is presumed to remain undisturbed at onset of flashback and is therefore known from the analytical solution of fully developed laminar flows in a circular

tube or two-dimensional channel. In addition, flame characteristics are described by one-dimensional flame propagation theory. By combining the premixed flame and flow characteristics, the critical condition at which flashback occurs is obtained.

4.2 Effect of Burner Diameter and Temperature

In their original work, Lewis and Von Elbe also investigated the effect of burner diameter on flashback behavior of fully developed laminar flows of natural gas air mixtures [47]. The tube burner was constructed from Pyrex with inner diameters varying from 0.315 to 1.55 *cm*. Between each test the burner was cooled down to the room temperature as the cooling effect was found an important parameter for obtaining consistent data. The inner surface of the Pyrex burner was also coated with various salts and metals to examine the effect of the surface material inside the tube. Because the heat conductivity of the burner is several times larger than that of a gas, the overall heat loss to the atmosphere is determined by the gas conductivity, not that of the burner. Hence, it was concluded that the wall temperature (up to 100 °C) had little effect on flashback propensity [47]. This is in good agreement with the study by Miller and Setzer [74], who concluded that the effect of premixer wall temperature is negligible below 115 °C. They did note that higher tip temperatures cause a strong effect on flashback limits. The effect of burner material on the tip temperature is discussed in more detail in Section 5.2.

Figure 12 presents the critical flow rates at which flashback occurs as a function of the mixture composition for burner diameters from 0.333 to 1.55 *cm*. The dashed curve shows the data at the beginning of the flame upstream propagation and the solid curve indicates “full flashback.” The data between these two curves (solid points) represents situations in which the flame cone was partially drawn into the tube. It was concluded that this resulted from some local high temperature regions on the burner rim due to “some irregularity” that was not further defined or discussed. The partially propagated flame increases the pressure inside the tube and causes a deflection of the streamlines in the opposite direction. The flame at this condition was described as a “tilted flame” in which the dynamic symmetry of the flame

was destroyed. The flame can propagate upstream unless the flame is quenched because of the tube cooling. The velocity gradient in this region, where the flame propagates into the burner, decreases due to the streamline distortion and consequently the flame flashes back below the critical flashback value. The velocity gradient of laminar flow in Eq. 5 was modified for tilted flames in [47] and later used by Grumer for predicting turbulent flashback limits [77].

The flow velocity at which flashback occurs increases with an increase in the burner diameter. Indeed, for a given volumetric air flow rate, the flow bulk velocity decreases with square of the diameter resulting in lower equivalence ratio or fuel percent. The points at zero flow rate show the inflammability limit of natural gas for upstream propagation.

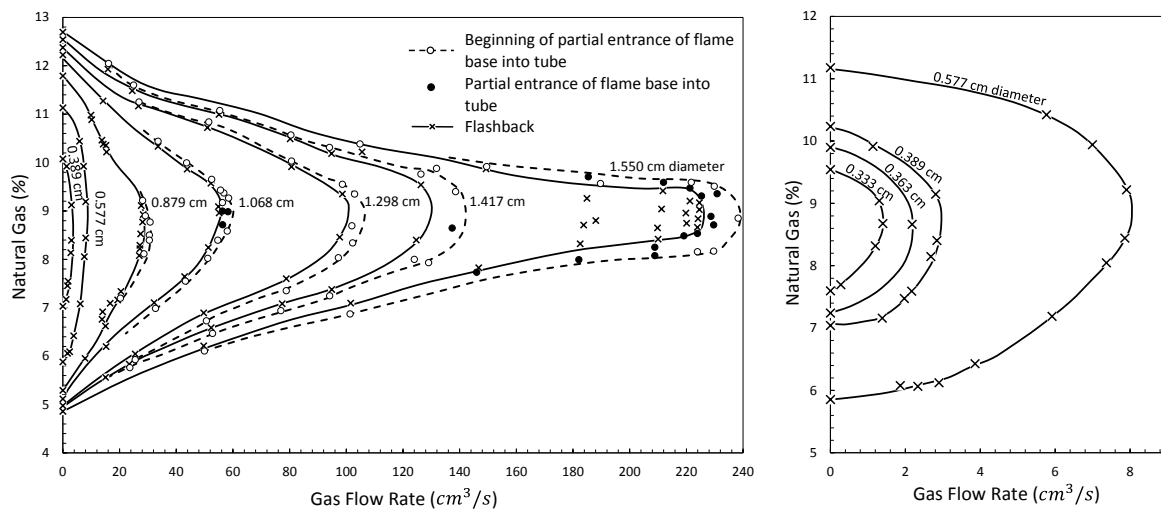


Figure 12: Flashback limit of natural gas as a function of volumetric flow rate for different burner diameters, reproduced from [47].

The variation of critical velocity gradient calculated from Eq. 5 is illustrated in Figure 13 as a function of natural gas concentration in air. The critical velocity gradient can correlate the effect of burner diameter for the diameter range studied. However, some discrepancy for the burner diameter of 1.55 *cm* is observed near stoichiometric conditions. This might be due to incipient turbulence at these conditions causing an increase the burning velocity [47]. The flame propagation is fully arrested for the limiting diameter for 0.315 *cm*. It is also observed that the burners with small diameters close to this value (0.333

and 0.389 cm) do not follow the general trend. This was explained by the fact that the flow velocity profile within the wall and the penetration distance for small burner diameters is not linear.

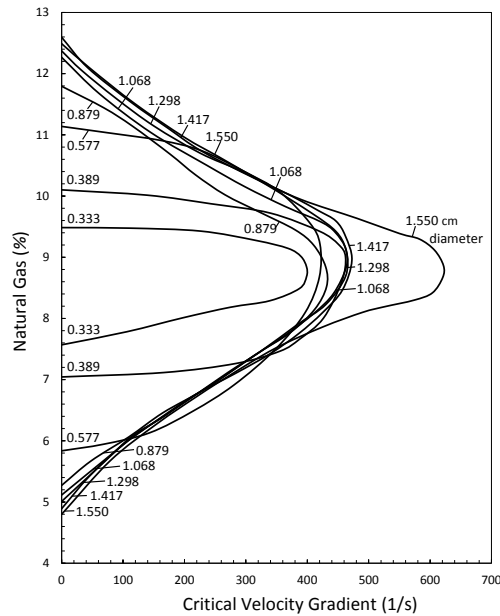


Figure 13: Critical velocity gradient of boundary layer flashback for natural gas at different burner diameters, adapted from [47].

Figure 14 presents another view of the effect of the burner diameter on flashback propensity at a lean condition (7.5% natural gas) [47]. At sufficiently large diameter, the critical velocity gradient remains approximately constant. The lack of diameter dependence was also observed by Kurz [73] for laminar flames for various hydrocarbon air mixtures shielded with inert gases of helium and nitrogen. However, the ratio of penetration distance to the burner diameter increases as the burner diameter decreases and finally reaches a point at which this value is no longer small enough to apply one of the underlying assumptions made for the critical velocity gradient concept. An underlying assumption of the critical velocity gradient concept is based on estimating the flow velocity at the penetration distance by the tangent to the velocity profile at the wall. This assumption is not valid when the penetration distance is comparable to the burner diameter; hence, the effect of the velocity profile curvature should be considered. With further decrease in burner diameter, the laminar burning velocity across the exit is

entirely affected by the quenching distance. For burner diameters less than 0.389 cm in Figure 14, the burning velocity is reduced, even at the burner axis, leading to a sudden decrease of critical velocity gradient. The flame is eventually quenched by the smallest diameter as indicated in Figure 14.

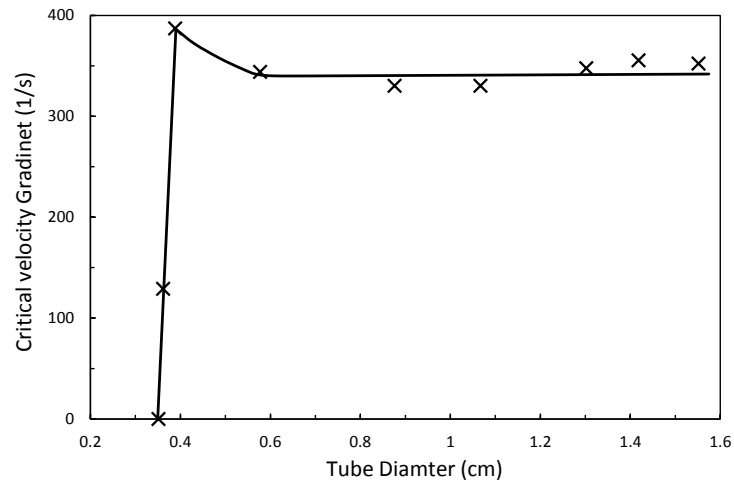


Figure 14: Critical velocity gradient variation of natural gas as a function of burner diameter at 7.5% natural gas-air mixture, reproduced from [47].

The qualitative effect of the burner diameter on critical velocity gradient shown in Figure 13 depends on the fuel concentration. For instance, the critical velocity gradient slightly increases with an increase in burner diameter at rich conditions. Hence, a general conclusion for variation of the critical velocity gradient with the burner diameter cannot be made regardless of the fuel percentage in the mixture. Furthermore, observations obtained from different studies are not entirely consistent. Figure 15 shows the flashback propensity of methane/air mixtures for two burner diameters at atmospheric pressure and room temperature [93]. The critical velocity gradient decreases with an increase in burner diameter at rich conditions. The burner tip temperature in the experiment [93] was not controlled during each test and values were not reported. This may be the main reason for such a difference in flashback limits compared to Figure 13.

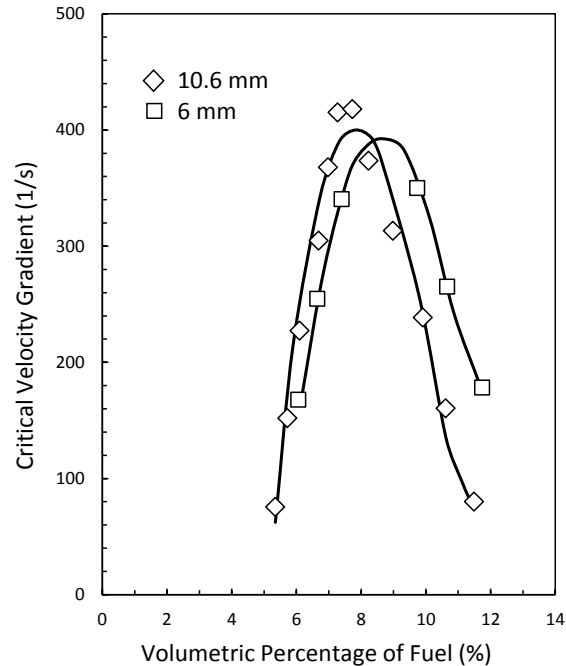


Figure 15: Critical velocity gradient of methane/air flames for two burner diameters, reproduced from [93].

In other work from von Elbe's group, the flashback limit of methane and propane with the mixture of oxygen and nitrogen was determined for laminar flames at atmospheric pressure and room temperature [51]. The flow was fully developed and the fuel/air mixture was injected through a Pyrex tube unless a constraint due to the flame temperature occurred. Near stoichiometric conditions for cases with oxygen used as an oxidizer, quartz, porcelain or copper were used interchangeably. The burner tube was cooled frequently between tests to avoid an increase in the temperature. Figure 16 indicates the variation of critical velocity gradient with the methane concentration in nitrogen/oxygen mixtures at burner diameters ranging from 1.397 to 18.07 mm. The burner diameter range was selected such that the Reynolds number remained less than 2000 for all cases. For methane/air reaction (labeled by the ratio of 0.21 in Figure 16), results of different burner diameters match well with the general trend line. However, when the burner diameter is comparable to the flame quenching distances, the deviation is pronounced as seen from the top curve in Figure 16. An increase of oxygen in the mixture does not change the dependency of the critical velocity gradient on burner diameter.

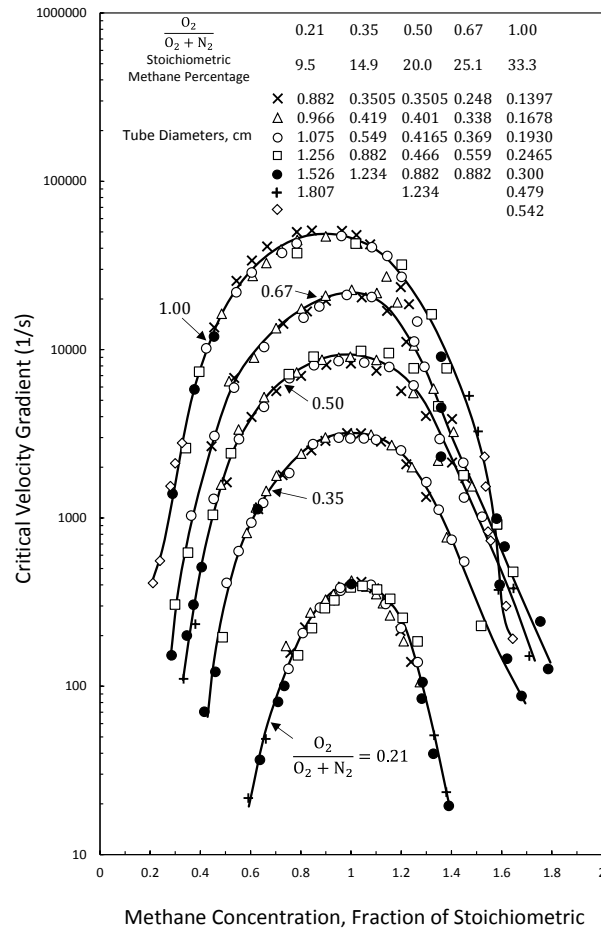


Figure 16: Critical velocity gradient variation of methane/oxygen-nitrogen mixtures at different tube burner diameters, reproduced from [51].

Figure 17 illustrates the variation of critical velocity gradient for propane/oxygen-nitrogen mixtures with different burner diameters. The propane/air mixture is not significantly influenced by changing the burner diameter. However, when the diameter decreases to a sufficiently small value, the quenching effect becomes more important. As discussed relative to Figure 13, for small diameters, the physical assumption of the linear velocity profile close to the wall used in the critical velocity gradient concept is not valid. A significant deviation from the solid curve (shown by diamond and cross shapes) is observed for propane/oxygen mixture in Figure 17. The effect of dilution at the burner rim for fuel rich mixtures becomes more important as oxidizer other than air is used.

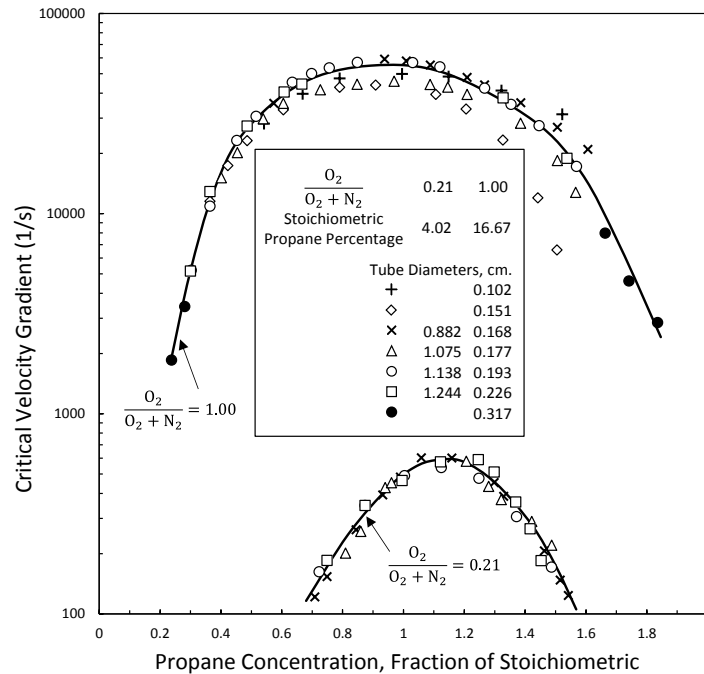


Figure 17: Critical velocity gradient variation of propane/oxygen-nitrogen mixtures at different tube burner diameters, reproduced from [51].

A more recent experiment studied the effect of burner diameter on laminar flashback of H_2 -CO and H_2 - CH_4 mixtures with variation of hydrogen from 5% to 25% in both mixtures [96]. Figure 18(a) indicates the variation of the critical velocity gradient of 25% H_2 -75%CO mixture at three different burner diameters, i.e. 6, 7, and 10.6 mm. In this study, the effect of burner diameter is negligible at lean conditions and becomes more pronounced as the equivalence ratio increases. Furthermore, the largest diameter in Figure 18(a) has the lowest critical velocity gradient for fuel rich mixtures. This observation was not reported by Lewis and Elbe [47] for methane/air mixtures and may be due to the thermal coupling effect between the burner head and flame.

The effect of burner diameter observed by Dam et al. on flashback propensity of 25% H_2 -75% CH_4 mixture is also illustrated in Figure 18(b). The effect of diameter is pronounced at lean condition. In addition, the 7 mm burner tube has the lowest critical velocity gradient. It is noted that the addition of 25% H_2 to the carbon monoxide or methane (Figure 18(a) and (b)) does not show the similar behavior.

The methane mixture is less affected by the hydrogen addition for fuel rich mixtures. The results suggest that the heat transfer to the burner rim and the quenching distance are complicated functions of fuel composition and influenced by the burner diameter at small diameters. The description of such a different characteristic is described in Section 4.4. It appears that no general conclusion can be made regarding the variation of critical velocity gradient with diameter according to the study by Dam et al. [96].

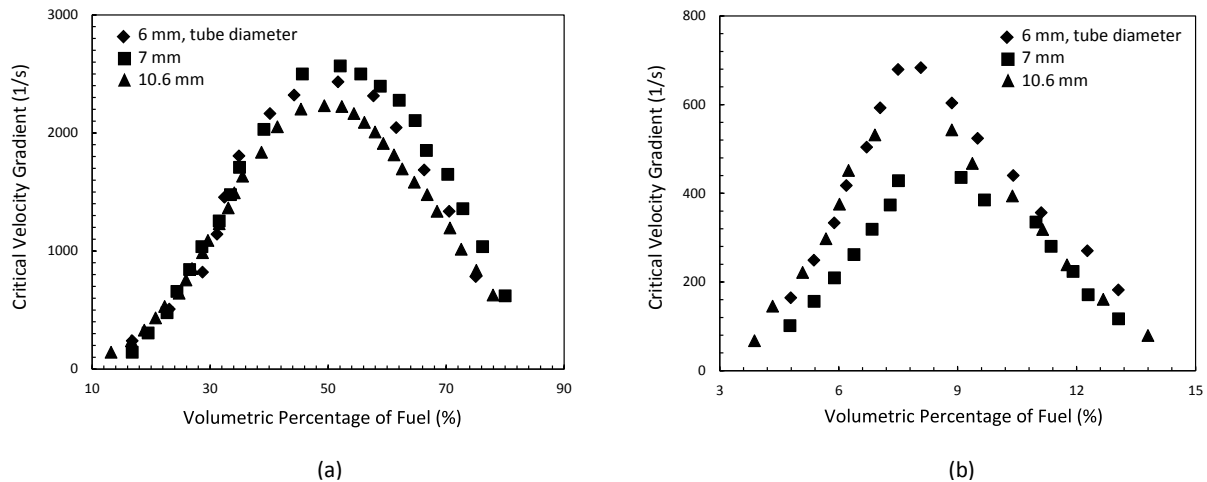


Figure 18: Critical velocity gradients as a function of fuel concentration for different burner diameters (a) 25%–75% H_2 –CO mixtures and (b) 25%–75% H_2 – CH_4 mixtures, reproduced from [96].

Early studies concluded that the critical velocity gradient is independent of burner diameter for sufficiently large values. However, it is noted that the effect of burner diameter cannot simply be extended to all equivalence ratio ranges and various fuel compositions. The size of the burner has an important effect on heat transfer between the burner rim and flame front. The complicated heat balance at the burner rim including convection of the gas flow mixture, radiation of products, and conduction of the burner material cannot be predicted by the simple critical velocity gradient model. Consequently, the variation of the burner diameter impacts differently on flashback limits for various fuel/air compositions. Furthermore, the heat balance between the burner rim and flame front is significantly affected by the equivalence ratio, determining the temperature of the stabilized flame. This is the main reason why experiments examining different fuel compositions do not show consistent results. In order to isolate the

effect of the burner diameter, the burner tip temperature needs to be monitored or controlled carefully. A small disturbance in flame front shape such as tilted flame can significantly change the tip temperature resulting in a different flashback limit. The size of burner (at sufficiently large values) does not significantly influence the flashback limits in terms of critical velocity gradient if the burner is cooled during the experiment and flashback is gradually approached such that the burner rim temperature remains at quasi-steady heat balance. However, this behavior changes when the quenching distance is comparable to the burner with small diameters.

4.3 Effect of Preheat Temperature and Operating Pressure

Dugger studied the effect of preheat temperature on flashback propensity of propane/air flames under laminar conditions [63]. The temperature of the reactants was varied from 300 to 620 K. The burner tube was constructed from brass with an inner diameter of 1.56 cm. The tube wall temperature was controlled such that it approximately remained constant at the temperature of the reactants. The Reynolds number values were less than 2000. Figure 19 presents the variation of critical velocity gradient with equivalence ratio at four different preheat temperatures of 306, 422, 506, and 617 K and illustrates that the critical velocity gradient increases with an increase in the preheat temperature. The laminar flame speed increases with temperature raised to an exponent which is a function of fuel composition [145–147]. Furthermore, the quenching distance decreases as the preheat temperature increases [44] and this effect is more pronounced in the experiment of Dugger [44] since the burner wall temperature was controlled to be the same as the preheat temperature.

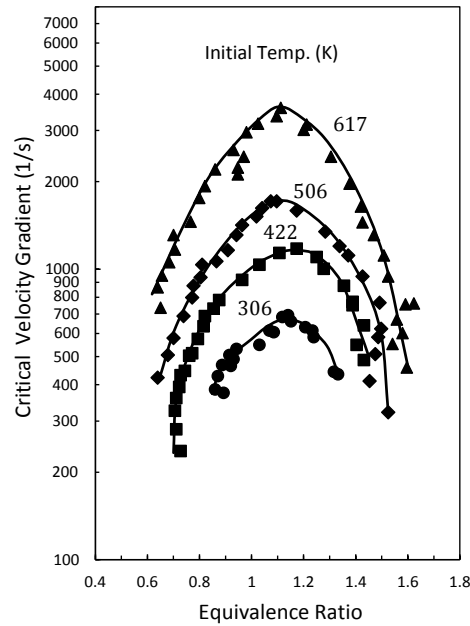


Figure 19: Effect of preheat temperature on flashback propensity of propane/air mixture, reproduced from [63].

To isolate the effect of laminar flame speed on flashback propensity, laminar propane/air mixture burning velocities are calculated from Eq. 28 as proposed by Metghalchi and Keck [145]. This expression is valid in the pressure range from 0.4 to 40 *atm*:

$$S_L = S_{L0} \left(\frac{T_u}{T_{u0}} \right)^\alpha \left(\frac{P}{P_0} \right)^\beta \quad \text{Eq. 28}$$

in which $T_{u0} = 298 \text{ K}$ and $P_0 = 1 \text{ atm}$. S_{L0} , α , and β are fitted constants presented in Table 1 [145]. Assuming an equivalence ratio of 1.2 in Figure 19, critical velocity gradients at four preheat temperatures are $g_{c0} = 640$, $1.79g_{c0}$, $2.426g_{c0}$ and $4.87g_{c0}$, respectively. Using Eq. 3 (in which $g_c \propto S_L$) and the laminar flame speed in Eq. 28, critical velocity gradient values can be calculated with respect to the reference preheat temperature of $T_0 = 306 \text{ K}$ and the reference critical velocity gradient of $g_{c0} = 640$.

Table 1: Constant values in Eq. 28 [145].

Equivalence ratio	$S_{L0} \text{ (cm/s)}$	α	β
1.2	33.8	2.02	-0.17

$$g_c = g_{c0} \left(\frac{T}{T_0} \right)^\alpha \quad \text{Eq. 29}$$

Based on Eq. 29, the critical velocity gradients at three preheat temperatures of 422, 506, and 617 K are $1.91g_{c0}$, $2.76g_{c0}$ and $4.12g_{c0}$, respectively. These predicted values at 422 and 506 K preheat temperatures are higher than the actual values obtained from measurements, while the predicted value at 617 K is lower. Consequently, the effect of preheat temperature on laminar burning velocity alone does not account for an increase in critical velocity gradient as its impact on quenching distance must also be accounted for.

A similar approach was used by Khitrin et al. [82] to study flashback propensity of methane/oxygen and methane/air mixtures with the preheat temperature varied from 20 to 400 °C. The study was conducted for quartz and metal tube burners with different diameters ranging from 18 to 25 mm. The temperature of the burner rim was maintained at the mixture preheat temperature to within 10 °C. Figure 20 illustrates the variation of the critical velocity gradient with equivalence ratio at five preheat temperatures. The slope of the critical velocity gradient is a function of equivalence ratio and exhibits more sensitivity as preheat increases.

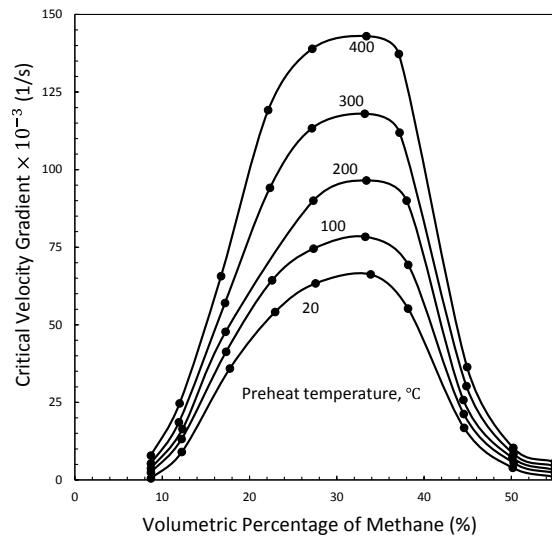


Figure 20: Critical velocity gradient of the methane/oxygen mixture at different preheat temperatures, reproduced from [82].

The measured data from Khitritin [82] are plotted in Figure 21 using the Peclet number (Eq. 13) as a non-dimensional parameter. The solid line shows the variation of $Pe_{flow} = 0.01(Pe_{flame})^2$, in which the constant differs from the dashed line suggested by Putnam and Jensen [52] representing $Pe_{flow} = 0.0125(Pe_{flame})^2$. Nevertheless, good agreement between the data obtained from various preheat temperatures and the correlations developed in previous studies is noted. This illustrates that the critical velocity gradient model is able to predict the preheat temperature effects. The difference between these two constants (i.e. 0.01 and 0.0125) may be due to the laminar flame speed measurement and its calibration.

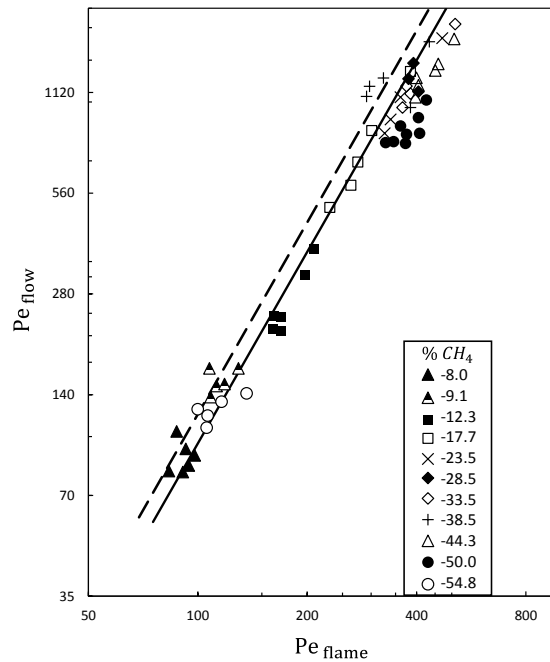


Figure 21: Relation between Pe_{flow} and Pe_{flame} for flashback propensity of methane/oxygen and methane/air mixtures, reproduced from [82].

The effect of preheat temperatures on flashback limit of methane/air mixtures was studied by Grumer and Harris [62] for sharp-edged short ports, which are common in gas appliances. Figure 22 shows the variation of critical velocity gradient as a function of preheat temperature for mixtures with given fractions of stoichiometric F (i.e. the fuel percentage divided by the stoichiometric fuel percentage). The

velocity profile for this burner was not fully developed and the critical velocity gradient was determined using an experimentally derived equation from [59,148]. It is noted that the variation of critical velocity gradient with preheat temperature shown in Figure 22 is better captured by exponential function rather than power law relation, which is generally used for laminar burning velocity correlations.

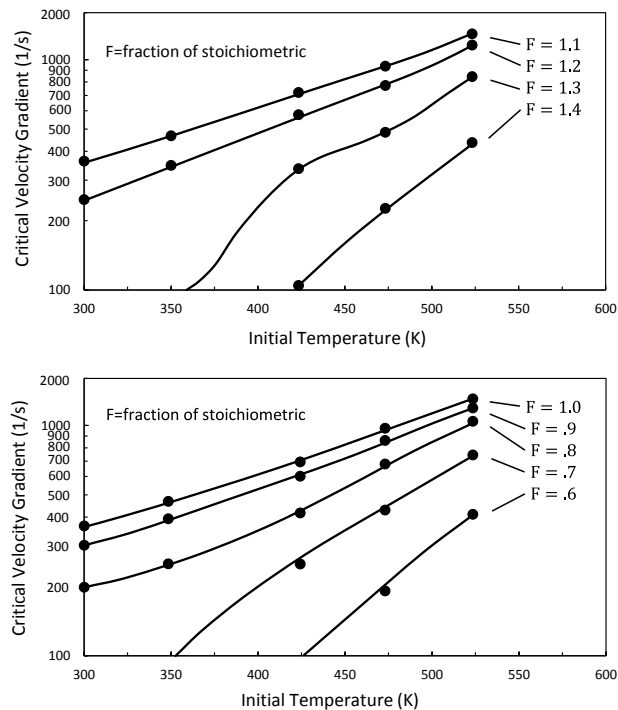


Figure 22: Critical velocity gradient of methane/air mixtures as a function of preheat temperature at various fractions of stoichiometric mixtures, reproduced from [62].

Fine conducted boundary layer flashback experiments for hydrogen/air mixtures at sub-atmospheric pressures varied from 0.15 to 1 atm [67,71]. The original model by Lewis and Von Elbe [47] was modified to take into account the effect of pressure. Flames were stabilized at the burner rim using burners with diameters of 0.546, 1.016, and 1.459 cm. The flow was fully developed at the exit section. The tip temperature was noted as a contributor to flashback and was thus controlled by water cooling to eliminate its effect. The pressure was increased slowly to approach flashback at constant mass flow rate and preheat temperature. The burning velocity was measured by determining the total flame area from

Schlieren images. The equivalence ratio was varied from 0.8 to 2.25. These tests were mainly obtained for rich conditions due to the high uncertainty at lower mass flow rates and characteristic of hydrogen flames.

Figure 23 shows the variation of the critical velocity gradient with pressure at equivalence ratio of 1.8 for three burner diameters studied by Fine [67,71]. The critical velocity gradient increases as the ambient pressure increases, resulting in higher flashback propensity. The critical velocity gradient dependency on pressure can be described as:

$$g_c \propto P^{n_f} \quad \text{Eq. 30}$$

in which n_f is the pressure exponent of critical velocity gradient variation. n_f varied between 1.27 and 1.42 for equivalence ratios of $0.95 \leq \phi_f \leq 2.25$. Fine observed that Eq. 30 was in better agreement with data points at equivalence ratios between $1.3 \leq \phi_f \leq 1.8$. The variation of critical velocity gradient is not linear at the very low pressures shown in Figure 23, which is due to the enhanced quenching effect.

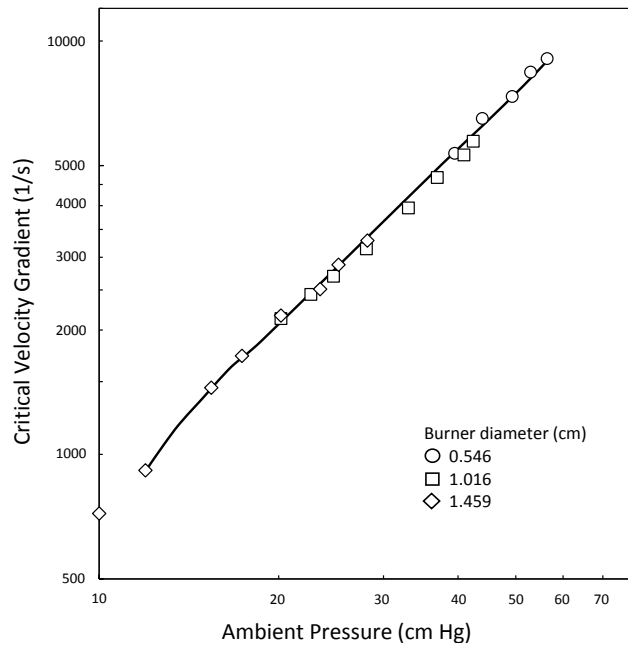


Figure 23: Critical velocity gradient variation as a function of ambient pressure for hydrogen/air flames at equivalence ratio of 1.8, reproduced from [67,71].

Fine [67,71] also measured burning velocity as a function of pressure at different equivalence ratios and a similar relation to Eq. 30 was suggested:

$$S_L \propto P^{n_u} \quad \text{Eq. 31}$$

n_u varied between 0.208 and 0.256 for the equivalence ratio between $1.10 \leq \phi_f \leq 1.9$. This correlation suggests that the laminar burning velocity increases with an increase in pressure. This is consistent with other studies of the burning velocity of hydrogen/air flames at sub-atmospheric pressure in the literature. For example, Figure 24 illustrates the variation of laminar flame speed with pressure at different hydrogen concentrations [149]. For equivalence ratios less than 0.79 ($X_{H_2O} = 0.25$), the reaction order is less than 2 and a negative pressure dependence is observed, while for equivalence ratio higher than 0.79, the reaction order is larger than 2 with positive pressure dependence. This indicates that the laminar flame speed increases with an increase of sub-atmospheric pressure at fuel rich conditions [149] consistent with the positive exponents proposed by Fine for Eq. 31.

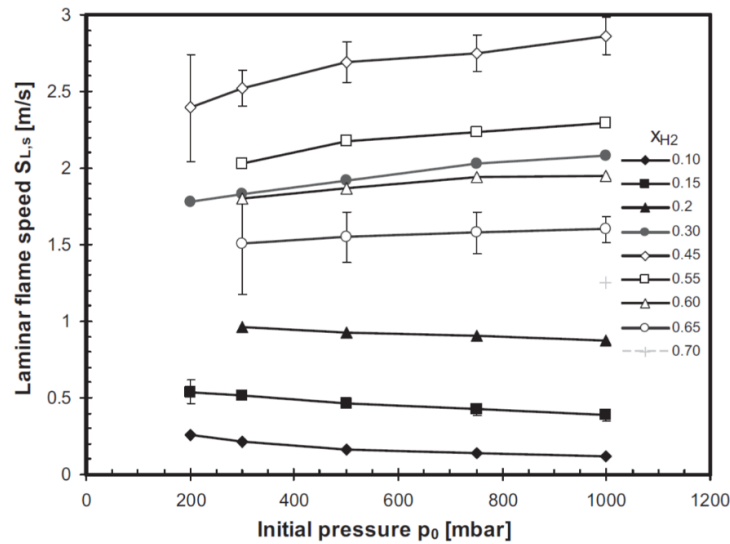


Figure 24: Laminar flame speed of hydrogen/air mixtures vs pressure at various hydrogen mole fractions [149].

Based on equations Eq. 30, Eq. 31 and using Eq. 3, the variation of the quenching distance with pressure would be:

$$\delta_q \propto P^{n_q} \quad \text{Eq. 32}$$

in which n_q is equal to $n_u - n_f$. Considering the average values for pressure exponents of critical velocity gradient and laminar flame speed, $n_q = -1.12$ was suggested [67,71]. The comparison of the pressure

exponent of quenching distance ($n_q = -1.12$) to the experimental results of Drell and Belles [75] for parallel plates indicates good agreement as illustrated in Figure 25. Drell and Belles reported pressure exponents of 1.051, 1.138, and 1.097 for the quenching distance between two parallel plates at equivalence ratios of 0.5, 1, and 2, respectively [75].

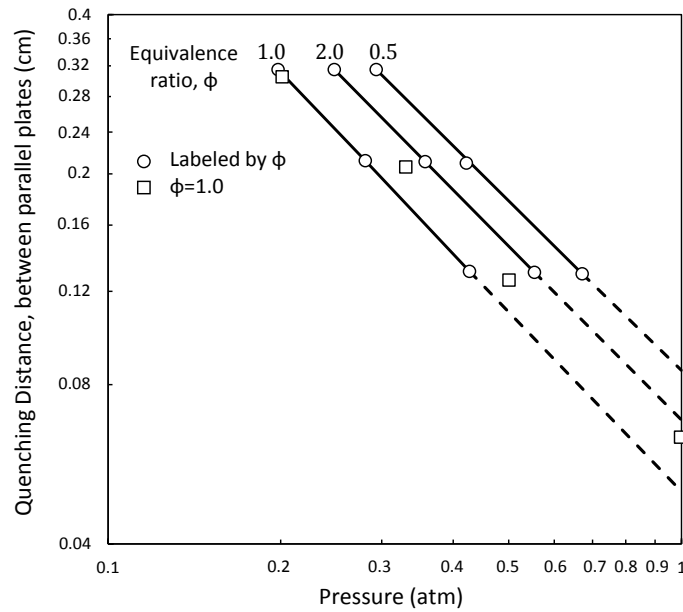


Figure 25: Effect of pressure on quenching distance of hydrogen/air mixtures, circle (data from [113]), square (data from [85]), reproduced from [75].

The variation of critical velocity gradient with equivalence ratio at different pressures found by Fine is indicated in Figure 26 [67,71]. The slope of the critical velocity gradient variation increases with an increase in pressure. This indicates that the flashback propensity is more sensitive to equivalence ratio variation at higher pressures. A general correlation was found to fit all the measured data of hydrogen/air flames at reduced pressures:

$$g_c = 2.6 \frac{S_L}{\delta_q} \quad \text{Eq. 33}$$

In this equation, δ_q is the quenching distance between two parallel plates. The quenching distance of hydrogen/air mixture in Eq. 33 was obtained from the tabulated data in [120]. The constant coefficient in Eq. 33 was found to be 3 from the study by Von Elbe et al. [48].

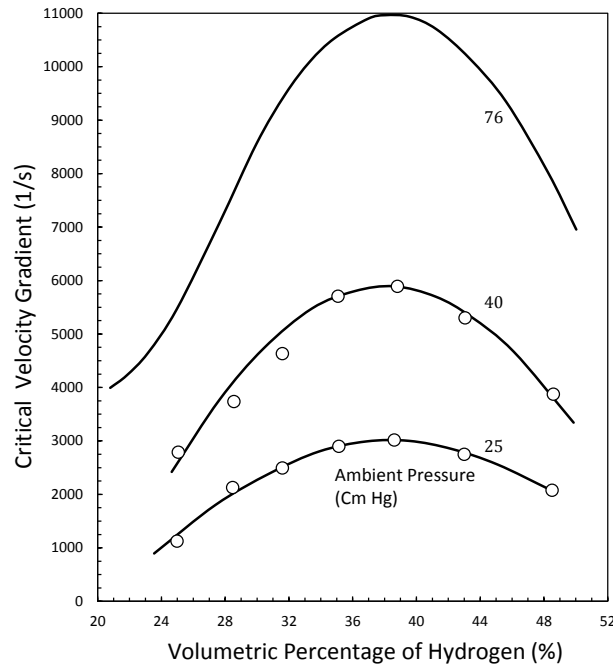


Figure 26: Critical velocity gradient as a function of fuel concentration for hydrogen/air flames at different pressures, top curve (atmospheric data from [48]), reproduced from [67,71].

Further work by Fine [70] investigated the effect of pressure on flashback propensity of hydrogen/oxygen-argon and hydrogen/oxygen-helium in which the concentrations of oxygen, argon, and helium were 20.6, 79.4, and 79.4, respectively. The experiment setup was similar to that used in previous studies [67,71]. The equivalence ratio was varied in the range of $0.9 \leq \varphi \leq 2.25$ for the hydrogen/oxygen-argon mixtures and $1.1 \leq \varphi \leq 1.5$ for the hydrogen/oxygen-helium mixtures. The burner head diameter used in this experiment was 0.546 and 1.016 *cm*. Figure 27 presents the variation of critical velocity gradient as a function of pressure for hydrogen/oxygen argon and hydrogen/oxygen-helium. The flashback data of argon diluted mixture are in a good agreement with results for helium diluted mixture. The pressure exponents of critical velocity gradients for hydrogen/oxygen-argon mixtures are 1.59 and 1.49 at equivalence ratios of 1.1 and 1.5 respectively. The exponent values are 1.50 and 1.48 for the hydrogen/oxygen-helium mixtures at these same equivalence ratios, respectively, which are

slightly different compared to the hydrogen/oxygen-argon mixtures. The average pressure exponent for equivalence ratios between $0.9 \leq \varphi \leq 2.25$ was 1.51 for both argon and helium diluted mixtures.

The burning velocity of the mixture diluted by helium is much larger than the burning velocity of the hydrogen/oxygen-argon mixtures [70,150]. This suggests that the quenching distance of the helium diluted mixture increases in a manner that the critical velocity gradient remains approximately constant. This is consistent with the observation that the flame for the hydrogen/oxygen-helium mixture anchored at a larger distance from the burner rim [70]. Furthermore, results for two burner diameters follow the same trend line as indicated in Figure 27, except at richest condition with equivalence ratio of 2.25 (see Figure 28). For the 1.016 cm diameter tube, a transition from laminar to turbulent flow for the hydrogen/oxygen-argon mixture occurs at critical velocity gradient about 8000 resulting in a departure from the trend line as shown in Figure 27(a).

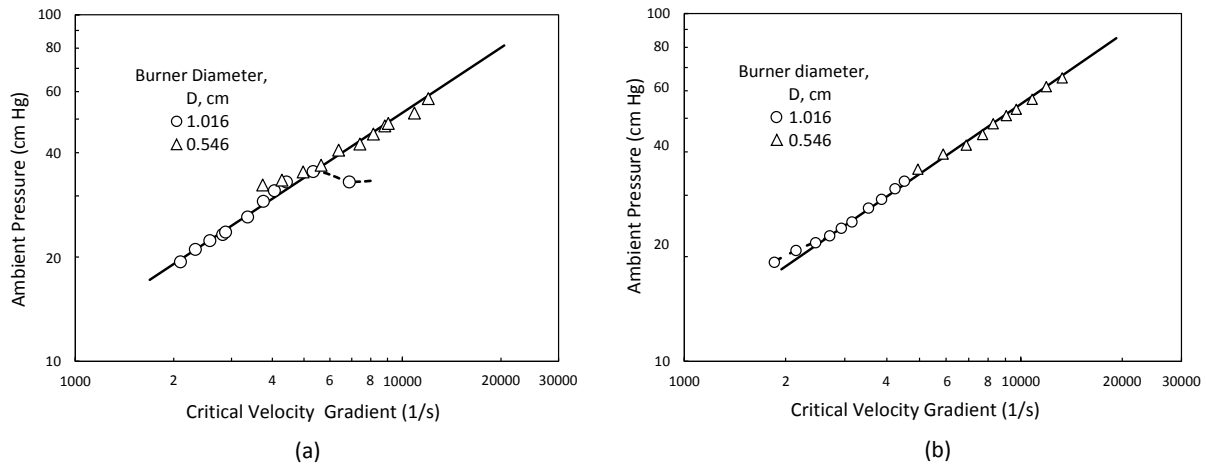


Figure 27: Variation of critical velocity gradient vs pressure at equivalence ratios of 1.1 (a) hydrogen/oxygen-argon and (b) hydrogen/oxygen-helium, reproduced from [70].

The variation of critical velocity gradient with equivalence ratio for hydrogen/oxygen-argon observed by Fine [108] is shown in Figure 28. An increase in pressure increases the velocity gradient at which flashback occurs, as previously mentioned. Data points for both burner diameters are coincident as no significant difference was observed with the exception of equivalence ratio of 2.25. The diameter

dependence at the richest condition is attributed to diffusion effects at the burner rim. The maximum critical velocity gradient occurs at about 39 percent hydrogen for both helium and argon diluted mixtures. This value agrees well with previous studies conducted for hydrogen-air flames [47,67,71].

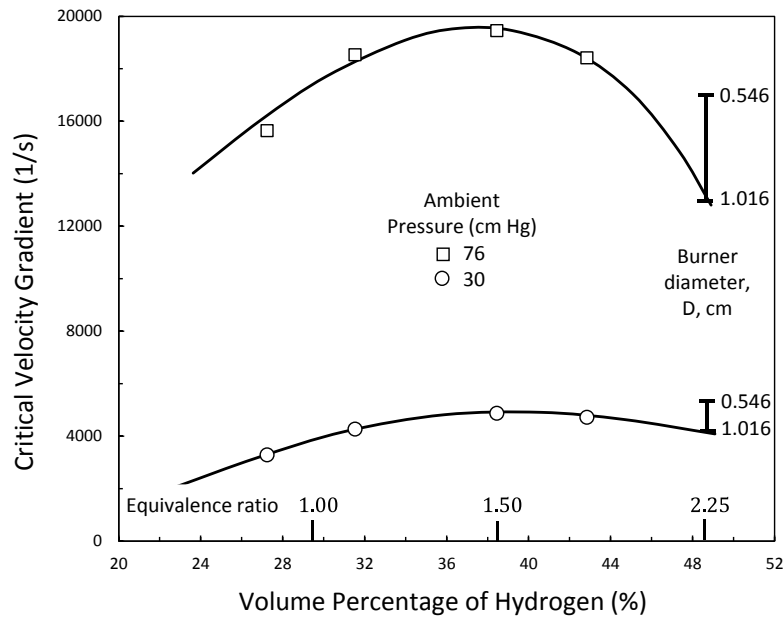


Figure 28: Critical velocity gradient of hydrogen/oxygen-argon mixture as a function of fuel concentration at two different pressures, reproduced from [70].

The results from the literature indicate that flashback limits are strongly affected by the preheat temperature and operating pressure. An increase of preheat temperature increases the laminar flame speed causing flashback to occur at higher air flow rate for a given equivalence ratio. Furthermore, the propagation of the flame front into the tube is a function of the burner tip temperature. Since the burner temperature remains approximately constant at the preheat temperature, the quenching distance significantly decreases at high preheat temperatures. In order to determine the effect of preheat temperature on flashback propensity, both laminar flame speed and quenching distance need to be considered for various fuel compositions.

All laminar flame flashback studies in the literature are limited to one atmosphere or sub-atmospheric operating pressures. The effect of sub-atmospheric pressure on critical velocity gradient is

primarily determined by the fuel composition and equivalence ratio. In fact, the laminar flame speed of different fuels can increase or decrease with an increases of pressure at lean or rich conditions, and this pressure dependence is weak in many cases [149,151,152]. As a result, the quenching distance plays a crucial role in changing the flashback limit. The quenching effect can be significant when determining the overall variation of flashback limits regardless of an increase or decreases of laminar burning velocity with pressure. Despite the insight offered by the studies discussed in this section, for practical devices operating at elevated pressure or with high hydrogen content fuels, the flow is usually turbulent, hence application of these results is dubious.

4.4 Fuel Composition

With the general ability of flashback tendencies of laminar flames to be characterized with the critical velocity gradient concept, experimental efforts were directed towards using the concept to help determine the stability limits of various fuel compositions.

4.4.1 Hydrocarbon Fuels

Early comprehensive studies were conducted by Grumer to investigate the effect of individual fuels and their mixtures at room temperature and atmospheric pressure [54,64]. The measured flashback data were later used by Grumer to predict the interchangeability of fuels [58]. Boundary layer flashback was investigated for natural gas, methane, ethane, and propane with air as an oxidizer. Burner diameters ranging from 0.390 to 1.358 *cm* were used. The laminar flow was fully developed at the burner exit and the burner head temperature was controlled to remain at room temperature before starting each test.

Figure 29 indicates the variation of critical velocity gradient as function of “fraction of stoichiometric” (i.e. the fuel percentage divided by the stoichiometric fuel percentage) for natural gas, methane, ethane, and propane. Flashback curves in Figure 29 qualitatively peak at the same fraction of stoichiometric for methane, natural gas, and ethane. However, the flashback curve of propane slightly shifts to the right as previously verified by Harris et al. [51] and shown in Figure 17.

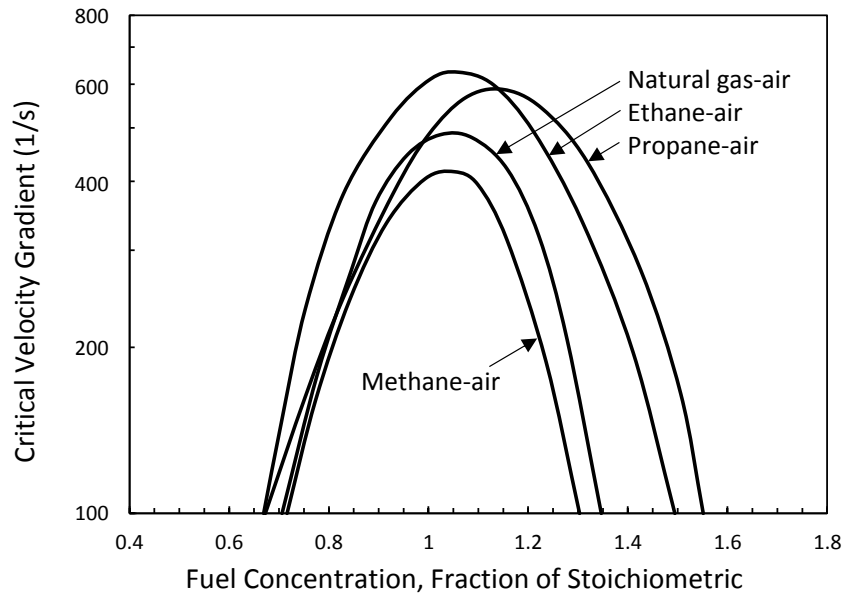


Figure 29: Flashback limits of different hydrocarbons as a function of fuel concentration, reproduced from [54,64].

Methane/air mixtures have the lowest flashback propensity of the hydrocarbon fuels studied. The critical velocity gradient for natural gas is slightly higher than that of methane due to the presence of higher hydrocarbons in the mixture. Ethane has a higher critical velocity gradient than propane at lean conditions, while the opposite is true at rich conditions. In order to interpret the difference in flashback limits, the laminar burning velocity of these fuels, as determined by Davis and Law [153] are compared in Figure 30. As shown, the laminar burning velocity of methane is lower than other fuels. The laminar burning velocity of ethane is slightly higher than propane within the stoichiometric and rich conditions, which can be also observed in [154,155]. Therefore, the critical velocity gradient behavior shown in Figure 29 does not follow the qualitative variation of the laminar burning velocity for propane, and ethane.

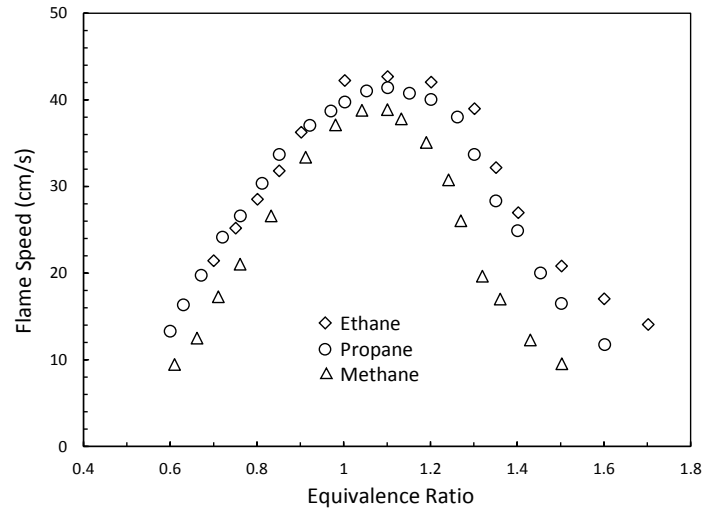


Figure 30: Laminar burning velocity of methane, ethane, and propane at atmospheric pressure and room temperature, reproduced from [153].

Figure 31 shows the flashback propensity of methane-propane and methane-ethane as a function of fuel volume concentration as studied by Davu et al. [93]. The addition of 25% propane to methane increases the maximum critical velocity gradient from 400 1/s to 490 1/s. Similarly with an addition of 5% ethane, the critical velocity gradient has the same amount of increase. This is consistent with the previous observation associated with Figure 29 regarding the higher peak of critical velocity gradient for ethane.

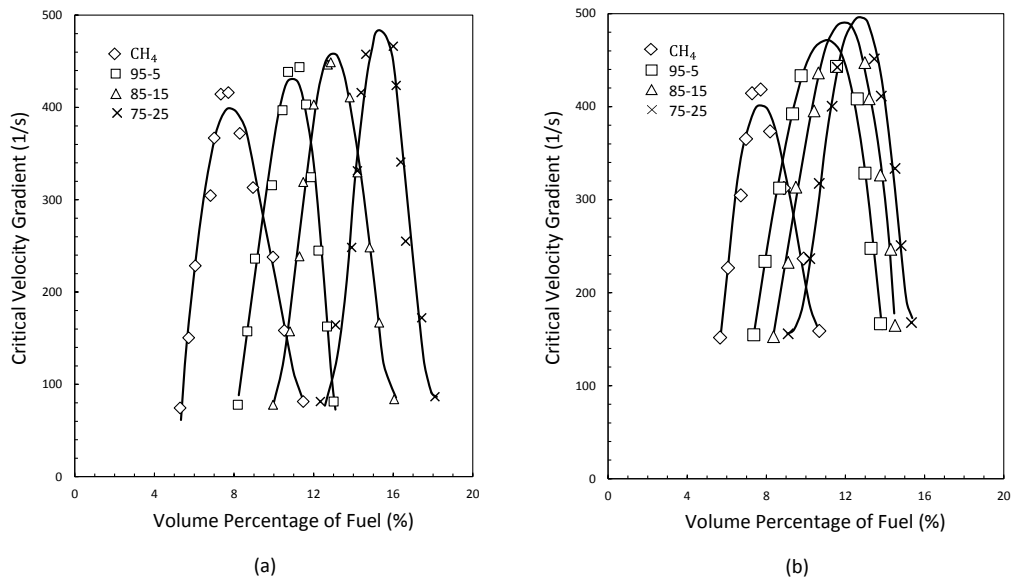


Figure 31: Variation of critical velocity gradient for (a) methane-propane mixtures and (b) methane-ethane mixtures, reproduced from [93].

4.4.2 Hydrogen Rich Fuels

Further discussion is now directed at hydrogen/air mixtures owing to the known high flashback propensity of hydrogen. Hydrogen has higher burning velocities and a smaller quenching distance than hydrocarbons [68,156]. The thermal and mass diffusivity of hydrogen are considerably larger compared to those for hydrocarbon fuels. Furthermore, reaction kinetics of hydrocarbons are basically limited by $CO \rightarrow CO_2$, which is relatively slow. The reaction kinetics of hydrogen is fast due in part to the lack of this step which must occur in all hydrocarbons fuels [141,157]. As a result, the flashback propensity of hydrogen is generally much higher than hydrocarbon fuels.

Von Elbe and Mentser [48] were the first to measure the flashback propensity of hydrogen at atmospheric pressure for different burner diameters varied from 0.108 to 0.33 *cm*. Water-cooling was used to control the temperature of the burner head. Figure 32 indicates the critical velocity gradient variation of hydrogen for the laminar flame [48]. The solid curve shows the flashback limit where the flame fully propagates upstream. The region between the dashed curved (beginning of partial entrance) and solid curve represents the partial entrance of the flame front into the burner, described as a tilted flame in [47]. The region of tilted flame is more extensive than that observed for natural gas flames as previously discussed in Section 4.2. These regions invoke perturbations to the stable flame and consequently led to flashback except at very lean or rich conditions [48]. Critical velocity gradient values of hydrogen shown in Figure 32 are significantly larger than those for methane shown in Figure 13. The maximum critical velocity gradient of hydrogen/air flames is more than twenty times that of methane/air flames and corresponds to ~40 percent hydrogen concentration (equivalence ratio of 1.59).

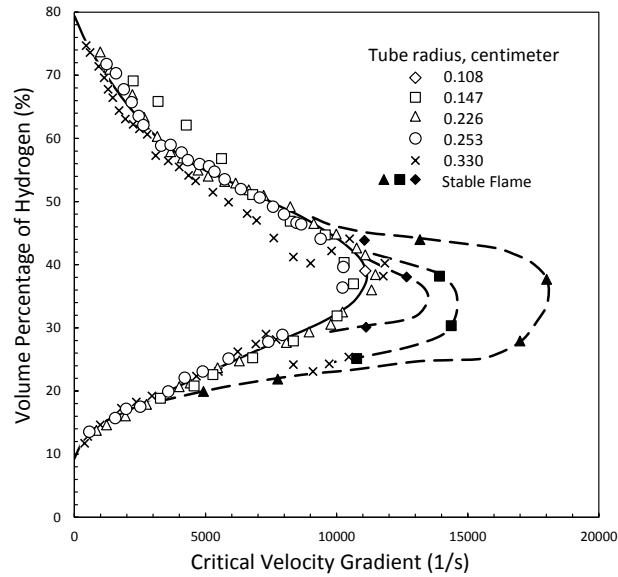


Figure 32: Critical velocity gradient of laminar hydrogen flame for different burner diameters, reproduced from [48].

Figure 33 illustrates the laminar burning velocity of hydrogen/air mixture at room temperature and atmospheric pressure obtained from Koroll et al [30]. The maximum laminar burning velocity occurs at 42 percent hydrogen concentration and is in good agreement with the maximum critical velocity gradient observed in Figure 32.

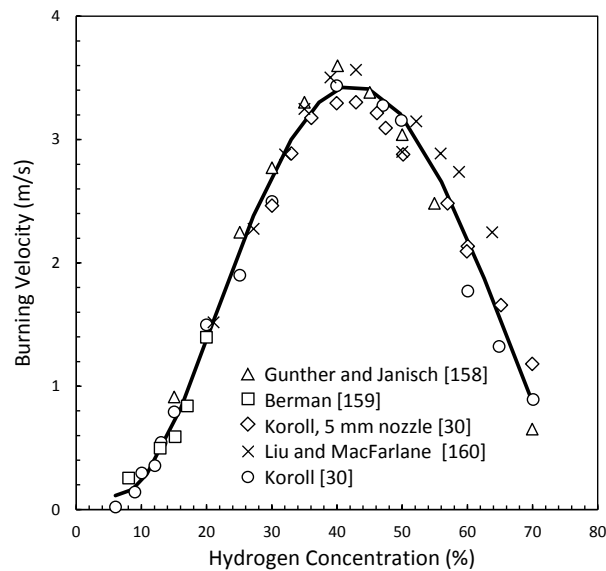


Figure 33: Laminar burning velocity of hydrogen/air mixture at 25 °C and atmospheric pressure measured by (Günther and Janisch [158]; Berman [159]; Liu and MacFarlane [160]), reproduced from [30].

4.4.3 Fuel Mixtures

Early studies demonstrated that the flashback propensity of binary mixtures is not generally a linear function of the individual components, especially when hydrogen or carbon monoxide is one of the components [55,57]. To better understand the behavior of binary mixtures, a comprehensive study was conducted by Grumer and Harris to investigate the flashback propensity of methane-hydrogen, carbon monoxide-hydrogen, and carbon monoxide-methane [57]. Experiments were carried out for several ratios of the two components. Since it was difficult to make a stability diagram for all different fuel compositions, results were presented in terms of critical velocity gradient variation as a function of hydrogen or carbon monoxide addition at given fraction of stoichiometric (F).

Figure 34 indicates the variation of critical velocity gradient for hydrogen-methane/air flames. From 0 to 50% hydrogen concentration, the ratio is defined by H_2/CH_4 and from 50 to 100% hydrogen the ratio is denoted by CH_4/H_2 . The results are presented for both fuel lean and rich conditions as labeled by the fraction of stoichiometric in Figure 34. Adding a small amount of methane to hydrogen decreases flashback propensity significantly; however, adding a small amount of hydrogen to the methane does not drastically increase the critical velocity gradient.

This observation is consistent with the variation of the laminar flame speed for hydrogen-natural gas mixtures [161]. Figure 35 indicates the variation of the normalized laminar burning velocity as a function of hydrogen concentration in the mixture at equivalence ratio ranging from 0.6 to 1.4. Based on the definition in [161], the burning velocity increment can be defined as :

$$\frac{(S_{Mixture} - S_{NG})}{(S_{H_2} - S_{NG})} \quad \text{Eq. 34}$$

in which, S denotes the laminar flame speed. For a given equivalence ratio, the laminar burning velocity increases exponentially (solid line). As a result, addition of methane to pure hydrogen fuel has more pronounced effect than the addition of hydrogen to pure methane according to Eq. 35 [161]:

$$\frac{(S_{Mixture} - S_{NG})}{(S_{H_2} - S_{NG})} = 0.00737 \exp\left(\frac{x}{20.38}\right) + 0.00334 \quad \text{Eq. 35}$$

x is volume percent of hydrogen in the fuel mixture.

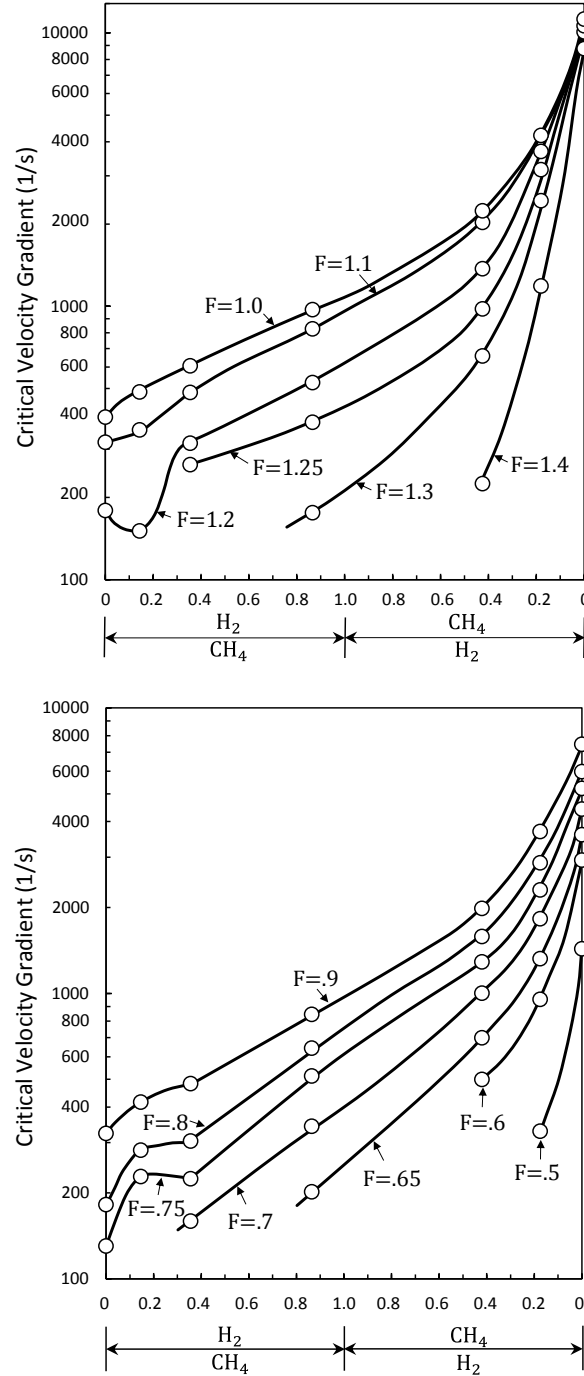


Figure 34: Critical velocity gradient of hydrogen-methane/air mixtures at different fuel concentrations (F), reproduced from [57].

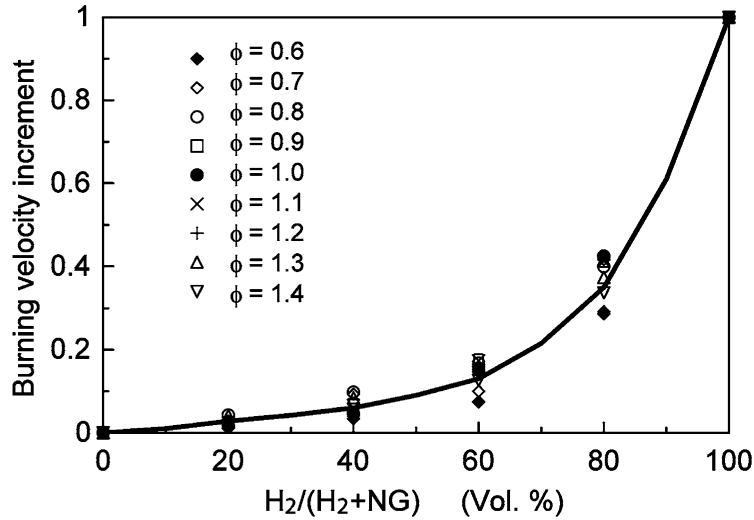


Figure 35: Normalized unstretched laminar burning velocity of hydrogen-natural gas/air mixture as a function of hydrogen concentration at different equivalence ratios [161].

Figure 36 illustrates the flashback propensity of carbon monoxide-hydrogen [57]. The flashback region of carbon monoxide-hydrogen mixture extends from stoichiometric fraction of 0.6 to 2.2, which is wider than that for hydrogen-methane mixtures (Figure 34). If Figure 36 is plotted in normal axis as a function of hydrogen concentration, the critical velocity gradient linearly varies for fraction of stoichiometric less than 1.4. However, when the hydrogen concentration is high, the critical velocity gradient increases rapidly with further increase in hydrogen content. This observation is consistent with the normalized laminar burning velocity measurement (see Eq. 36) by Dong et al. as shown in Figure 37 [162]. They concluded that the variation of laminar burning velocity is divided into three regions as seen in Figure 37: (1) rapid increase for hydrogen concentrations below 20%, (2) linear variation, and (3) rapid increase at hydrogen concentrations above 80%. Based on this study, the laminar burning velocity varies linearly for equivalence ratio from 0.7 to 2.1 [162]:

$$\frac{(S_{Mixture} - S_{CO})}{(S_{H_2} - S_{CO})} = -0.05914 + 0.00875x \quad \text{Eq. 36}$$

in which, S denotes the laminar flame speed. The variation of critical velocity gradient at some rich conditions in Figure 36 is different from the overall trend, which is not predicted by the laminar burning velocity variation. There is no explanation for this observation in [57].

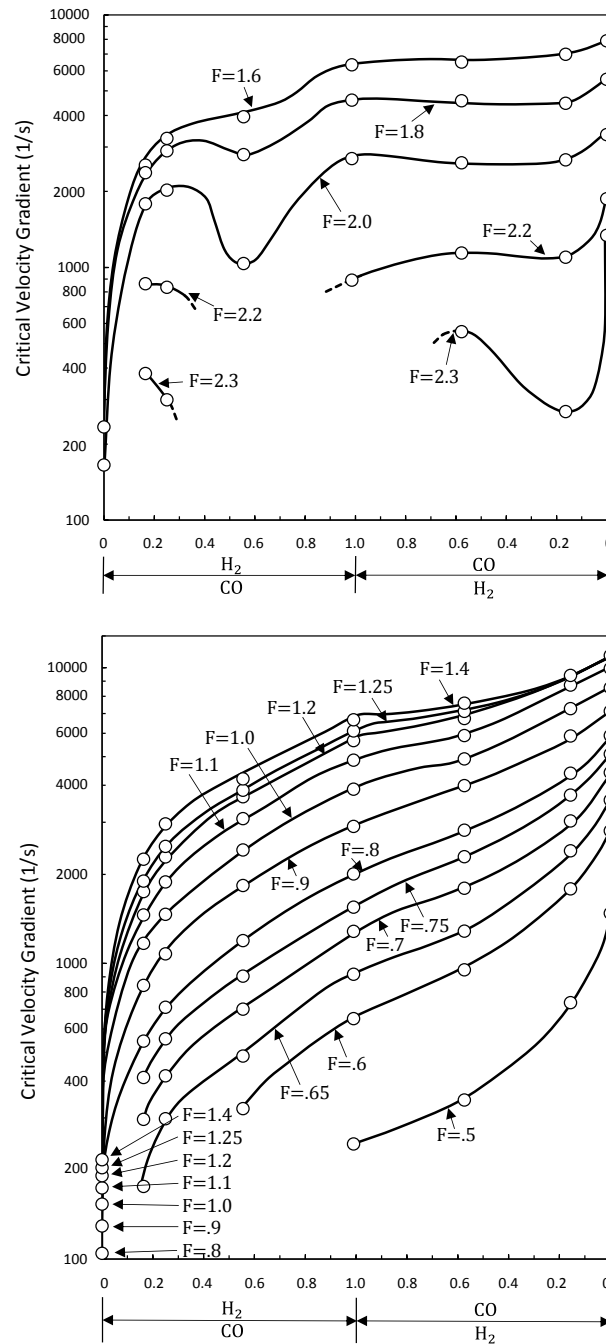


Figure 36: Critical velocity gradient of carbon monoxide-hydrogen/air mixtures at different fuel concentrations (F), reproduced from [57].

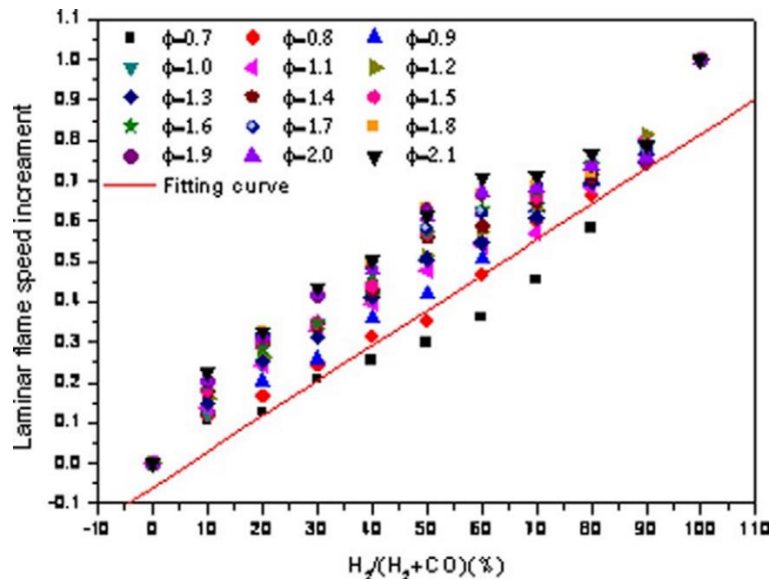


Figure 37: Normalized laminar burning velocity of carbon monoxide-hydrogen/air mixture vs hydrogen concentration at different equivalence ratios [162].

To further explore the subtleties of fuel mixture effects, Figure 38 presents the variation of critical velocity gradient for carbon monoxide-hydrogen/air mixtures as a function of fuel volume concentration. The addition of hydrogen to carbon monoxide significantly increases the critical velocity gradient. The effect of hydrogen concentration on critical velocity gradient is pronounced at lean conditions according to Figure 38(a) (data from Grumer and Harris [57]). However, this conclusion is not consistent with the recent experimental study by Dam et al. [96]. Indeed, as indicated in Figure 38(b), the effect of hydrogen addition in lean mixtures is small compared to its effect when added to fuel rich conditions. Furthermore, the addition of hydrogen shifts the peak of critical velocity gradient to lean conditions in Figure 38(a), while the opposite is observed in Figure 38(b). It is noted the maximum critical velocity gradient of hydrogen occurs at 40% fuel concentration as previously observed in Figure 32 [48], which is in agreement with the qualitative trend observed in Figure 38(a).

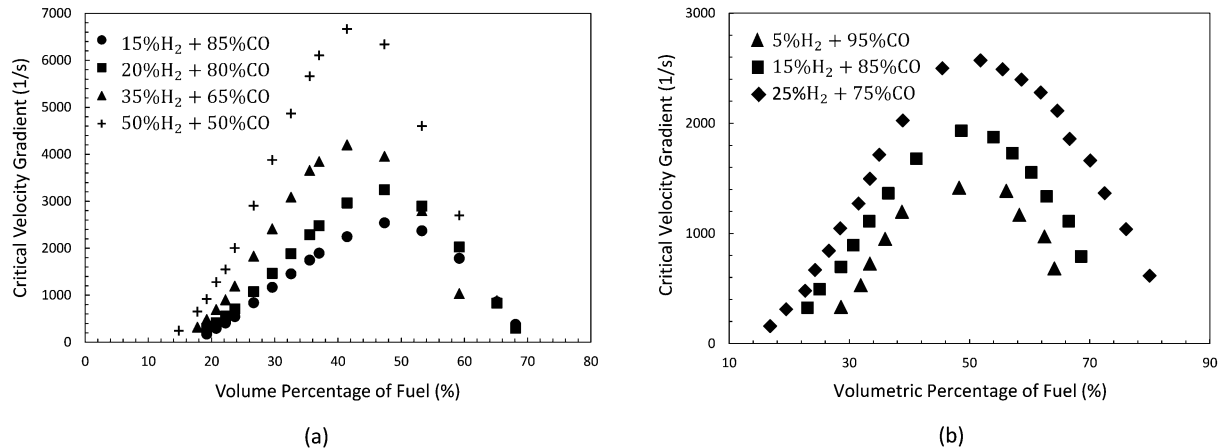


Figure 38: Critical velocity gradient variation as a function of fuel concentration for carbon monoxide-hydrogen/air mixtures (a) data from Grumer and Harris [57] and (b) reproduced from Dam et al. [96].

Figure 39 presents the critical velocity gradient variation of carbon monoxide-methane/air mixtures. Adding small amount of methane to carbon monoxide increases the flashback propensity drastically. However, the addition of small amounts of carbon monoxide to methane has only a slight effect. The molecular weight of methane is smaller than that of carbon monoxide and the addition of methane would be even smaller in terms of mass fraction. This illustrates that the dissimilar behavior resulting from adding a second fuel component to a single gas cannot be explained by difference in mass fraction [57].

The critical velocity gradient variation in Figure 39 is in accordance with the laminar burning velocity measurements by Scholte and Vaags [163] (shown in Figure 40). Figure 40 shows the burning velocity of methane-carbon monoxide mixtures as a function of the fuel/air ratio. An addition of a small amount of methane significantly increases the laminar burning velocity. With adding more methane, the laminar burning velocity peaks at about 90%CO-10%CH₄ and finally drops to the methane burning velocity. It is interesting to note that a similar point exists in Figure 39 (i.e. $CH_4/CO = 0.1$) at which the critical velocity gradient is maximum. Furthermore, Figure 39 illustrates that methane has a higher flashback propensity than carbon monoxide for fraction of stoichiometric less than 1.2, which is consistent with the laminar burning velocity variation reported in [163].

To create a stability diagram for a specific fuel binary compositions, one can select the fuel component ratios in Figure 34, Figure 36, Figure 39 and obtain critical velocity gradient at different fuel concentrations (see Figure 38a).

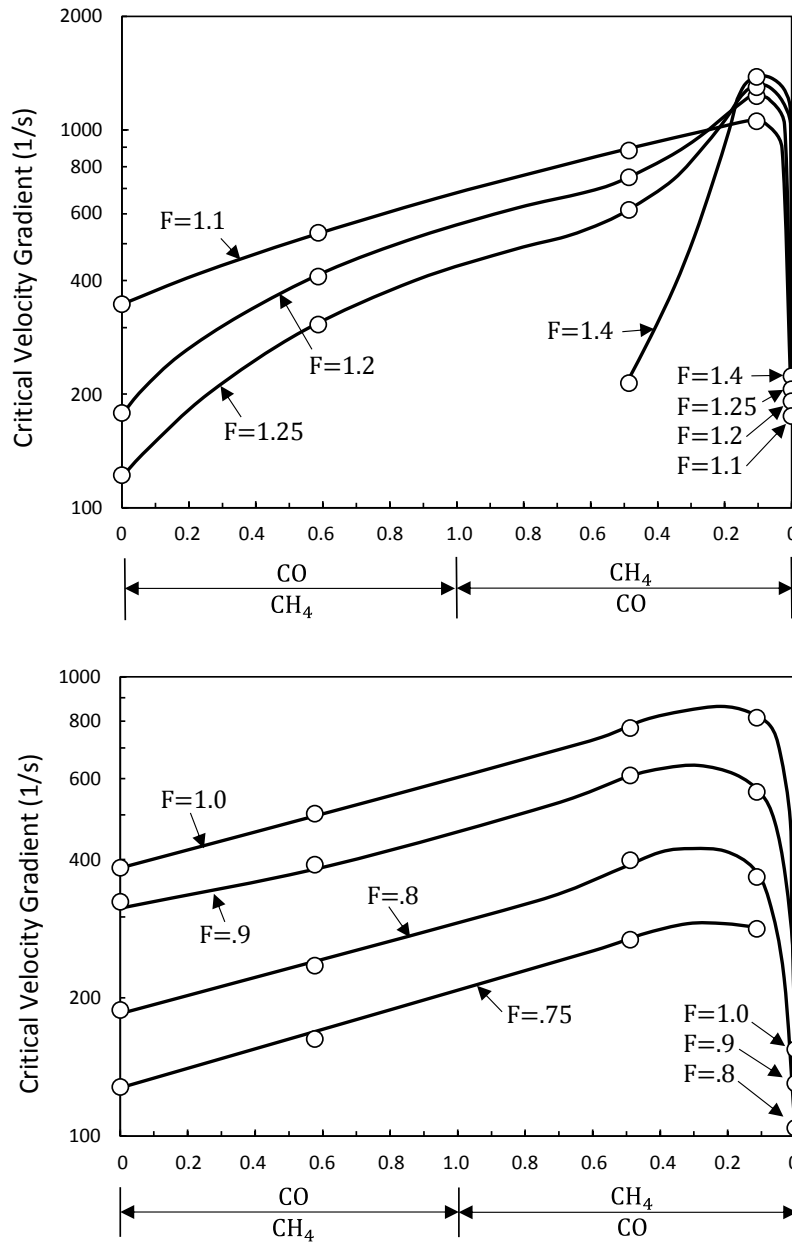


Figure 39: Critical velocity gradient of carbon monoxide-methane/air mixtures at different fuel concentrations (F), reproduced from [57].

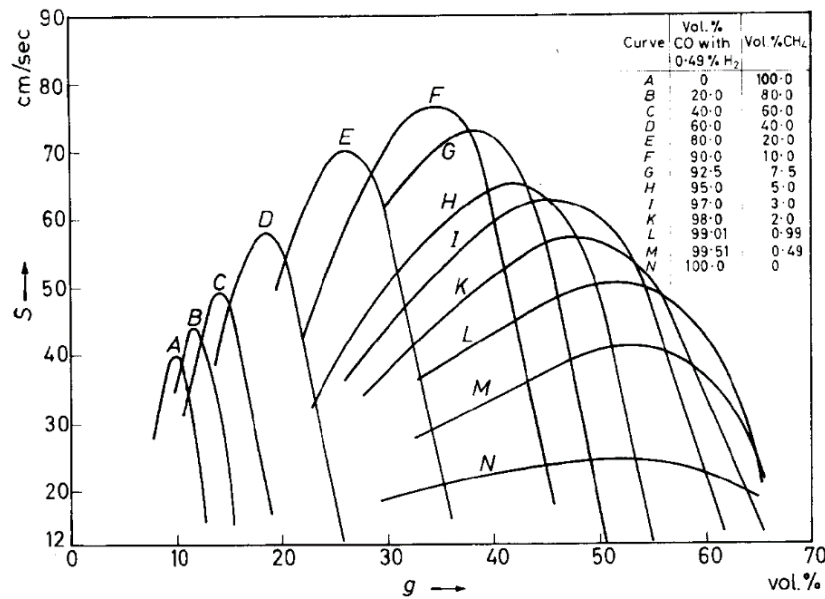


Figure 40: Burning velocity of carbon monoxide-methane/air mixtures as a function of fuel/air ratio [163].

The flashback propensity of air mixed with hydrogen-ethylene, propane-hydrogen, and hydrogen diluted with nitrogen was also investigated by Grumer et al. [65]. Figure 41 and Figure 42 present experimental results for these mixtures. Adding small amounts of ethylene or propane to hydrogen significantly decreases the flashback propensity of the mixture. Adding small amount of hydrogen to ethylene or propane increases the critical velocity gradient to a small extent.

Figure 43 illustrates the variation of laminar burning velocity for propane-hydrogen/air mixtures as a function of equivalence ratio at various hydrogen concentrations [164]. X_h denotes the volume percent of hydrogen in the fuel mixture. For a given equivalence ratio, small increase of propane in the pure hydrogen results in a significantly lower laminar flame speed and consequently much lower flashback limit as concluded in [55,65].

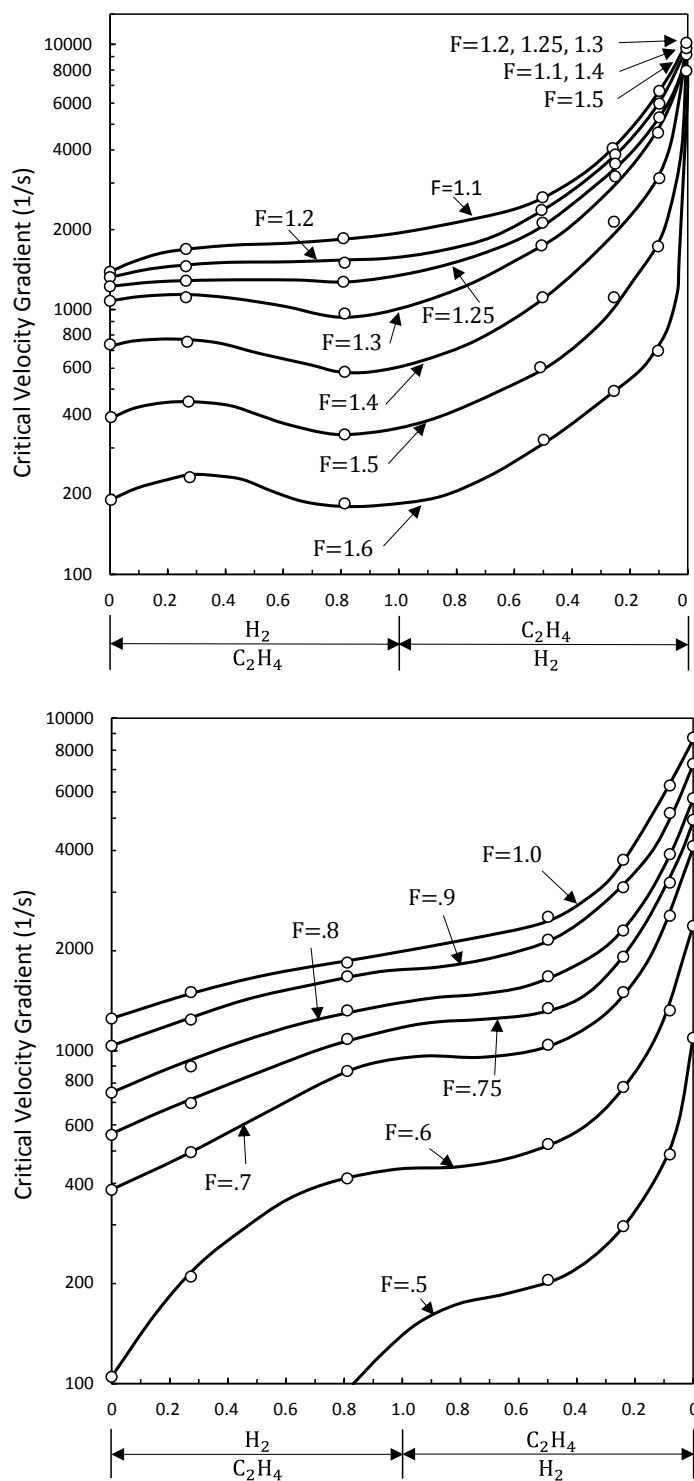


Figure 41: Critical velocity gradient of ethylene-hydrogen/air mixtures at different fuel concentrations (F), reproduced from [65].

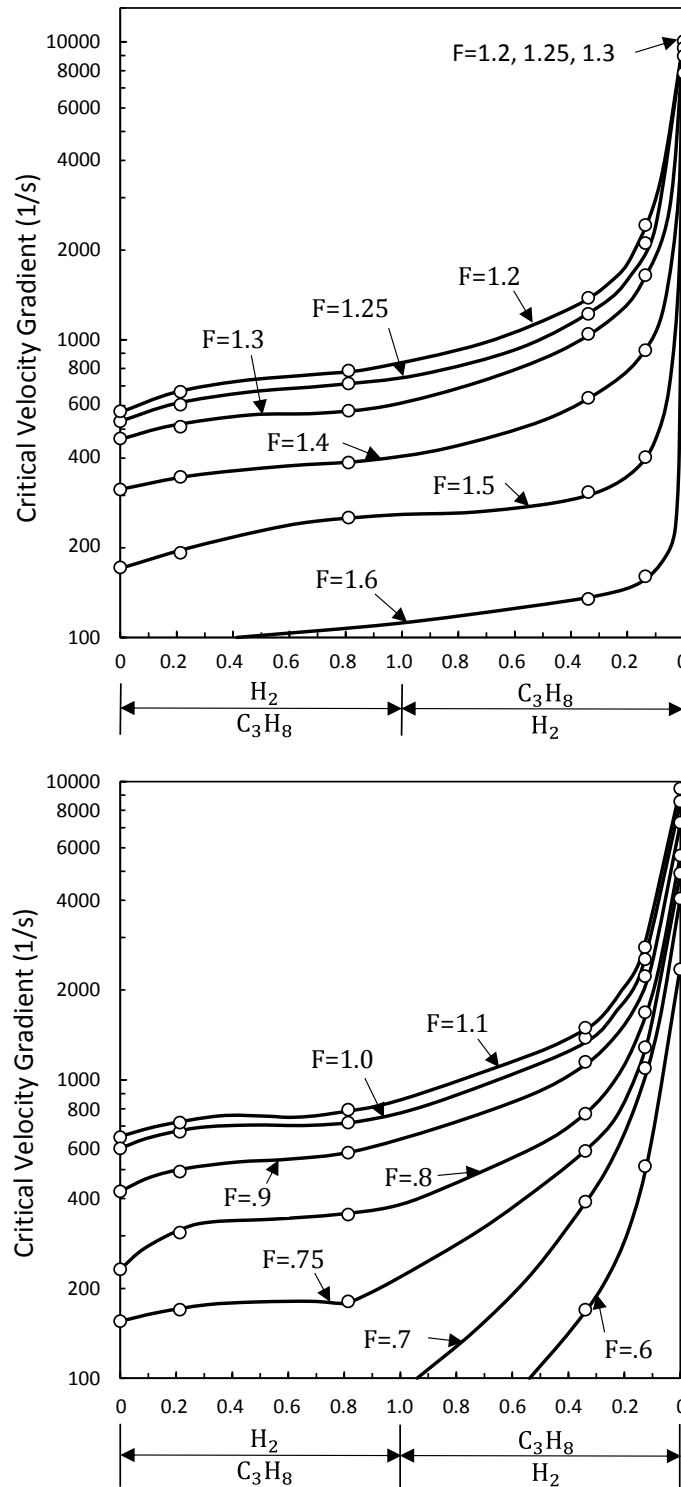


Figure 42: Critical velocity gradient of propane-hydrogen/air mixtures at different fuel concentrations (F), reproduced from [65].

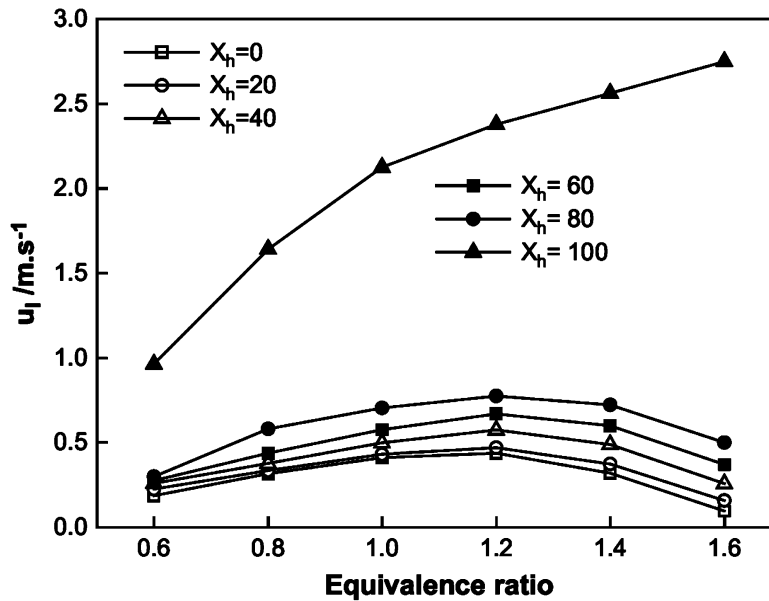


Figure 43: Laminar burning velocity of propane-hydrogen/air mixtures as a function of equivalence ratio at different hydrogen concentrations [164].

These results again indicate that the flashback propensity of multicomponent fuels is not a linear combination of the critical velocity gradient of single components involving hydrogen or carbon monoxide [57]. What has been observed is that for certain fuel components, a weighted average critical velocity gradient can be successfully applied. These fuels include binary and ternary hydrocarbon fuels including mixtures of alkanes and alkenes such as methane, ethane, propane, butane, and ethylene as studied in [65,72]. Figure 44 illustrates the flame stability diagram of mixture containing 79.4% methane-20.6% ethylene. The solid curve represents the calculated data from the linear combination of critical velocity gradients (Eq. 37), which is in a good agreement with experimental results. This method was also used for and 78.6 ethylene and 21.4% methane and a good agreement was obtained. Based on this conclusion, the critical velocity of a multicomponent mixture can be calculated from Eq. 37.

$$g_{y+z+\dots} = n_y g_y + n_z g_z + \dots \quad \text{Eq. 37}$$

in which, y and z are fuel components and n is fractional volume concentration of each component. Values of g_y , g_z are obtained from critical velocity gradients of single component fuels.

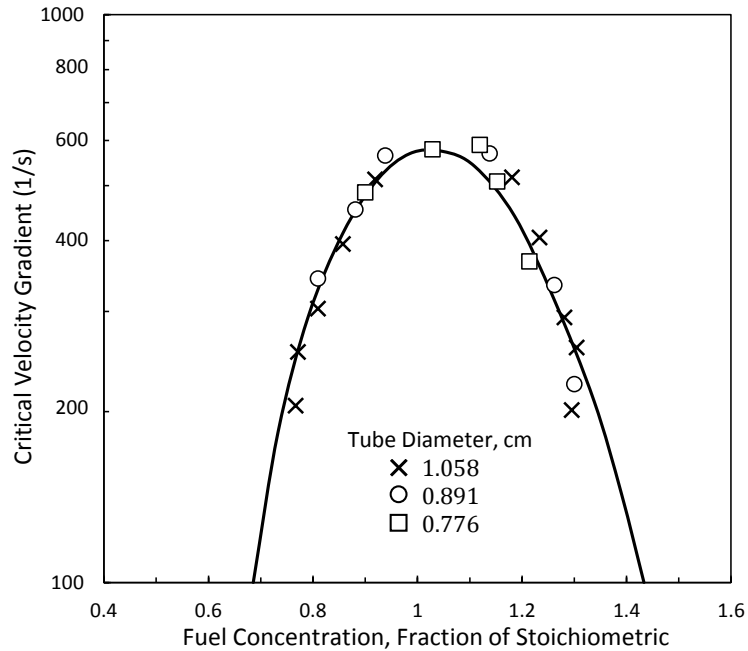


Figure 44: Boundary layer flashback propensity of 79.4% methane-20.6% ethylene vs fuel concentration, reproduced from [65].

When hydrogen and carbon monoxide are present in multicomponent fuels, a modified version of Eq. 37 was developed to calculate the critical velocity gradient of ternary fuel mixtures. Indeed, the critical velocity gradient of a multicomponent fuel is calculated from flashback limits of binary fuel mixtures, which are available for different fuel compositions as previously discussed. For a mixture including three gases α , β , and γ , a combination of two binary fuels $\alpha + \beta = n_y$ and $\alpha + \gamma = n_z$ are considered in which the value of α is divided between β and γ in the ratio of β/γ . The selection of α , β , and γ for a ternary fuel mixture was based on experimental results. For a fuel mixture containing hydrogen, carbon monoxide, and methane, the binary fuel components are selected as CO-CH₄ and H₂-CH₄. Figure 45 illustrates the evaluation of the implemented method for three fuel components. The curve lines represent calculated values from Eq. 37 and predict experimental results fairly well as shown in Figure 45(a). But, clear deviation is observed in Figure 45(b) at rich condition.

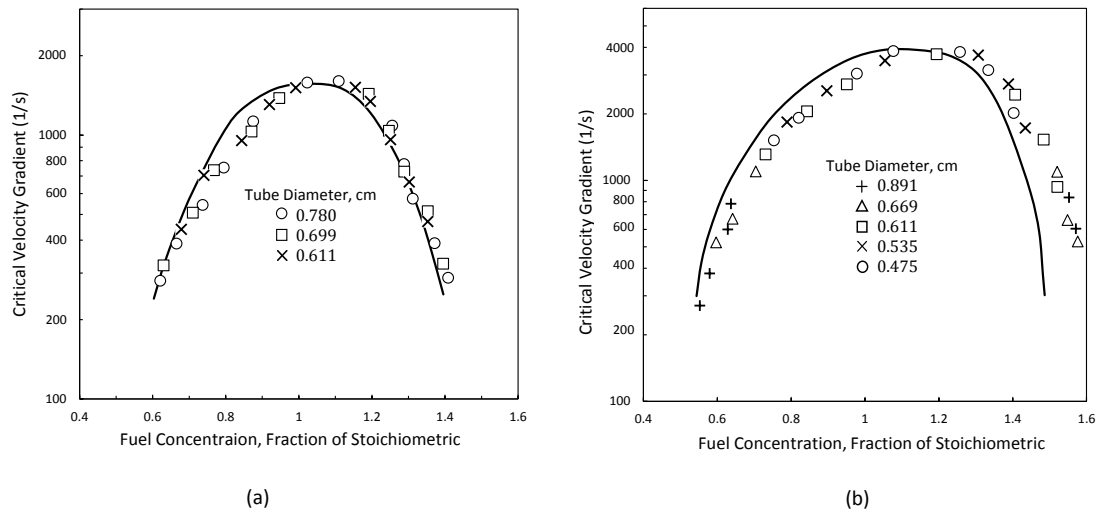


Figure 45: Boundary layer flashback limit for different burner diameters (a) 29.4% CH₄, 31.7% CO, 38.7% H₂ mixture and (b) 10.3% CH₄, 34.0% CO, 55.7% H₂, reproduced from [57].

Eq. 37 was also implemented to predict the flashback propensity of multicomponent fuels as summarized in Table 2. Carbon dioxide and nitrogen were treated as diluents. This means that associated critical velocity gradients of these constituents are considered to be zero in Eq. 37. The critical velocity gradients of these fuels were calculated by assuming different single and binary mixtures as indicated in Table 3. As mentioned earlier, the selection of α was not arbitrary and determined experimentally. Since the critical velocity gradient of these single and binary mixtures are known from [57,65], the flashback propensity of the multicomponent mixture can be calculated. The comparison of calculated results with those from the experiment showed a good agreement [65].

A similar correlation was developed in [87] to calculate the critical velocity gradient of a fuel mixture consisting of hydrogen, methane, carbon monoxide, carbon dioxide, and nitrogen. The final equation was presented based on the volume fraction of fuel components in the mixture, in which coefficient and exponents were obtained from the experimental data obtained. The error of the correlation was less than 6 percent for studied cases.

Table 2: Composition of multicomponent fuels studied in [65].

	Fuel 1	Fuel 2	Fuel 3	Fuel 4	Fuel 5
Methane	37.4	29.1	32.1	42.6	37.5
Ethane	-	-	-	2.2	-
Ethylene	33.4	26.2	28.4	18.1	20.4
Propane	-	22.1	-	1.9	-
Propylene	-	0.2	-	0.2	-
Butane	-	-	-	0.2	-
Butylene	-	-	-	0.1	-
Hydrogen	15.2	11.8	12.5	17	17.5
Carbon monoxide	-	-	-	9.1	3.9
Nitrogen	14.0	10.6	27.0	3.4	13.3
Carbon dioxide	-	-	-	5.2	7.4

Table 3: Listing of complexes for use in Eq. 37 [65].

	Flashback complexes
Fuel 1	(C ₂ H ₄ + H ₂) and (CH ₄)
Fuel 2	(C ₂ H ₄ + H ₂) and (CH ₄) and (C ₃ H ₈)
Fuel 3	(C ₂ H ₄ + H ₂) and (CH ₄)
Fuel 4	(C ₂ H ₄ + H ₂) and (CH ₄ + CO)
Fuel 5	(C ₂ H ₄ + H ₂) and (CH ₄ + CO)

A systematic study of fuel composition effects on flashback was conducted by Van Krevelen et al. [78]. They developed a method to provide a generalized flame-stability diagram for all fuel mixtures. A generalized flashback diagram was introduced in Figure 46 by using (1) non-dimensional critical velocity gradient, and (2) reduced gas concentration. Based on this model, the critical velocity gradient (g_c) is normalized by the peak value of the flashback curve (g_c)_M in Eq. 38 (see Figure 46).

$$(g_c)_R = \frac{g_c}{(g_c)_M} \quad \text{Eq. 38}$$

The reduced gas concentration (F_R) is obtained from Eq. 39 to create a scaled parameter for fuel variation.

$$F_R = 1 + \alpha(F - F_M) \quad \text{Eq. 39}$$

in which, F is the fuel gas concentration (fraction of stoichiometric) and F_M corresponds to the location at which critical velocity gradient is maximum. α is defined as a reduction factor, which is equal to the reciprocal of the width of flashback curve and normalized such that it is equal to one for methane.

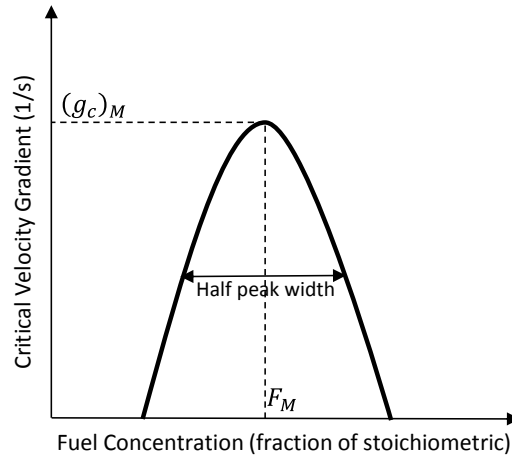


Figure 46: Schematic of boundary layer flashback curve for a fuel/oxidizer mixture, adapted from [78].

Measured values of $(g_c)_M$, F_M , and α for various fuels are summarized in Table 4 (data obtained from [57,65,78]). The generalized flashback curve has a peak value $(g_c)_R = 1$ at $F_R = 1$. Using characteristic values in Table 4 and generalized flashback curve in Figure 47 (semi-log plot), one can obtain the critical velocity gradient at desired fuel concentration for studied fuels using Eq. 38 and Eq. 39.

This method was extended to characterize the flashback propensity of mixtures diluted with inert gases including nitrogen and carbon dioxide. The characteristic values of binary and multicomponent fuels $(g_c)_M$, ϕ_M , and α were determined as functions of inert-free mixture and diluents concentrations. Some correlation were developed for hydrogen, carbon monoxide, methane, ethane, ethylene, and propane with combination of either nitrogen or carbon monoxide [78].

A similar approach was used by Van Krevelen et al. for binary and multicomponent fuel mixtures and characteristic values were tabulated in [78]. As previously pointed out, the characteristic parameters for mixtures of hydrocarbons are additive [57,65]. Consistent with earlier studies, it was concluded that this cannot be extended to mixtures of carbon monoxide and hydrogen with hydrocarbons [78]. Furthermore, the behavior of the carbon monoxide was mentioned to be abnormal in many respects. Hence, “depression factors” were defined to capture this effect.

Table 4: Characteristic flashback values of different fuel compositions [78].

Fuel gas	$(g_c)_M$ (1/s)	F_M	α
CH ₄	400	1.00	1.00
C ₂ H ₆	650	1.13	0.80
C ₃ H ₈	580	1.12	0.60
C ₂ H ₄	1400	1.10	0.63
C ₃ H ₆	800	1.14	0.68
C ₆ H ₆	720	1.09	0.63
H ₂	10500	1.20	0.46

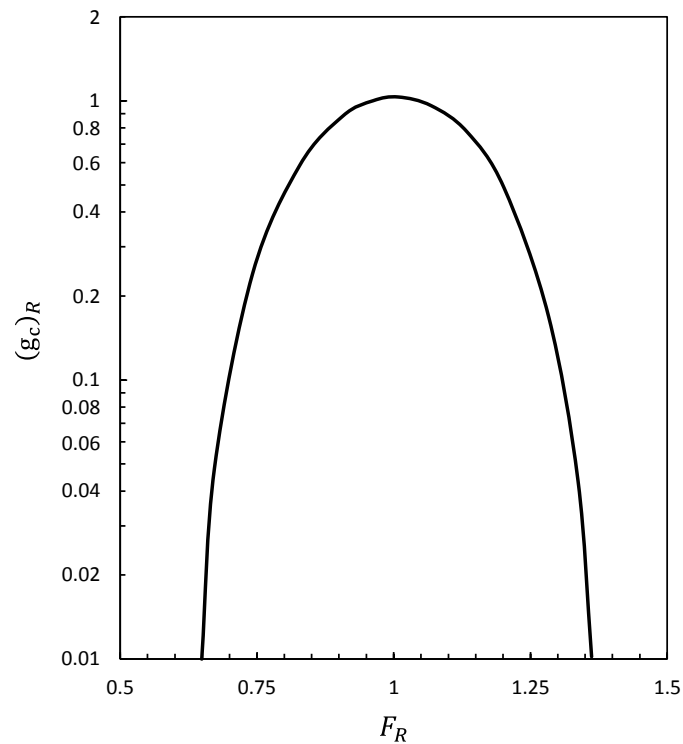


Figure 47: Generalized boundary layer flashback curve, reproduced from [78].

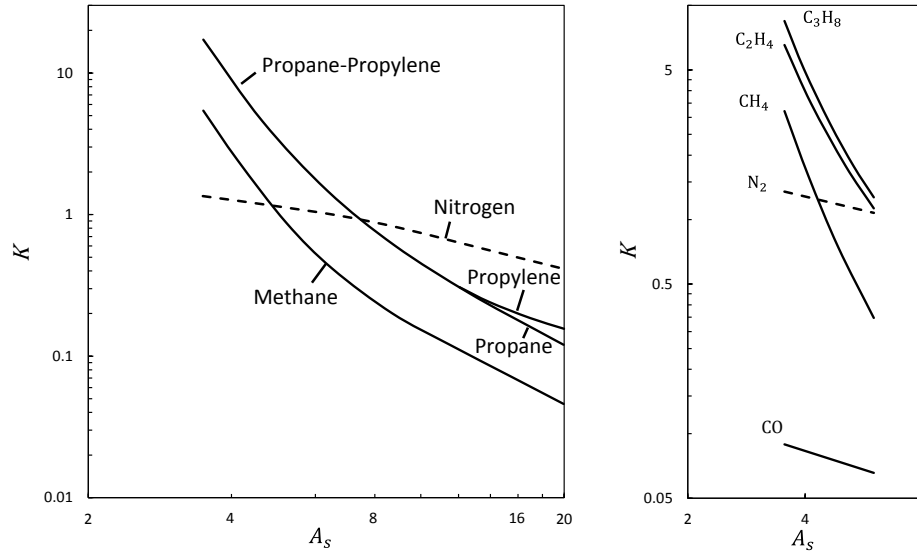
In further work to predict fuel composition effects when hydrogen is present, Caffo and Padovani developed a simpler method to predict the flashback propensity in multicomponent fuels [81]. In their correlation, the critical velocity gradient of multicomponent fuels was calculated by assigning a constant coefficient to hydrocarbons and a variable coefficient to hydrogen. Thus, the critical velocity gradient of inert-free mixtures was introduced as:

$$g_c = a_{H_2} \times \%H_2 + \sum a_i \times \%C_i \quad \text{Eq. 40}$$

where a_i is a coefficient for hydrocarbons present in the mixture. a_i is fixed for each component depending on the equivalence ratio. a_{H_2} is the hydrogen coefficient represented as a function of the mixture stoichiometric air coefficient (A_s) and the concentration of other hydrocarbons ($\%C_i$) accompanying the hydrogen. Based on the experimental results, a_{H_2} was presented in the form of:

$$a_{H_2} = \sum K_i \times \%C_i \quad \text{Eq. 41}$$

in which K_i is determined from the mixture stoichiometric air coefficient (A_s). This variation is indicated in Figure 48 for various fuels and nitrogen as a diluent. The overall value of a_{H_2} is primarily affected by (A_s) according to Eq. 41. The values of a_i are also tabulated in Table 5. This result was also extended to carbon monoxide and different coefficients were tabulated in [81]. It is noted that all coefficients in Table 5 and reference [81] were obtained for stoichiometric conditions and cannot be generally extended to other equivalence ratios. The study by Caffo and Padovani indicates that the critical velocity gradient of the hydrogen and hydrocarbons mixtures can be expressed as a function of the mixture stoichiometric air coefficient. This method was evaluated for various fuel compositions at stoichiometric condition with hydrogen concentration varied from 8% to 56% in the fuel mixture and the results matched experimental data within 5% [81].

Figure 48: Coefficient K as a function of stoichiometric mixture coefficient, reproduced from [81].Table 5: Values of a coefficient for hydrocarbons [81].

Hydrocarbon	CH_4	C_3H_8	C_3H_6
a coefficient	4.54	6.06	7.01

Correlations for flashback propensity of multicomponent fuel mixtures under laminar conditions were also developed by Putnam et al. [85]. They examined different fuel compositions as summarized in Table 6 [85]. Flashback propensity of each mixture was determined for various equivalence ratios and the value of $g_c \delta_q / S_L$ was calculated for each case and reported in Table 6. Each group labeled by (a, b, and c) has a constant adiabatic flame temperature and a constant product composition at stoichiometric condition. The burning velocity was measured by determining the flame front angle stabilized above the burner rim. Quenching distance was reported as the minimum width between two plates in which the flame can propagate. The critical velocity gradient was calculated from Eq. 5. The equivalence ratio corresponding to the peak critical velocity gradient also corresponded to the minimum quenching distance value. However, the peak of the burning velocity (S_L) occurred at lower equivalence ratios compared to that corresponding to the maximum g_c and minimum δ_q condition. This deviation was attributed to the burning velocity measurement procedure for the atmospheric test conditions. The

further details of this experiment including the burning velocity and quenching distance data can be found in [84].

Table 6: Fuel compositions studied for boundary layer flashback in [85].

Mixture number	CO	H ₂	CH ₄	CO ₂	O ₂	N ₂	$g_c \delta_q / S_L$
1a	0.590	0.295	0	0.115	0	0	2.36
b	0.566	0.147	0.096	0.151	0.039	0	2.40
c	0.542	0	0.193	0.187	0.078	0	1.90
2a	0.295	0.590	0	0.115	0	0	3.20
b	0.147	0.373	0.205	0.191	0.083	0	2.52
c	0	0.156	0.410	0.268	0.166	0	1.90
3a	0.442	0.442	0	0.115	0	0	2.50
b	0.360	0.221	0.171	0.179	0.069	0	2.40
c	0.277	0	0.342	0.243	0.138	0	1.92
4a	0.257	0.128	0	0.115	0	0.500	2.60
b	0.245	0.064	0.042	0.140	0	0.511	2.58
c	0.232	0	0.083	0.164	0	0.521	2.15
5a	0.128	0.257	0	0.115	0	0.500	2.60/2.35
b	0.064	0.162	0.087	0.167	0	0.521	2.85
c	0	0.067	0.173	0.218	0	0.541	1.80
6a	0.192	0.192	0	0.115	0	0.500	2.35
b	0.155	0.096	0.073	0.158	0	0.518	2.75
c	0.118	0	0.145	0.201	0	0.535	-

The final results were plotted in the form of concentrations ratio defined by $(H_2 - CH_4)/(H_2 + CH_4 + CO)$ as shown in Figure 49. The value of $g_c \delta_q / S_L$ approaches 2 as methane concentration increases and reaches 2.5 with an increases of hydrogen. Since the measured quenching distance between two parallel plates is around twice of the penetration distance [47], $g_c \delta_p / S_L$ would be 1-1.25, which is consistent with that obtained by Lewis and Elbe [47].

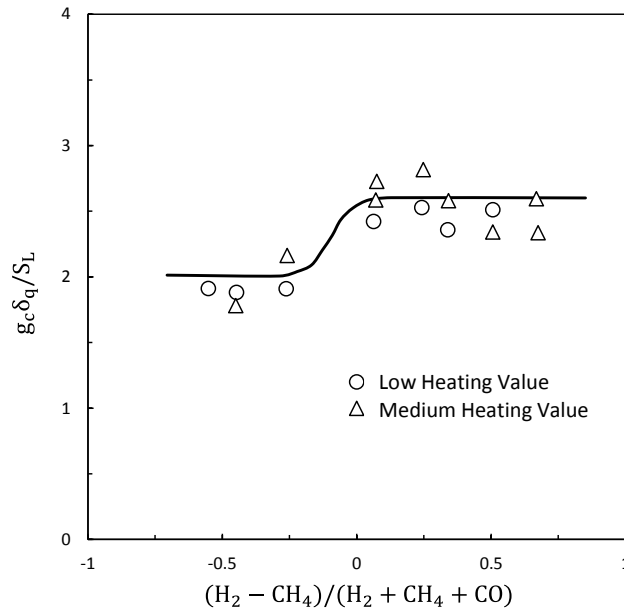


Figure 49: Variation of $g_c \delta_q / S_L$ as a function of fuel concentration ratio (see Table 6), reproduced from [85].

In summary, fuel composition can significantly impact the flashback limit as illustrated for single, binary, and multi-component fuels. A direct relationship between laminar burning velocity and flashback propensity is clearly observed for all fuel mixtures as predicted by simple models. Different studies have developed various correlations for the flashback limit of multicomponent fuels. Many of the studies observed that special consideration is required when hydrogen or carbon monoxide is present in the mixture. The empirical equations include various coefficients which are not known or validated for all operation conditions and experiment configurations. However, qualitative comparisons made in Section 4.4 can be served as a tool to investigate the interchangeability of fuel gases [54,58].

4.5 Numerical Analysis

Numerical simulations of boundary layer flashback have provided important details about the influence of the flame on the flow field. One-dimensional numerical simulations have been conducted for flame propagation normal to the wall (see head-on quenching in Figure 7) by different studies [123,125,165–169]. However, only a few studies have addressed laminar boundary layer flashback, and these are summarized in this section.

The first numerical study was performed by Lee and T'ien [86] in 1982 for propagation of a premixed laminar flame through a tube. The analysis was limited to small diameters due to the computational cost. This pioneering simulation showed the importance of the flame back-pressure due to flame-flow interaction. The steady state Navier-Stokes and energy equations were solved numerically for the laminar flow with one-step chemical reaction and second order finite rate Arrhenius kinetics. A parabolic velocity profile for fully developed laminar flow was assumed at inlet condition with a variable magnitude in order to produce stationary flame representing the flashback condition. The wall temperature was considered equal to the unburned mixture temperature ($T_0 = 298$). The effect of the tube radius on flashback limits was studied for ten cases at equivalence ratio of 1 as indicated in Figure 50. The reference length (L) was defined by the flame thickness, which is the ratio of thermal diffusivity to the one-dimensional laminar burning velocity ($\alpha/S_L = 4.629 \times 10^{-3} \text{ cm}$). The reference value of $S_L = 38 \text{ cm/sec}$ was used as a scaling parameter. As the tube radius decreases, the critical velocity gradient increases and then abruptly reduces. This trend line is similar to the experimental results of Lewis and Elbe [47] in Figure 14. At first, when the tube diameter decreases, the quenching distance moves closer to the tube axis and the critical velocity gradient assumption based on the linear velocity profile close to the wall is not valid. With further decrease of the tube diameter, the flame within the entire tube is influence by the quenching effect.

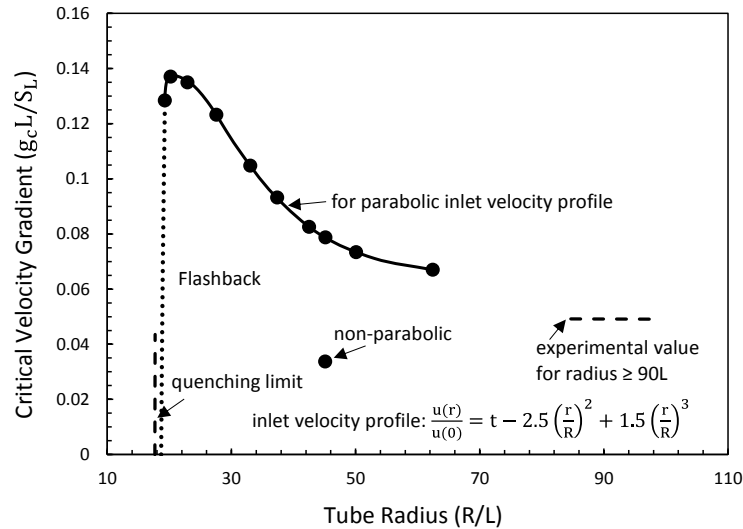


Figure 50: Critical velocity gradient as a function of tube radius, reproduced from [86].

Figure 50 also shows that the calculated critical velocity gradient reaches a plateau independent of the tube radius since the flame stabilization zone remains approximately unchanged at sufficiently large diameters. This value seems slightly high compared to measured critical velocity gradient value in the experimental data of [113] (denoted by dashed line in Figure 50). The quenching limit was interpolated from calculated results (indicated by dotted line) and the tube radius at which quenching occurs is lower than reported experimental values in [113]. It is noted that the experimental data in [47,113] were obtained for an unconfined flame stabilized in ambient air, while the numerical analysis was conducted for a confined flame.

The flame front is bent towards the unburned reactants through nearly the entire channel since the tube size was relatively small (recall the smallest diameter used by Lewis and Elbe). The shape and characteristic of the flame front are similar to the observation based on experimental results in Figure 5 [47,48,63]. The flame front is strongly curved inside the tube and one-dimensional flame sheet cannot provide a realistic values for the laminar flame speed, especially close to the tube wall (see the isothermal line in Figure 51).

The static pressure increases normal to the flame front and then decreases, which is in agreement with velocity vectors in Figure 51. The unburned mixture decelerates as it approaches the flame front in Figure 51. When the temperature of the mixture increases across the flame front, the density decreases and flow velocity increases accordingly. Interaction between velocity, pressure, and heat release leads to the distortion of the streamline before entering the flame zone, previously observed in experimental results of [170,171]. Divergence of the velocity vectors continues through the reaction zone and, as the flow passes the flame, the streamlines become parallel to the tube axis. The calculated burning velocity along the tube axis was higher than the one-dimensional laminar flame speed due to the interaction of the flame and pressure field [86].

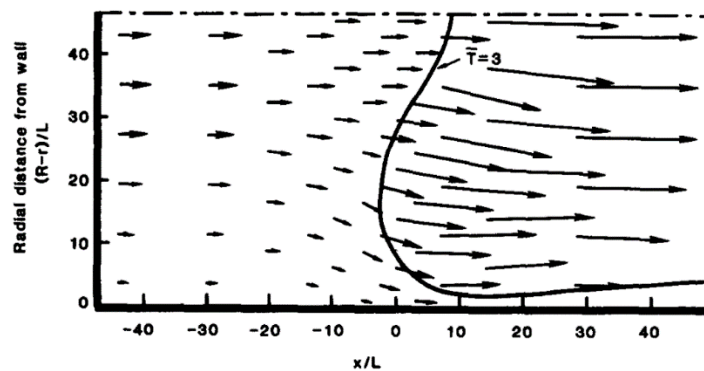


Figure 51: Velocity vectors at $T_0 = 298$ for the tube radius of $R = 46.5 L$ [86].

Lee and T'ien [86] only considered the confined flame configuration. However, in practical applications, the flame is generally stabilized downstream of the burner rim. Mallens and Goey conducted a numerical study for unconfined methane/air flames and compared results with the experimental data [89]. The experimental setup by Mallens and Goey consisted of a rectangular slit burner with cross section of $55 \text{ mm} \times 2b$, in which $2b$ was varied from 4 to 10 mm. The burner rim temperature was kept constant at 50°C by water cooling. A similar setup was used to measure the flashback propensity of flames stabilized downstream of circular tubes with diameters from 4 to 10 mm. The tip temperature was not controlled during this experiment as it was believed to have a negligible effect on flashback limits. The

range of equivalence ratio covered in both experiment was between 0.9 and 1.1 and flow was fully developed at the burner exit.

Mallens and Goey also conducted a steady two-dimensional simulation of both laminar premixed flames considered in their experimental study. Governing equations of motion of motion and energy were formulated with stream-vorticity functions and solved considering one-step chemical reaction with the Arrhenius type source term. A parabolic velocity profile was used at inlet condition. The detailed description of the numerical method implemented is available elsewhere [89,172–174]. In order to simulate the flashback limit by the steady code, the bulk flow velocity was decreased incrementally (5 percent reduction at each step) to reach the condition when the flame did not stabilize at the burner rim and propagated upstream. This critical flow rate was reported as a flashback limit for numerical results.

The comparison of numerical and experimental results is illustrated in Figure 52. The critical velocity gradient variation as a function of burner width and diameter are shown for both slit and tube burners. The experimental data agree well with the numerical results and the difference is less than 10% for all widths and diameters studied [89]. The deviation is more significant for the tube burner, possibly due to the low mass flow rate and associated uncertainty in measurements which can be up to 15 percent. The experimental results of Harris et al. [51] are shown as solid squares for two tube burner diameters of 8.82 and 9.66 mm. As suggested by the numerical results in Figure 52, the slit burner has a lower critical velocity gradient than the tube burner, which is consistent with the conclusion made by Putnam and Jensen [52]. The difference between the two burner configurations reduces to less than 5 percent for burner diameters larger than 6 mm. This trend is not predicted by experimental results probably due to the measurement uncertainty. Furthermore, the effect of tip temperature can be significant for the tube burner since it was not cooled in this experiment.

The critical velocity gradient decreases with an increases of burner diameter or channel width, which is consistent with the numerical results by Lee and T'ien [86]. As the burner diameter decreases,

the quenching distance is comparable to the burner diameter and consequently the velocity at which flashback occurs can no longer be estimated by a linear velocity profile.

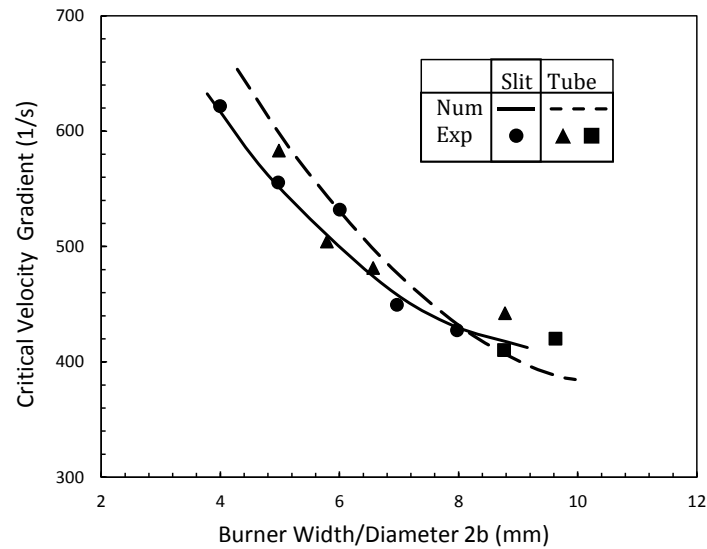


Figure 52: Critical velocity gradient variation as a function of burner width/diameter: the solid triangle and circle show experimental results for the tube burner and slit burner, respectively. Solid squares are experimental results by Harris et al. [51], reproduced from [89].

Figure 53 presents the variation of critical velocity gradient as a function of equivalence ratio for the 5 mm diameter burner. The numerical results match well with experimental data within equivalence ratio of 0.9 and 1.1. However, the numerical model does not follow the experiment trend line outside this interval. The deviation between numerical and experimental results is significant for the lowest and highest equivalence ratios studied. This is attributed to inaccuracies in the calculated burning velocity resulting from the one-step chemical reaction as well as inaccuracies in flow rates when setting the equivalence ratio in the experiment. The maximum critical velocity gradient occurs at equivalence ratio of 1, corresponding to experiment by Harris et al. [51].

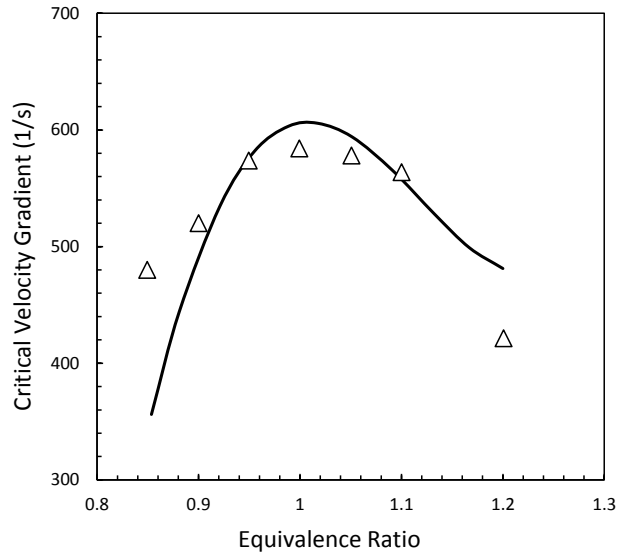


Figure 53: Critical velocity gradient variation with equivalence ratio for 5 mm burner diameter, experimental results are shown by triangles, reproduced from [89].

Kurdyumov et al. [90] carried out a simplified analysis of boundary layer flashback. They investigated the propagation of a flame close to the wall inside a tube, the diameter of which was much larger than the preheated flame thickness. With this condition, the computational domain was modeled by the flow over a flat plate and the so-called thermal-diffusive approximation was used. Neglecting the density and viscosity variation with temperature, the laminar velocity profile ($u = Ay, v = 0$) remains undisturbed through the entire domain. As a result, energy and deficient reactant mass conservations were solved without fluid dynamic coupling in non-dimensional forms for adiabatic and isothermal wall boundary conditions. A one-step chemical reaction with Arrhenius kinetics was used while taking into account the effect of Lewis number of different fuels. The Karlovitz number was used to present flashback limits and defined as $(\alpha g / S_L^2)$, which is inversely proportional to the Damköhler number ($Ka = 1/Da$). Based on this definition, flashback occurs if the velocity gradient drops below a critical value. For Lewis number less than unity, the diffusion of the fuel is enhanced, resulting in an increase in flame temperature and, consequently, higher burning velocity. Hence, the critical velocity gradient increases with a decrease

in Lewis number leading to higher Karlovitz number. The Lewis number effect is more pronounced for the adiabatic wall boundary condition.

As pointed out, the governing equations of motion were not solved in this analysis and consequently the flame-flow interaction was not captured by the simulation. The interaction of heat release, pressure, and velocity distribution plays a crucial role in boundary layer flashback inside a tube or channel as discussed in 5.4.

A two-dimensional simulation of the flame propagation in a tube was also carried out in [91,92] considering the thermo-diffusive approximation. Figure 54 shows the flame propagation velocity in a tube normalized by the planar laminar burning velocity for various Lewis numbers [92]. The isothermal and adiabatic boundary conditions are denoted by solid and dashed lines, respectively. The circle symbols show the tube radius at which the flame quenching occurs. For Lewis number less than unity, the calculated flame speed is larger than that of the adiabatic planar burning velocity due to the cellular flame structure. It is noted that two solutions exist for the isothermal wall, upper branch (stable) and lower branch (unstable), separated by the quenching radius. For Lewis number larger than unity, the flame propagation approaches the planar burning velocity monotonically by an increase of the tube radius. For unity Lewis number, the curve has a “mushroom-shaped” as observed in [88,175] and the stable branch approaches the planar flame speed asymptotically. When the tube diameter is sufficiently large compared to the quenching radius, three dimensional analysis is required and calculated burning velocity changes significantly [92]. However, if the tube diameter is similar to the quenching diameter, two dimensional analysis successfully predicts the results since the flame structure is essentially two dimensional.

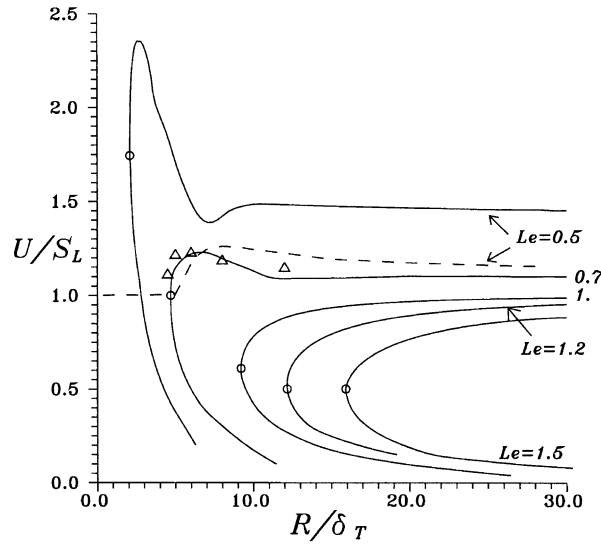


Figure 54: Non-dimensional flame velocity as a function of tube radius normalized by preheat zone thickness, solid line (isothermal wall), dashed line (adiabatic wall), circle (flame quenching), triangle (three-dimensional solution for isothermal wall and $Le=0.7$)[92].

Flame propagation in the boundary layer was studied numerically and experimentally by Kurdyumov et al. [94] and qualitative agreement was observed. The experiment was conducted with propane/air and methane/air mixtures at equivalence ratio of 0.53 and 0.57, respectively. The tube diameter was 21.4 mm and laminar flow was fully developed with the Reynolds number less than 150 for all cases. The flame was stabilized downstream of the burner similar to the experiment by Lewis and Elbe [47]. Kurdyumov et al. implemented laser tomography to capture the streamlines through the steady flame [94]. A tilted flame flashback was observed in this experiment and Figure 8(b) indicates the flame anchoring inside the tube burner. The flame front is strongly curved close to the tube wall as previously illustrated in Figure 5 [47,48,63]. The flame thickness was estimated around 0.23 mm and since it is significantly smaller than the tube diameter, the flame structure was considered two-dimensional in the laser sheet plane near the wall. Figure 55(a) shows the tomographic cut of streamlines across the flame zone. The parabolic velocity profile is also indicated in Figure 55(a) within 9 mm from the tube wall. As the flow approaches the flame front, the streamline departs from the parallel path along the tube wall due to back pressure of the flame front.

The critical velocity gradient of methane/air mixture at equivalence ratio of 0.57 was 50 1/s, which is in good agreement with Lewis and Von Elbe's results [47] shown in Figure 13. The value for the propane/air mixture at equivalence ratio of 0.53 was found to be 39 1/s, which is in agreement with results from [113].

The steady Navier-Stokes and energy equations were solved numerically along one-step chemical reaction and Arrhenius law [94]. The two dimensional numerical simulation was similar to the study in [86], but final equation were simplified by asymptotic techniques for scale parameters of the problem. The effect of Lewis number was studied as a parameter, while the Prandtl number was assumed to be a constant value at 0.72. The dependency of viscosity, thermal conductivity, and fuel diffusivity on the temperature were represented by a power law function. Streamlines obtained from numerical results is similar to the experimental results as indicated in Figure 55(b). Streamlines are deviated as approaching the flame front and then become parallel to the tube wall, which is consistent with observation in Figure 55(a).

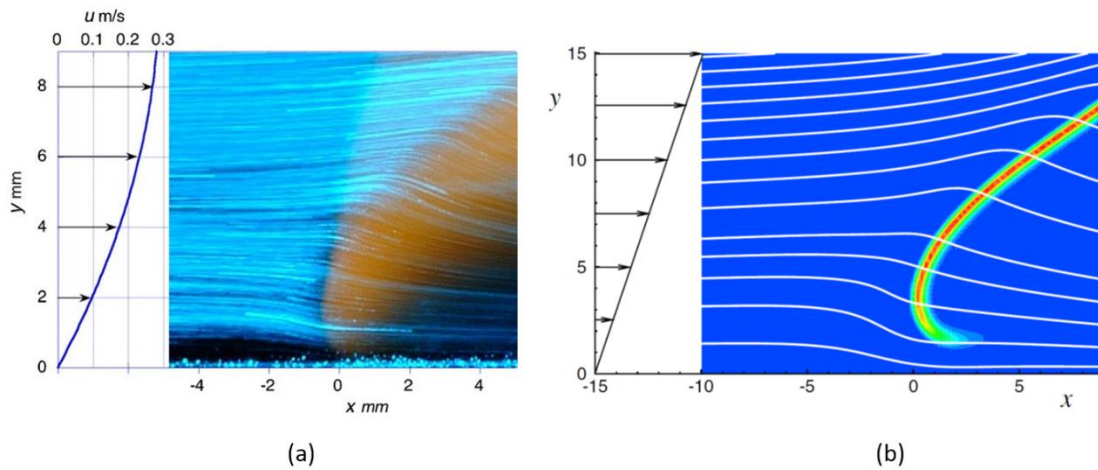


Figure 55: (a) Tomographic visualization of streamlines across the laminar flame and upstream velocity profile and (b) streamlines pattern and flame front location for the stabilized flame near the tube wall [94].

Figure 56 illustrates the critical Damköhler number as a function of Lewis number for isothermal and adiabatic wall boundary conditions. The experimental results are shown in Figure 56 for comparison.

With a decrease in the Lewis number, the Damköhler number decreases, corresponding to increased flashback propensity. The experimental results of methane/air with $Le \approx 1.8$ and propane/air with $Le \approx 0.97$ mixture show that the flashback occurs at critical conditions of $Da = 9.8$ and $Da = 6.5$, respectively, which are in agreement with the numerical results shown in Figure 56. For Lewis numbers less than unity, the differential diffusion and curvature effects cause a higher flame temperature and burning velocity. It is important to realize that the real boundary condition of the tube wall is the combination of both isothermal and adiabatic cases and the experimental data fall within these two limits (i.e. solid and dashed lines) as indicated in Figure 56.

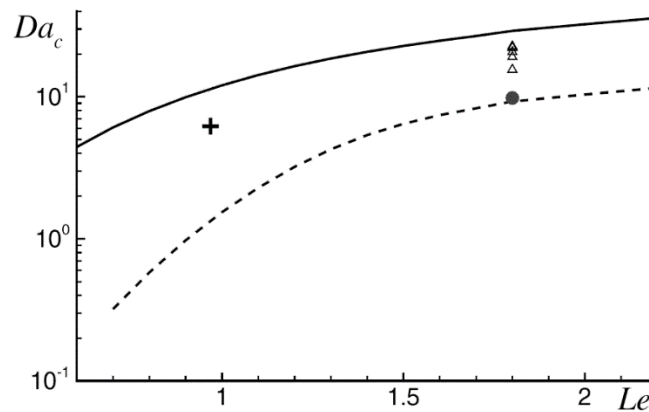


Figure 56: Critical Damköhler number as a function of Lewis number: solid line (isothermal wall), dashed line (adiabatic wall), triangle (experimental data points from [113]), circle (experimental result for propane), cross (experimental result for methane) [94].

The flame-wall interaction of the laminar flow has been numerically studied in the literature through one-dimensional and multidimensional analyses. The one-dimensional numerical modeling of laminar flame-wall interaction was mainly focused on head-on quenching in which the premixed flame propagates perpendicular to the wall. However, these studies cannot be applied to boundary layer flashback since the flame propagates parallel to the wall and the sidewall quenching occurs. In addition, reaction mechanisms used in most of these studies are limited to the single-step model, which is not able to provide realistic results due to the lack of recombination reactions near the wall.

The multidimensional numerical simulation of wall-flame interaction is expensive in terms of computational time. As a result, much of the existing research in the literature feature single-step chemical reaction mechanisms or are simplified by the thermal-diffusive approximation. The latter case neglects fluid dynamics coupling in which the flow is not affected by the thermal expansion resulting from the presence of the flame.

4.6 Key Points and Status

- For unconfined flames, the interaction between the approaching reactants and the flame is small which is a major assumption underlying the critical velocity gradient concept. In this case, the critical velocity gradient concept to be successfully applied to predict flashback. The flow-flame interaction is significant in confined flames and thus the concept is less able to capture the behavior and a new physical framework is needed to account for these cases. Work in recent years has established this framework (e.g., [99]).
- Generalization of the concept to flows that are not fully developed has yet to be completely evaluated. Most studies specifically generate fully developed flow to allow analytical expressions to be used to estimate the velocity gradient near the wall. The role of developing flow in impacting flashback remains an open question.
- The role of the quenching distance is very important and yet different concepts are used to establish the value of this parameter. The quenching distance is measured between two parallel plates (inside a tube) or approximated by the flame thickness. These concepts may not be able to provide realistic values for the boundary layer flashback in which sidewall quenching occurs.
- Burner diameter plays a role in flashback once it falls below a certain diameter, which is comparable to the quenching distance. Above this diameter, burner diameter appears to be unimportant based on studies isolating this parameter. From a practical view, for uncooled burners, thermal coupling between the flame and the burner results in a complex interaction between factors impacting heat transfer (e.g., flame temperature, flame proximity to the burner rim, product composition, and burner diameter). This interaction is not captured by models of the flashback process in any comprehensive manner.
- The reactant temperature plays a strong role on flashback of laminar flames. This is due to temperature effect on both the flame propagation rates (they increase with temperature) and quenching distance (reduces with increased temperature). The relative changes in these two parameters is a function of fuel composition which leads to difficulty in generalizing the effect in a manner that can be readily captured with a simple model. These effects, when also considered in the heat transfer impacts mentioned in regards to burner diameter effects, require careful consideration in any general model.
- Only a few studies have considered the impacts of operating pressure on laminar flame flashback. The laminar flame speed dependence on pressure has been established for many conditions in recent years and generally decreases weakly with an increase of pressure. The quenching distance, which is proportional to the flame thickness, decreases roughly linearly with pressure and plays the main role for increased flashback propensity at high pressure.

- Fuel composition plays a complex role on flashback behavior, but straightforward correlations have been developed to capture these effects especially for burners with controlled temperature. If hydrogen or carbon monoxide is involved in the mixture, additional consideration is required. The addition of small amounts of these species can have a strong impact on flashback propensity for a given condition.
- Numerical simulation of flashback in laminar jet flames has been carried out on a limited basis. Highly simplified chemistry and two dimensional approaches have been used with some success, but general agreement with experimental results is not as close as desired.

5. Turbulent Flames

As discussed in Section 4, boundary layer flashback of laminar flames has been extensively investigated in the literature and the critical velocity gradient concept is able to capture the overall behavior. However, the flame-wall interaction in turbulent flows is significantly more complex [176]. Considering the nature of turbulent flashback which, to a large degree, is a function of turbulent Reynolds number, spatially and temporally resolved diagnostics are required to capture the detailed processes. Due to computational complexity and the inherent temporal nature of turbulent flashback, simulations of the phenomena are relatively few with recent direct numerical simulations by Gruber et al. [100,108] one notable example. Due to the lack and/or cost of appropriate computational and experimental resources, many researchers have relied on the basic theory developed for laminar flames by Lewis and Von Elbe [47] in an attempt to estimate flashback in turbulent flames. As is discussed in this section, application of the basic theory has been made with some success although some completely new approaches have been considered in recent years.

5.1 Prediction Models

The laminar burning velocity and quenching distance can be measured by different techniques [177,178] as implemented by many researchers. However, extending laminar boundary layer flashback results to turbulent flames requires additional consideration. Also, in practice, the velocity profile at the burner head is not always fully developed due to the short length of premixer or presence of a nozzle or other feature in the premixer passage. The simple critical velocity gradient concept may not apply to such cases

as evidenced by the study of Yamazaki and Tsuji [80] for a converging nozzle with different perforated plates. Nevertheless, the concept of critical velocity gradient originally developed for laminar flames has been frequently used for characterizing turbulent boundary layer flashback propensity in the literature.

Khitrin et al. studied the turbulent flashback propensity for methane-air, methane-oxygen, and hydrogen-air mixtures [82]. Figure 57 presents the flashback velocity of laminar and turbulent flames as a function of hydrogen concentration in the mixture. Different points on the solid line show various burner diameters as labeled in Figure 57. Two dashed lines represent the flashback of laminar hydrogen flame from [48]. The velocity at which flashback occurs is much higher for turbulent flames. According to Figure 57, flashback velocity under turbulent conditions is not strongly affected by burner diameter while the flashback velocity of laminar flame is a function of diameter which is also consistent with Figure 12 and Eq. 5.

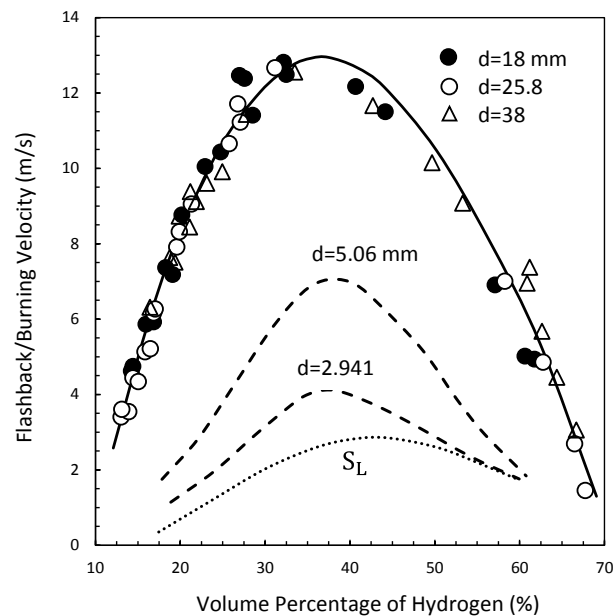


Figure 57: Flashback propensity of turbulent hydrogen flame (solid curve) and laminar hydrogen flame (dashed curves [48]). Dotted line shows the laminar burning velocity, reproduced from [82].

Using Eq. 3 and the calculated critical velocity gradient at the wall from Eq. 8, the following non-dimensional equation is obtained:

$$Pe_{flame} = \left(\frac{\delta_p}{D} \right) Re^{1.8} Pr \quad \text{Eq. 42}$$

in which $Pe_{flame} = DS_L/\alpha$ and Pr is the Prandtl number of the unburned mixture. This equation was obtained based on the laminar critical velocity gradient model neglecting flame-flow interaction. The extension of critical velocity gradient model to turbulent flame was justified by assuming that the flashback of turbulent flame occurs in the viscous sublayer ($y^+ < 5$) where the velocity profile is linear. This basic model was applied to the experimental data for methane/oxygen and methane/air mixtures under turbulent conditions [82]. A Prandtl number of 0.9 was assumed for methane/oxygen and methane/air mixtures and the penetration distance was considered proportional to the one-dimensional flame thickness ($\delta_p = K\alpha/S_L$) [82]. With these considerations, the final form of Eq. 42 is reduced to:

$$Re = \text{const. } Pe_{flame}^{1.11} \quad \text{Eq. 43}$$

Figure 58 illustrates the experimental data with respect to the critical velocity gradient model on a log-log plot. While the trends are captured to some extent, the deviation from the solid line (Eq. 43) is fairly significant in terms of predicting flashback propensity.

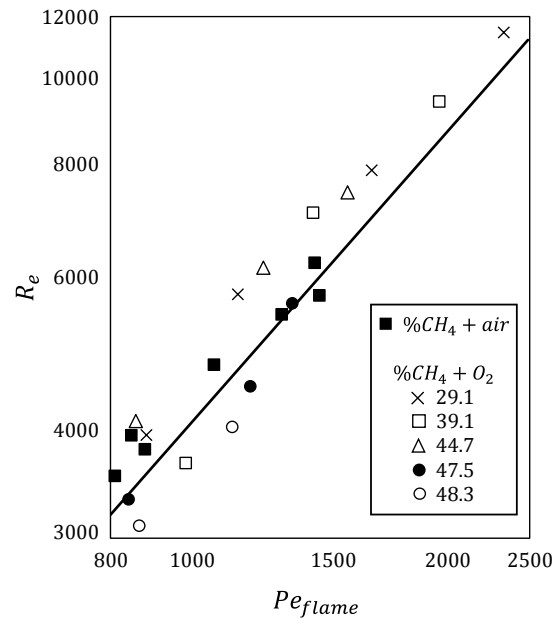


Figure 58: General prediction model (Eq. 43) for turbulent flashback propensity of methane/oxygen and methane/air mixtures, reproduced from [82].

In this case, the critical velocity gradient model is not able to correctly predict the turbulent boundary layer flashback due to the simplifying assumptions used in the original for the model, including (1) linear velocity profile near the wall and (2) use of the unstretched laminar burning velocity. However, the velocity gradient at the wall can still characterize the local flow velocity and can be used along with the local turbulent burning velocity for boundary layer flashback prediction as discussed in Section 5.4.2.

5.2 Effect of Flame Configuration and Boundary Layer Heating

The effect of thermal coupling between the burner and flame front was noted in the literature even in the initial experiments conducted by Lewis and von Elbe [47]. To reach flashback limits, the equivalence ratio is required to be gradually increased such that the burner rim temperature remains in balance with the heat transfer from flame front, i.e. a quasi-steady process. It has been observed that, if flashback is approached quickly in an uncooled burner rim, the critical velocity gradient determined may be significantly different from the result with a slow approach. The first comprehensive study to investigate the effect of tip temperature on turbulent boundary layer flashback was conducted by Bollinger and Edse [66]. The experiment was carried out for hydrogen/oxygen flames at atmospheric pressure. The effect of burner material, burner diameter, and tube wall thickness were studied. Experiments for burners with small diameter were carried out in a closed room. Some other tests with larger diameter tubes were conducted in an open firing pit. The tip temperature was monitored using thermocouples mounted on the burner head. The flashback was approached slowly to ensure that the thermal equilibrium was maintained as the equivalence ratio increased.

Figure 59 indicates the variation of critical velocity gradient as a function of hydrogen concentration for various burner diameters and thicknesses. Curves 2, 3, and 5 represent the effect of the burner diameter on critical velocity gradient. Results from burner tubes with diameters of 0.483 and 0.613 *cm* are similar at lean conditions. However, the dilution of the excess hydrogen with surrounding air leads to a significant deviation between results of curve 3 and 5 at rich condition. Also, the trend line of curve

2 differs significantly from two other burner tube diameters. This effect is more pronounced for the curve 1 with thinner wall as illustrated in Figure 59. The ratio of quenching distance to burner diameter increases as the burner diameter decreases. Indeed, once the quenching distance magnitude is comparable to the burner diameter, the velocity at which flashback occurs cannot be approximated by the linear velocity profile close to the wall. This observation is consistent with previous studies [47,48] for laminar flames.

The critical velocity gradient of burners with thicker walls (0.025 in) is consistently higher than that for burner tubes with thinner walls (0.009 in), i.e. curve 2 vs 1, curve 3 vs 4 in Figure 59. The effect of dilution via entrainment is stronger for thinner tube walls, leading to lower burning velocity at the burner rim and consequently a more stable flame. Hence, while the thicker tube wall may decrease tip temperature, they are more prone to flashback due to the reduced dilution effect according to [66].

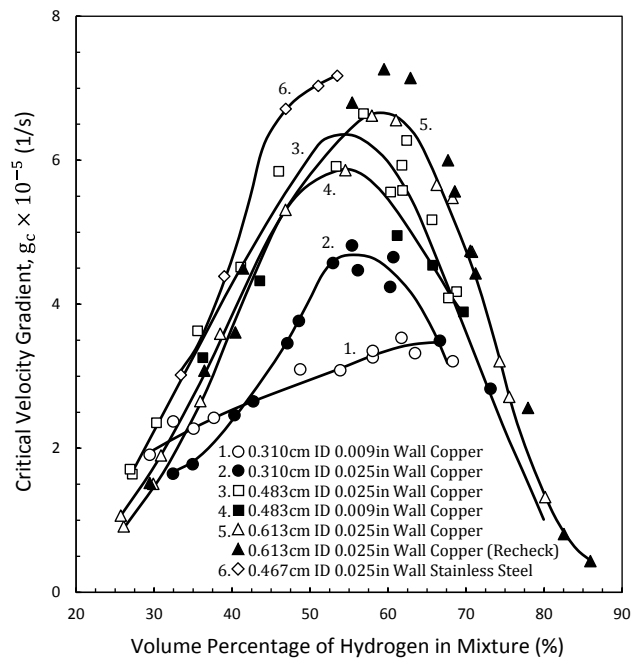


Figure 59: Effect of burner diameter on critical velocity gradient of hydrogen-oxygen flames for uncooled burners in closed room, reproduced from [66].

Figure 60 and Figure 61 illustrate the effect of the burner tube material on critical velocity gradient and tip temperature for a constant burner thickness. The flashback propensity for the copper burner is much lower than the other studied burner materials. This is due to the significantly lower tip temperature of the copper burner as indicated in Figure 61. Comparing the thermal conductivity of burner materials studied in [179], it is observed that the tip temperature is proportional to the thermal conductivity as indicated in Figure 61. The heat transfer from the flame to the burner balances with the heat loss to the ambient air and the incoming unburned mixture. The heat conduction along the tube wall is determined by the burner material and consequently the heat loss increases with an increase in thermal conductivity of the burner head, leading to lower burner tip temperature. However, the critical velocity gradient variation does not simply follow the thermal conductivity and tip temperature variations for stainless steel, Inconel, and Monel as illustrated by Figure 60. The thermal conductivity of these materials is approximately the same order of magnitude. The nickel burner has much higher thermal conductivity than stainless steel, Inconel, and Monel resulting in lower tip temperature and critical velocity gradient.

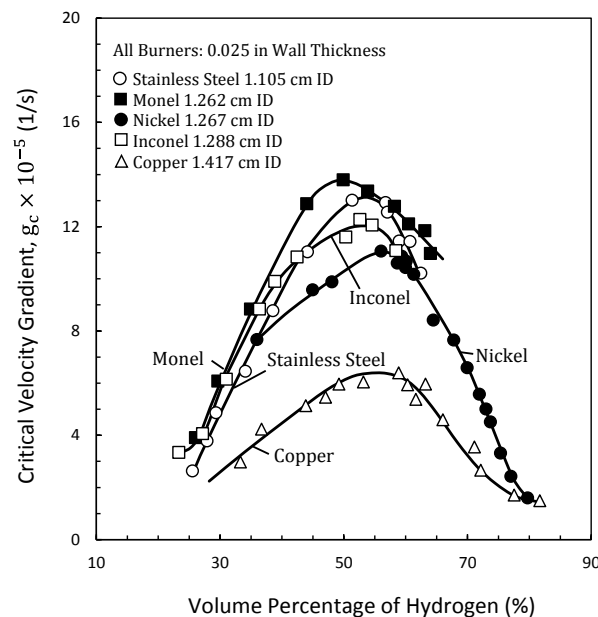


Figure 60: Critical velocity gradient variation with hydrogen concentration for uncooled burners constructed from different materials, reproduced from [66].

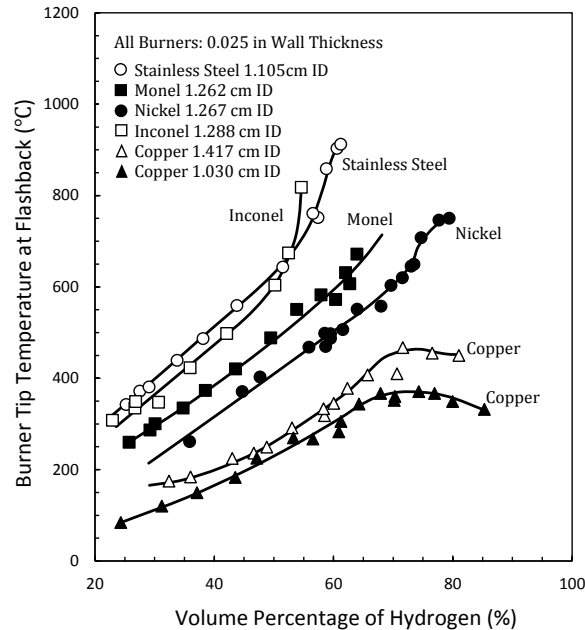


Figure 61: Tip temperature variation as a function of hydrogen concentration for uncooled burners constructed from different materials, reproduced from [66].

Flashback propensity of high hydrogen content fuels was studied for an unconfined flame (recall Figure 8) by Shaffer et al. [102]. Different fuel compositions including hydrogen, carbon monoxide, and methane were investigated. The experiment was carried out at atmospheric pressure and room temperature for two constant adiabatic flame temperatures of 1700 and 1900 K. The flow was fully developed at the burner exit. The premixed reactants were injected through a quartz tube and flame was stabilized inside a quartz liner. To approach flashback, the fuel and air mass flow rates were decreased simultaneously such that the equivalence ratio remained constant, corresponding to a given adiabatic flame temperature.

The hydrogen data were obtained without an enclosure for comparison with results from [82,98]. A significant difference was observed for hydrogen with open flames and mainly attributed to the burner material of the injector as burner cooling was not implemented in this study. A quartz tube used in this experiment, while the flame in [98] was stabilized on top of a brass pilot burner. Final results of this study

were presented in the form of a least squares best fit correlation based on fuel compositions and adiabatic flame temperatures as:

$$g_c = (2.401.H_2 + 5.318.CO - 1.632.CH_4 - 0.264.AFT + 0.604.H_2.CH_4 + 0.00264.CH_4.AFT)^2 \quad \text{Eq. 44}$$

in which, the fuel concentration and adiabatic flame temperature are in percent and Kelvin, respectively. The effect of fuel composition can be isolated in this correlation and some conclusions were made by varying the fuel component concentration in Eq. 44. As carbon monoxide or methane are added to the pure hydrogen, the flashback propensity decreases to a different degree, which is consistent with the conclusion in [103]. The hydrogen-carbon monoxide mixture is found to be more prone to flashback than the hydrogen-methane for a given volume concentration of hydrogen in the mixture. This is in agreement with laminar flashback results shown in Figure 34 for H₂-CH₄ and Figure 36 for H₂-CO. It is noted that the addition of the methane to hydrogen decreases the laminar burning velocity exponentially (see Figure 35), while the laminar burning velocity of H₂-CO mixture decreases linearly with an increase of carbon monoxide (see Figure 37).

A similar correlation Eq. 45 was also developed for tip temperature variation as a function of fuel composition and adiabatic flame temperature [102]. The hydrogen content has a significant impact on increasing the tip temperature and, for constant hydrogen concentration, added CO content increases flashback propensity more than added CH₄.

$$T_{tip} = -1.58.H_2 - 3.63.CO - 4.28.CH_4 - 0.264.AFT + 0.38.AFT \quad \text{Eq. 45}$$

The thermal coupling between the burner and flame was investigated in more detail by Duan et al. [103]. The flashback propensity of enclosed flames was studied for the mixtures of hydrogen, methane, and carbon monoxide. Different preheat temperatures of 300, 450, 588, and 810K were examined for all fuel compositions mentioned. To investigate the effect of tip temperature on flashback propensity, burner heads constructed from stainless steel and quartz were used. Comparing the critical mass flow rate for stainless steel and quartz burners for a given fuel composition, it was concluded that the quartz burner

head increases the flashback propensity. This is attributed by Duan et al. [103,104] to an increase of tip temperature, resulting from the lower thermal conductivity compared to the stainless steel. A simple analytical heat transfer solution was derived to show the dependency of the burner tip temperature on thermal conductivity for non-preheated mixtures. Based on this analysis, the tip temperature is proportional to square root of the thermal conductivity for a jet flame operating at room temperature.

Some correlations were developed in [103] based on statistical analysis of the data to predict the critical velocity gradient as a function of fuel contents, thermal properties, laminar flame speed, thermal properties, and quenching distance. The quenching distance was approximated by the one-dimensional-flame thickness taking into account the effect of wall temperature.

Duan et al. studied the influence of burner materials and geometrical flame configurations on flashback propensity of hydrogen/air jet flames [104]. Different burners were investigated including open, confined, and enclosed flames (recall Figure 8). Two enclosures with inner diameters of 44.45 and 63.5 *mm* were used to evaluate the effect of enclosure size on tip temperature and flashback limits. Furthermore, a ceramic block with a slightly larger inner diameter than the injector diameter was connected to the injector, resembling the confined flame.

The tip temperature of the enclosed flame was significantly higher than the open flame as the recirculation of hot products influences the burner rim temperature. Comparison of the associated critical velocity gradient indicated higher flashback propensity of the enclosed flame for metal and quartz burner heads. The difference between flashback limits of open and enclosed flames is more pronounced for the quartz burner with lower thermal conductivity. As a result, the difference in flashback limits is mainly due to boundary layer heating, which can induce the upstream flame propagation.

Duan et al. [104] also concluded that enclosure size has a negligible effect on tip temperature of the stainless steel burner, while it significantly affects the quartz burner. The quartz burner has a higher tip temperature for a given adiabatic flame temperature and is more sensitive to the enclosure size.

Hence, the critical velocity gradient remains approximately constant for the stainless steel burner and decreases with an increase of enclosure size for the quartz burner. This indicates that the variation of critical velocity gradient is a result of change in the tip temperature and consequently the boundary layer heating. It also illustrates that a complex interaction between burner geometry and materials exists.

Figure 62 summarizes all results in terms of critical velocity gradient as a function of equivalence ratio. Note that the confined flame has significantly higher flashback propensity than unconfined flames regardless of the burner size and material. This conclusion was noted in [98] as will be discussed later. It was observed in [99] that the flashback mechanism for confined flames is affected by the flame-flow interaction in the near-wall region at the onset of flashback. Indeed, a recirculation region exists in the boundary layer before the flashback occurrence, enhancing the flame regression into the burner (further discussed in Section 5.4). Cooling the burner decreases the flashback propensity as seen from results of stainless steel burner in Figure 62. Furthermore, the injector diameter has a negligible effect on flashback limits of hydrogen flames, consistent with the findings of Khitrin et al. [82].

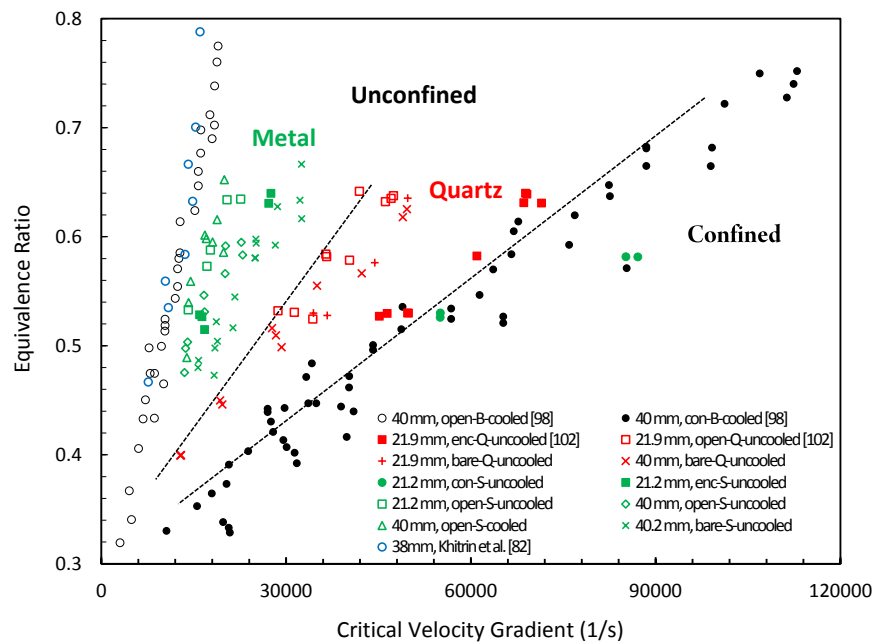


Figure 62: Summary of results for various injector configurations, burner diameters and materials [104].

Turbulent boundary layer flashback of methane-hydrogen/air in a channel with the presence of an adverse pressure gradient was investigated by Eichler and Sattelmayer [97]. The final results were presented in terms of critical velocity gradient and compared to the previous data in the literature for a tube burner. A diffuser with rectangular cross section was considered and the lower diffuser wall was adjusted to the 4 degree slope. The flow was fully developed at the burner exit.

The temperature of the lower diffuser wall was controlled by impinging the air through three jets and monitored by three thermocouples.

The flame tip location in the boundary layer was captured by OH* images. It was observed that the flame tip was strongly curved towards the wall channel as previously indicated in Figure 5. The critical velocity gradient was then calculated from the wall friction by the Reynolds averaged numerical simulation (RANS) of a 2D model.

The final results are summarized in Table 7 for methane and hydrogen at different mass flow rates. g_c denotes the mean critical velocity gradient calculated from the friction factor obtained by numerical simulation. Another set of experiments were conducted for each case to evaluate the effect of the wall heating on flashback propensity.

The critical velocity gradient results for confined flames are significantly higher than previous results in the literature for unconfined flames. The critical velocity gradient of natural gas at equivalence ratio of 0.8 in [77] was reported to be less 2000 1/s for turbulent flame, which is three times smaller than the value reported in Table 7 [97]. Such a significant difference reveals that the flashback characteristic is influenced by the flame configuration (i.e. confined and unconfined flames in Figure 8).

Table 7: Critical velocity gradient at different mass flow rates for methane and hydrogen, diffuser angle of 4 degree [97].

Fuel	$\dot{m}_{air}$ (g/s)	ϕ_{BF}	g_c (1/s)
CH4	40	0.8	5329
CH4	50	0.8	6952
H2	50	0.25	8098
H2	60	0.25	10,047
H2	100	0.29	31,899
H2	120	0.29	42,576

The significantly different flashback limit for confined flames was further studied by Eichler et al. [98] for hydrogen/air mixtures. A setup similar to that shown in Figure 63 was used [98]. The metal channel was followed by a ceramic tile and another ceramic block. Three configurations were studied. In the first and second configurations, the ceramic tile is flush with the lower wall of the channel and the ceramic block is offset by 4 mm. The corner of the ceramic tile in configuration 2 is replaced by the stainless steel material. In configuration 3, two small steps are formed between the channel, tile, and ceramic block with different heights, 0.5 and 3.5 mm; 2 and 2 mm.

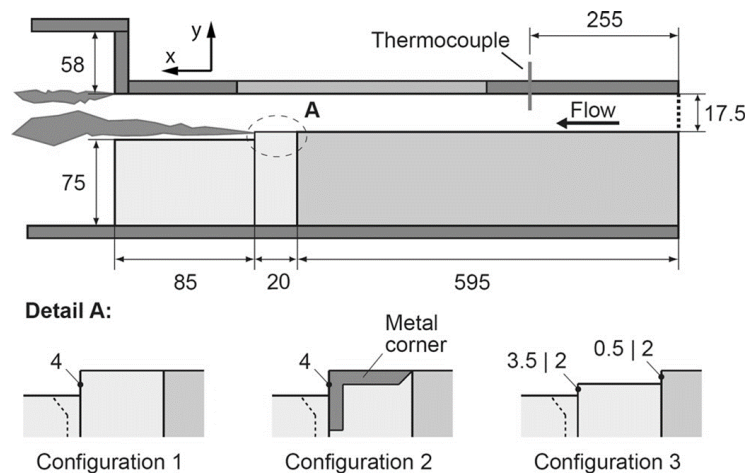


Figure 63: A schematic of boundary layer flashback setup used by Eichler et al. [98].

The channel length was less than the minimum value to create a fully developed flow at the exits section. As a result, the critical velocity gradient was calculated from a log law approximation in Eq. 47, in which the bulk flow velocity is related to the friction velocity [114].

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}} \quad \text{Eq. 46}$$

$$\bar{u} = u_\tau \left(\frac{1}{\kappa} \right) \ln \frac{hu_\tau}{\nu} + B - \frac{1}{\kappa} \quad \text{Eq. 47}$$

Where h is half channel height and constants $\kappa = 0.41$, $B = 5$ are given from Coles and Hirst [180].

As mentioned earlier, the flashback propensity of confined flames in a channel was significantly higher than the existing data for unconfined tube burners. However, this was only validated for the channel, and it was not clear if the confinement is the main reason for such an increase. Hence, a confined flame was also studied in [98] for a tube burner to compare results from both experiments. A pilot burner made from brass was used to stabilize the flame. A ceramic block was used above the pilot burner and caused the flame to creep upstream along the ceramic wall and finally stabilized at the burner rim, resembling the condition when the flame already exists in the tube. The pilot burner surface was air cooled to keep the temperature between 40 and 60 °C. Fully developed flow was not achieved and consequently the velocity profile was verified by PIV measurement and RANS simulations. It was verified that the velocity gradient at the tube wall can be estimated by the Blasius correlation in Eq. 8 with less than 4% error from the measured values.

Figure 64 presents all results obtained for flashback limits in the channel and tube burner at different equivalence ratios. The first key observation is that the flashback data points of the unconfined tube burner lie significantly lower than those for all confined cases. Furthermore, the results for confined flames in either the channel or the tube are qualitatively similar. These two observations suggest that the higher flashback limit of confined flames is a result of different flashback characteristics and can be verified by both channel and tube burners.

The flashback limit of configuration 1 (recall Figure 63) is generally higher for all ranges of equivalence ratio covered. The second configuration with 0.5 mm step has a lower flashback limit than the first configuration, but this difference vanishes as the equivalence ratio increases to 1. A few data

points were obtained for the configuration 3 with 2 mm step and the trend line indicates a lower flashback propensity compared to 0.5 mm step. Hence, an increase in step size decreases the flashback limit as indicated in Figure 64. For very lean conditions with equivalence ratio less than 0.4, the flame did not flashback through the entire channel and the flame oscillated between the intersection of ceramic tile and steel block, so this was mentioned as a reason for an offset of data points for three first sets. The flashback limit of the channel with metal corner is similar to results for the ceramic block and no clear difference was observed. It was also noted that the flashback data of the metal corner was limited due to observed thermoacoustic instabilities, which was a result of lower surface temperature compared to the ceramic case.

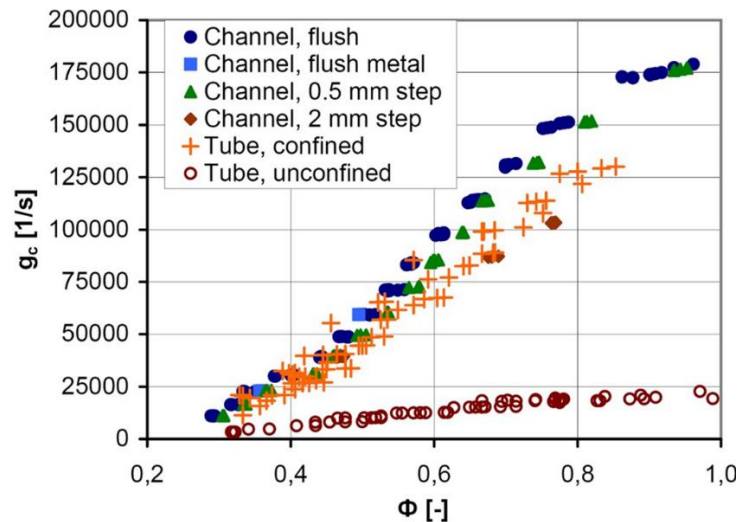


Figure 64: Boundary layer flashback limits of turbulent hydrogen flame for confined and unconfined configurations [98].

The thermal coupling between the burner rim and flame front plays an important role in boundary layer flashback of unconfined flames. As a result, an increase in tip temperature can significantly affect the boundary layer heating and triggering the flashback. Burners constructed from different materials show variation in flashback propensity limits when the burner is not cooled. The presence of the flame enclosure increases the tip temperature and consequently results in higher flashback propensity. Furthermore, a decrease of the enclosure size reduces the heat loss to the ambient air and the

entrainment of the fresh oxidizer, leading to higher tip temperatures. The flashback propensity of confined flames is consistently higher than unconfined flames as experimentally verified for flame propagation through circular tubes and channels. The significant difference in flashback limit reveals that the flame-flow interaction near the wall-region cannot be predicted by the critical velocity gradient concept for confined flames. This observation has been the main objective of recent numerical and experimental studies in the literature [99,100,106,108], discussed in Section 5.4.

5.3 Effect of Operating Pressure and Preheat Temperature

Only a handful of studies of turbulent flashback propensity for mixtures at elevated pressures have been reported in the literature. Edse carried out the first flashback experiments using a Bunsen burner flame at pressures up to 100 *atm* for hydrogen/oxygen mixtures [56]. The other few studies that exist are discussed in this section.

The effect of sub-atmospheric pressure on the flashback propensity of hydrogen was studied for laminar flow, transitional, and turbulent flows by Fine [71,76]. Air and 20.6% oxygen-79.4% argon were used as oxidizers in these experiments with hydrogen as the fuel and flashback propensity was examined for equivalence ratios between 0.8 and 3. The experiment setup by Fine [71,76] was similar to the previous work by the same group carried out under laminar conditions [67]. Burner tubes of 0.546, 1.016, 1.459, and 1.890 *cm* diameters were used. The turbulent flame speed was calculated by determining the mean flame surface from the captured images as explained in [71,181]. The longitudinal velocity fluctuation was measured at the center of the burner to find the range of Reynolds number at which the transition to turbulent flow occurred. This measurement was made in the absence of the flame and the transition range was found to be occurring between $1500 \leq Re \leq 2500$. In addition, the flashback data show a clear transition from a laminar regime to the turbulent regime at similar range of Reynolds numbers. The critical velocity gradient in transition regime was calculated using the laminar approach in Eq. 5.

Figure 65 indicates the variation of critical velocity gradient for hydrogen/air mixtures as a function of pressure at an equivalence ratio of 0.95. The pressure exponents of the critical velocity gradient (slope of solid lines in Figure 65) are 1.28 and 1.22 for laminar and turbulent flames, respectively. It was observed that the pressure exponent remained approximately constant for both laminar and turbulent flames and it was independent of the burner diameter. The average pressure exponent of critical velocity gradient for turbulent flashback was 1.31, which is similar to the 1.35 ± 0.08 , previously found for laminar conditions (Section 4.3). The pressure value stays constant at transitional region (horizontal dashed lines in Figure 65), while the Reynolds number is increased. The value of critical velocity gradient corresponding to the transition regime increases with a decrease in diameter. This transition for the diameter of 1.016 occurs at critical velocity gradient of 4,000, which is higher than the measured value for the cold flow.

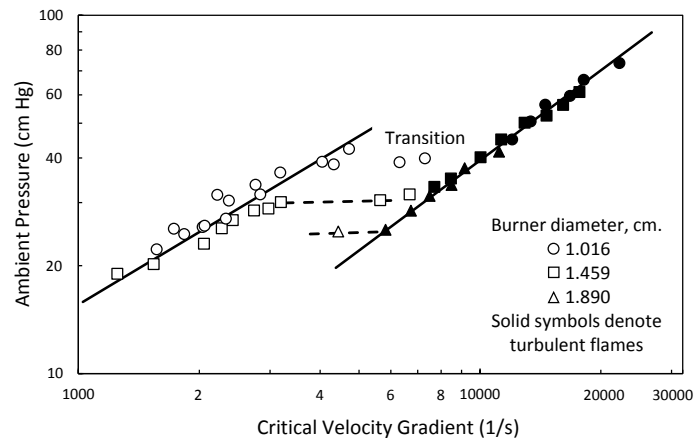


Figure 65: Critical velocity gradient of hydrogen/air mixture as a function of pressure at equivalence ratio of 0.95, reproduced from [71].

The critical velocity gradient of the turbulent flames was found to be approximately 2.8 times that of laminar flames [76]. Hence, the constant coefficient in Eq. 33 (Section 4.3) is multiplied by this value, resulting in 7.3 in the expression:

$$g_c = 7.3 \frac{S_L}{\delta_q} \quad \text{Eq. 48}$$

The comparison of the laminar and turbulent velocities indicated that the ratio of turbulent burning velocity to the laminar burning velocity is always less than 1.3 for this study. Consequently, an increase in critical velocity gradient of turbulent flames is not solely a result of higher flame speed. Indeed, the penetration distance of turbulent flame also decreases according to the Eq. 48 and thus also plays an important role in increasing flashback propensity for turbulent conditions.

Figure 66 illustrates the variation of critical velocity gradient for hydrogen/20.6% oxygen-79.4% argon at equivalence ratio of 1.5. The pressure exponent of critical velocity gradient is about 1.52 for turbulent flames, which is equal to the value reported for the laminar flame in [70] as discussed in Section 4.3.

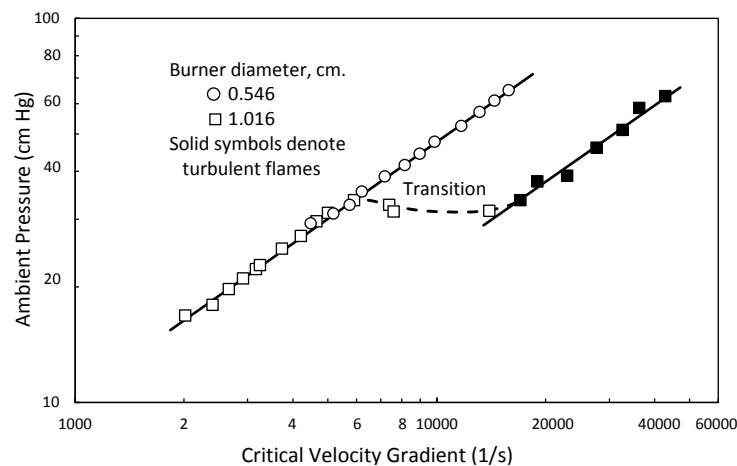


Figure 66: Critical velocity gradient of hydrogen/20.6% oxygen-79.4% argon as a function of pressure at equivalence ratio of 1.5, reproduced from [76].

Figure 67 illustrates the variation of critical velocity gradient as a function of fuel percentage for hydrogen/air at different sub atmospheric pressures. For a given hydrogen concentration, the critical velocity gradient increases with an increase in pressure. Furthermore, the slope of critical velocity gradient variation increases with an increase of pressure at lean and rich conditions. An important observation is that the maximum critical velocity gradient is independent of pressure and diluents, and remains constant at equivalence ratio of 1.5. The laminar flame speed also peaks at the same equivalence ratio of 1.5.

according to [67] (see Figure 26), stating that the critical velocity gradient is proportional to the reaction rate.

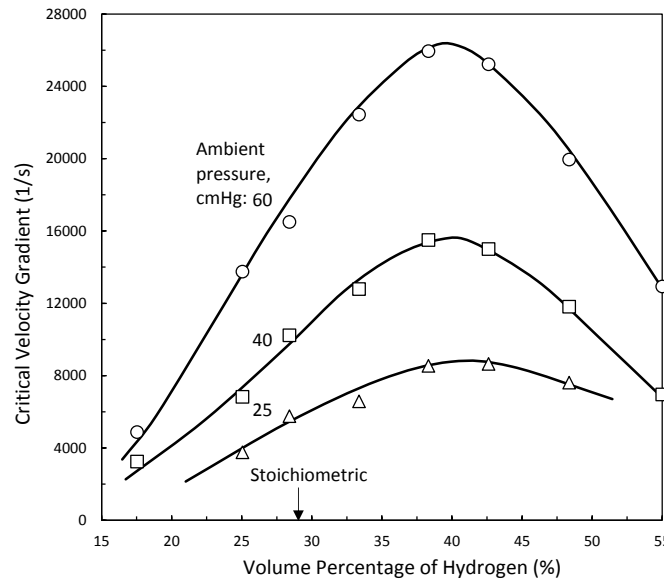


Figure 67: Critical velocity gradient of turbulent hydrogen/air flame as a function of fuel percentage at different pressures, reproduced from [71].

A similar study was conducted by Fine to examine the effect of sub-atmospheric pressure on flashback propensity of propane/oxygen, propane/50%oxygen-50%nitrogen, and propane/20.6% oxygen-79.4% argon [70,76] using an experiment setup similar to their previous laminar studies [67,71]. The turbulent burning velocity was determined based on the mean surface of the flame front as previously pointed out. Again, burner tubes with inner diameters of 0.546, 1.016, 1.459, and 1.890 cm were examined. The range of equivalence ratio studied for propane/50%oxygen-50%nitrogen varied between 0.6 and 1.45. Propane/oxygen and propane/20.6% oxygen-79.4% argon were studied at equivalence ratios of 0.85 and 1, respectively. Critical velocity gradient data were independent of the burner diameter, except for the equivalence ratio larger than 1.25.

The laminar critical velocity gradient for propane/50%oxygen-50%nitrogen peaked at the equivalence ratio of 1.05, while the maximum critical velocity gradient for turbulent flame was not observed within the range of equivalence ratio studied.

Figure 68 illustrates the critical velocity gradient variation of propane/50%oxygen-50%nitrogen as a function of pressure at equivalence ratio of 1. The flashback data are divided to four regions in Figure 68, including very low Reynolds number, laminar, transitional, and turbulent regimes. Laminar and turbulent flames shows a linear trend in logarithmic scales and the slope of these trends line (i.e. pressure exponent) are equal to 1.17 and 1.22, respectively. Furthermore, the data obtained at different burner diameter lie on the same line. In transition regime, the pressure remains constant (horizontal dashed lines) while the Reynolds number is increased and the range of Reynolds number found to be similar; $1500 \leq Re \leq 2500$, as previously observed in [71]. The transition to the turbulent flow for the burner diameter of 0.546 was not observed at pressures below one atmosphere. At very low Reynolds numbers, the slope of critical velocity gradient for the laminar flame drops below the slope of solid line as indicated in Figure 68. This is mainly due to the fact that the velocity gradient cannot be calculated by the linear velocity profile as the quenching effect is enhanced.

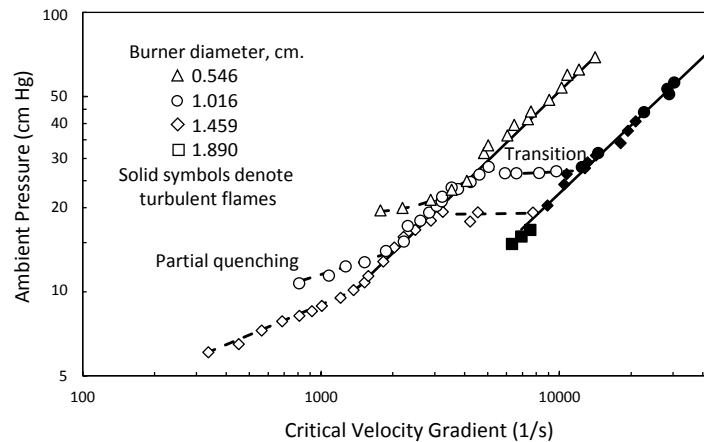


Figure 68: Critical velocity gradient of propane/50%oxygen-50%nitrogen mixtures as a function of pressure, reproduced from [70,76].

The average pressure exponents of critical velocity gradient for propane/50%oxygen-50%nitrogen in laminar and turbulent regimes were 1.13 and 1.11, respectively. These values are slightly lower than those of hydrogen flames as previously mentioned from [71]. For propane/oxygen, the pressure exponent of 1.4 was found for both laminar and turbulent flames. Propane/20.6% oxygen-79.4% argon data points

were not sufficient to find the pressure exponent under turbulent conditions, while the pressure exponent of 0.91 was found for the laminar flames.

Using the measured burning velocity and quenching distance in [182], the critical velocity gradient can be represented as the ratio of the laminar burning velocity to the quenching distance. This ratio for the laminar flame at equivalence ratio of 1 is written as:

$$g_c = 3.1 \frac{S_L}{\delta_q} \quad \text{Eq. 49}$$

The constant value of 3.1 is higher than the coefficient of 2.6 found for the laminar hydrogen flames. The ratio of critical velocity gradients $(g_c)_{turbulent}/(g_c)_{laminar}$ increases from 2.2 to 3.5 with variation of equivalence ratio from 0.6 to 1.25. Assuming the average value of 2.8, Eq. 49 for the turbulent flame would be:

$$g_c = 8.7 \frac{S_L}{\delta_q} \quad \text{Eq. 50}$$

The effect of preheat temperature (300 to 620 K) and sub-atmospheric pressure was studied by Fine [79]. Different fuels were examined in this experiment including hydrogen/air, propane/oxygen-nitrogen, ethylene/oxygen-nitrogen, and ethylene/air. Oxygen-nitrogen mixture contains 50% oxygen and 50% nitrogen by volume. A brass tube burner head with 1.016 cm in diameter was used. The tip temperature was kept within ± 10 K of the preheated gas temperature. For high heating value mixtures (such as propane) at high equivalence ratio, the tip temperature was higher than some low preheat temperatures. Hence, the burner was connected to the chamber to decrease the tip temperature through increasing the conduction heat loss.

Figure 69 illustrates the variation of critical velocity gradient as functions of pressure and temperature for hydrogen/air and ethylene/air. Each figure presents results for an equivalence ratio at which the corresponding critical velocity gradient is about the maximum. Flashback curves at a given preheat temperature are divided into three sections including laminar ($R < 1500$), transition ($1500 <$

$Re < 2500$), and turbulent regimes ($Re > 2500$). In Figure 69(a), dashed line represents the non-preheat data studied in [70,71,76], obtained for water-cooled burner head. The room temperature data in Figure 69(b) represent the experimental results from [49]. The pressure exponent of critical velocity gradient is independent of preheat temperature for the different mixtures studied. The trend line in Figure 69(b) has some curvature for low Reynolds numbers. This is due the enhanced flame quenching effect at these particular conditions. The variation of the critical velocity gradient as a function of preheat temperature is also independent of pressure. The average temperature exponent of critical velocity gradient, i.e. slope of the trend line in Figure 69(a) and Figure 69(b), are 1.52 and 1.59, respectively.

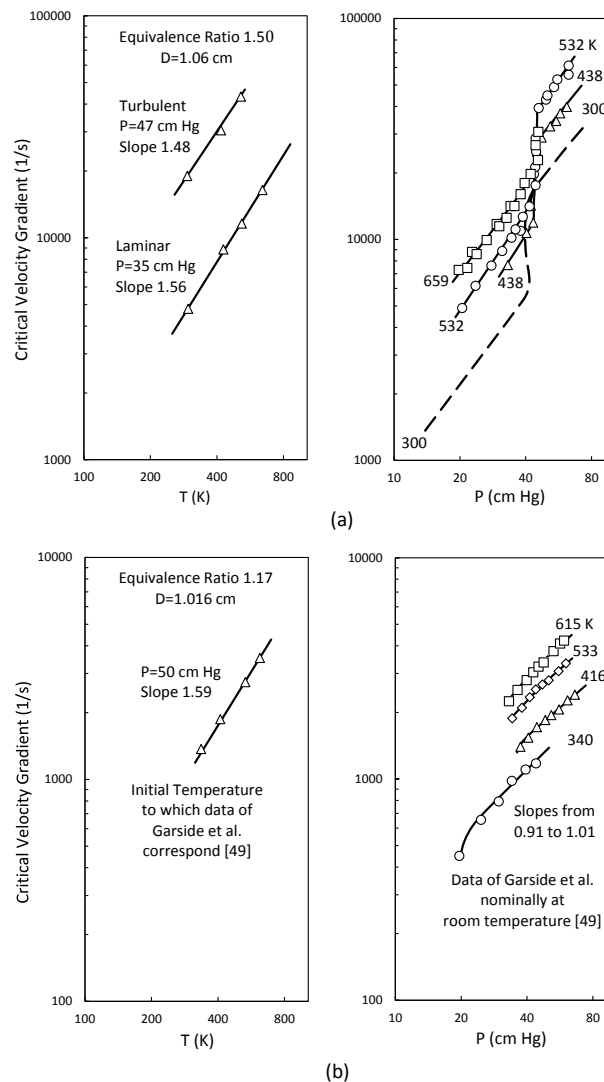


Figure 69: Critical velocity gradient as functions of pressure and preheat temperature for (a) H₂/air at $\phi = 1.50$ and (b) C₂H₄/air at $\phi = 1.17$, reproduced from [79].

The flashback propensity of hydrogen-carbon monoxide flames was studied by Daniele et al. [95] at elevated pressures up to 15 *atm*. H₂-CO mixtures in equal volumetric concentration were studied at equivalence ratios between 0.2 and 0.7. Fuel/Air mixtures were preheated between 577 K and 674 K and were injected into the combustor by a pipe with the inner diameter of 25 *mm*. The complete description of the high pressure facility used in this experiment can be found elsewhere [95,183]. Flashback was approached by gradually increasing equivalence ratio while the preheat temperature and pressure remained constant. Furthermore, mass flow rate was adjusted to obtain a constant velocity for each set of data. Flashback was controlled by both active and passive mechanisms. For passive control, nitrogen gas was injected into the boundary layer of the burner located 150 *mm* upstream of the injector outlet. The nitrogen mass flow rate remained constant at 1% of the air flow rate for all operating conditions such that the flame position was not affected in significant manner.

Figure 70 presents the equivalence ratio at which flashback occurs as a function of pressure. The velocity at the burner head was kept constant at 45, 60, and 65 m/s as labeled for each data set. An increase in pressure for a given preheat temperature and velocity at the burner head decreases the equivalence ratio at flashback significantly, corresponding to increased flashback propensity. The dependency of the equivalence ratio on the pressure can be represented by $\phi = P^{-0.5}$, which is the average of all pressure exponents in Figure 70. For a constant pressure and velocity, an increase in preheat temperature from 577 to 674 K decreases the equivalence ratio at flashback, indicating higher flashback propensity. It was noted that an increase of injector velocity from 45 to 60 *m/s* has a small effect on increasing flashback limit.

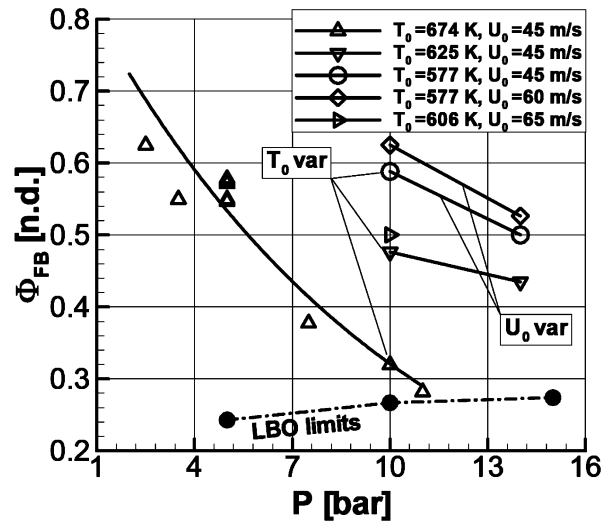


Figure 70: Flashback propensity of hydrogen-carbon monoxide mixture as a function of pressure at different preheat temperatures and velocities [95].

The turbulent burning velocity in this experiment was measured using the global consumption approach and it was observed that an increase in pressure had different impacts on turbulent flame speed depending on equivalence ratio [95]. Close to the flashback limit, the turbulent flame speed increases with an increase of pressure (pressure exponent of 0.12). Since this dependency is weak, the turbulent burning velocity alone cannot explain the significant increase in flashback propensity with an increase of pressure. Indeed, it was concluded that the dependency of flashback on pressure is mainly due to the quenching distance variation [95].

Lin et al. used the turbulent burning velocity to predict flashback propensity of high hydrogen content fuels at high pressure [105]. The experiment was conducted with hydrogen rich fuel gases including 70% H_2 -30% N_2 , 85% H_2 -15% N_2 , and pure hydrogen reacting with air. The bulk velocity at the injector and preheat temperature were kept constant at 40 m/s and 623 K ., while the pressure was varied up to 10 atm . The experiment setup was similar to the previous studies by that group [95]. The turbulent flame speed was calculated using the global consumption rate definition [17].

Lin et al. [105] observed that the flame length can be a good indication of flashback. Indeed, flashback occurred when the flame length reduced to twice the diameter of the injector. Furthermore,

the turbulent velocity correlated well with the flame length. The maximum turbulent burning velocity before flashback was on order of 10 m/s, which was close to the flow velocity in premixing tube at the boundary of viscous sublayer ($y^+ = 5$). As a result, the critical velocity gradient was used to predict flashback propensity. The laminar flame speed in Eq. 5 was replaced by the turbulent flame speed. Based on previous conclusions in the literature, the penetration distance was estimated by the quenching distance, which is proportional to the flame thickness (δ_f) [52]. The final correlation proposed by Lin et al. [105] to predict flashback limit is written as:

$$g_c = \frac{S_T}{Le \times \delta_f} \quad \text{Eq. 51}$$

Flashback occurs when the velocity gradient of the flow (g_f), estimated by the Blasius correlation in Eq. 8, is equal to the value of g_c . Fully developed flow was not achieved in this experiment. Nevertheless, the velocity gradient of turbulent flow was calculated from Eq. 8. Lewis number is used in Eq. 51 to capture the influence of hydrogen in the mixture. This usage was based on the observation that the effect of Lewis number was significant on flashback propensity as described in [90,92,94].

The model by Lin et al. [105] (Eq. 51) was evaluated against the experimental flashback data for 50% H_2 -50% CO in [95]. The model predicts the higher flashback propensity associated with higher preheat temperatures and pressures. The comparison of experimental results with predicted data shows a reasonable agreement. It should be noted that the experimental data were presented for 45 m/s, which is 5 m/s above the velocity of the predicted data. Considering this fact, the predicted data points would be slightly different.

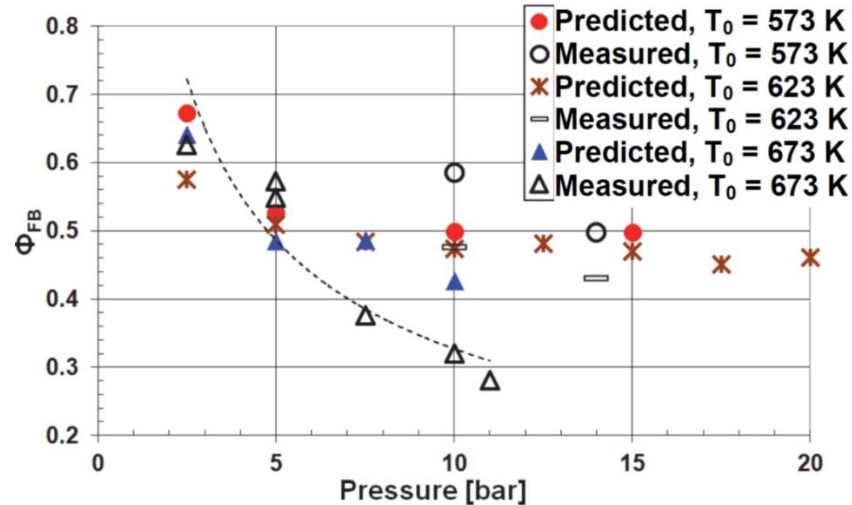


Figure 71: Predicted and measured equivalence ratios of 50%H₂-50%CO mixture at flashback as functions of pressure and preheat temperatures [105].

Another experimental study of boundary layer flashback of a 2.54 cm diameter, fully developed premixed jet flame at gas turbine premixer conditions was carried out by Kalantari et al. [110]. The flashback propensity of hydrogen jet flames was investigated at elevated pressure (3-8 atm) and temperatures (300-500 K). The range of Reynolds number studied in this experiment was $5 \times 10^4 < Re_D < 3 \times 10^5$. Three burner materials including stainless steel, copper, and zirconia ceramic were used to evaluate the effect of tip temperature on flashback propensity. The tip temperature was measured by two thermocouples located at the burner rim.

An increase in pressure causes higher flashback propensity and flashback limits were found to be more sensitive to equivalence ratio at higher pressures. Figure 72 indicates the variation of equivalence ratio as a function of pressure at different preheat temperatures and injector bulk velocities. An increase in preheat temperature for a given velocity increases the equivalence ratio at which flashback occurs. The higher mass flow rates results in higher equivalence ratio at flashback, but this cannot be used an effective method to avoid flashback since there is no significant difference between 30 and 40 m/s, which is according to velocity variation in Figure 70 [95]. The pressure exponent of equivalence ratio in Figure 72

($\phi_{FB} = P^{-m}$) is denoted by m . These values are similar to the average pressure exponent of carbon monoxide-hydrogen mixture reported by Danielle, et al. [95].

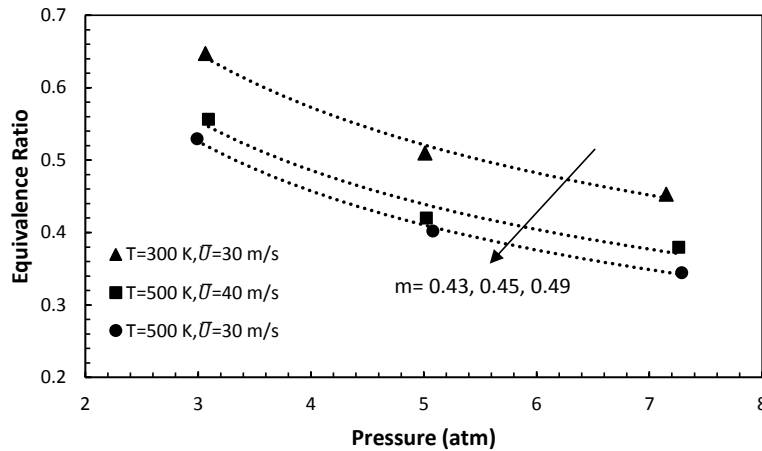


Figure 72: Variation of equivalence ratio vs pressure for the stainless steel burner head [110].

The variation of tip temperature as a function of equivalence ratio is illustrated in Figure 73 for different preheat temperatures and pressures. Trend lines indicate the variation of flashback propensity corresponding to velocity change at the burner head. An increase in preheat temperature for a given pressure increases the tip temperature at which flashback occurs, while the equivalence ratio at which flashback occurs decreases. Indeed, the higher tip temperature is a result of balance between an increase in preheat temperature and the decrease in equivalence ratio. For constant mixture temperature, an increase in pressure significantly decreases the equivalence ratio at which flashback occurs, while increasing the tip temperature at flashback. This observation is explained by considering the dependency of the quenching distance on pressure. As the pressure increases, the quenching distance decreases and consequently enhances the flashback propensity. Furthermore, the effect of tip temperature is enhanced with an increase of pressure. This observation is consistent with the experimental study by Edse [56] for hydrogen/oxygen mixture at high pressure. Edse observed significant scattering in flashback limits for a given condition at high pressure due to the tip temperature variation. Indeed, the repeatability of the flashback data was influenced remarkably by the duration in which flashback was approached [56].

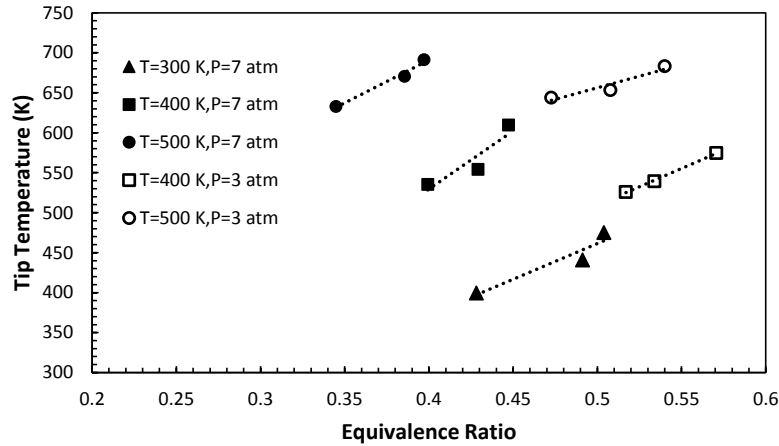


Figure 73: Variation of tip temperature with equivalence ratio for hydrogen at different pressures and preheat temperatures [112].

In an effort to systematically capture the effects of the interacting parameters, Kalantari et al., developed a correlation based on Buckingham Pi theorem [110] and calibrated using the experimental data at high pressure in order to predict the flashback limit [107,110]. The final correlation [110] was presented in non-dimensional form as follows:

$$Da = Const. Le^{1.68} \cdot Pe_f^{1.91} \cdot \left(\frac{T_u}{T_0}\right)^{2.57} \cdot \left(\frac{T_{tip}}{T_0}\right)^{-0.49} \cdot \left(\frac{P_u}{P_0}\right)^{-2.1} \quad \text{Eq. 52}$$

Damköhler number represents ratio of laminar flame speed square to thermal diffusivity and critical velocity gradient in Eq. 53. Peclet number is also defined as a ratio of injector diameter to the penetration distance, which is proportional to the flame thickness.

$$Da = \frac{S_L^2}{\alpha \cdot g_c}, \quad Pe_f = \frac{d \cdot S_L}{\alpha} \quad \text{Eq. 53}$$

Lewis number is calculated based on the thermal diffusivity of the mixture and mass diffusivity of hydrogen into the mixture. The unburned preheat temperature, tip temperature, and operating pressure are defined in three independent non-dimensional groups. The developed correlation was used for the experimental results at high pressure as well as the existing high pressure results for H₂-CO from [95], as depicted in Figure 74. Furthermore, the application of the correlation was evaluated for a commercial 65 kW micro turbine generator [112]. The atmospheric pressure result is for a combustor, removed from the

engine, configured to operate at atmospheric pressure and various preheat temperatures (i.e. 300 to 700 K) [101,112]. The engine data were obtained for a commercial engine operated at different engine loads. In both configurations, studies were carried out with various amounts of hydrogen added to either natural gas or carbon monoxide [112]. The predicted Damköhler numbers are in good agreement with the experimental result as indicated in Figure 74.

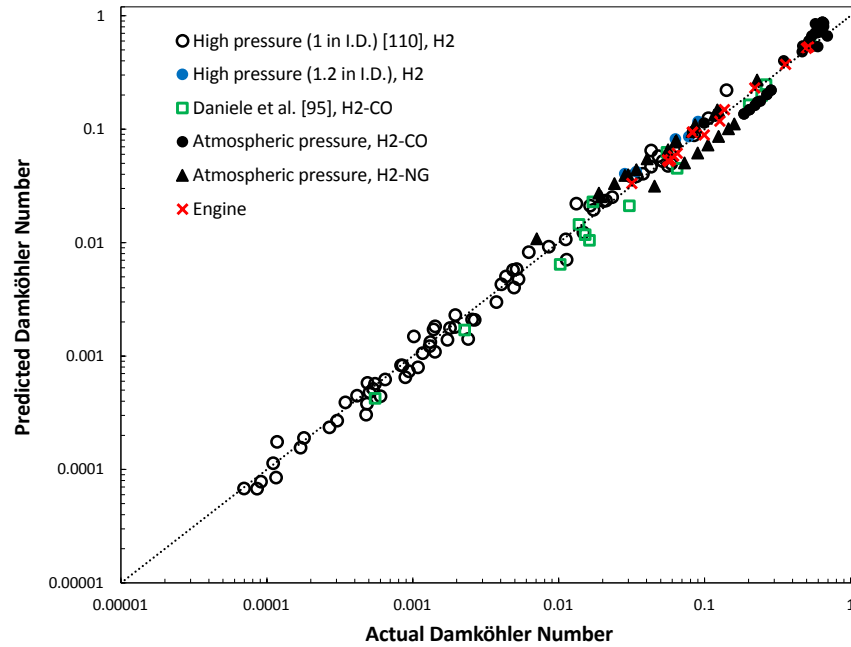


Figure 74: Predicted Damköhler number using Eq. 52 [112].

The turbulent flashback propensity of a mixture increases at high pressure as experimentally observed for various fuel compositions. It is thus apparent that the laminar flame speed cannot be used to determine the turbulent flashback limit and that the turbulent burning velocity is required to address the flame propagation at high Reynolds number. However, as pointed out in Section 2.1, turbulent burning velocity is a function of flow characteristics and can be determined based on different definitions [22]. While the ratio of turbulent burning velocity to laminar flame speed is considered a function of pressure based on consumption speed and flame angle methods [184–188], the dependency of the turbulent burning velocity magnitude on pressure was found to be weak in several studies [29,189,190].

Furthermore, the quenching distance decreases with an increase of pressure and was found to be an influential parameter for increased flashback propensity at high pressure.

5.4 Flame–Wall Interaction in Boundary Layer Flashback

A large number of experimental studies have investigated boundary layer flashback in the literature, primarily using the concept of critical velocity gradient for interpreting final results. In addition, a few numerical studies have been performed to study boundary layer flashback in more detail, mainly limited to laminar conditions. Until fairly recently, the underlying physics of the problem was based on the model (by Lewis and Elbe [47]) neglecting flame-flow interaction. However, the lack of experimental and numerical analyses taking into account high spatial and temporal resolution are still apparent. Section 5.4 discusses recent important studies having significant contributions to understanding of boundary layer flashback.

5.4.1 Confined flames

The confined flame situation refers to a condition in which the flame already exists in the boundary layer within a tube or channel. Flame can enter the premixing tube due to the operability issues such as pressure, velocity fluctuations, and auto ignition as indicated in Figure 8. The turbulent flame propagation through a channel has been studied experimentally and numerically with high spatial and temporal resolutions, achieved by micro-resolution particle image velocimetry (Micro-PIV) measurements and direct numerical simulation (DNS). Since there is no mechanism for flame anchoring through a channel, the flame rapidly propagate upstream [99,100].

Eichler and Sattelmayer investigated boundary layer flashback of premixed flame for confined flames using simultaneous micro-PIV and chemiluminescence measurements [99]. Flame-flow interaction near the wall-region at the onset of flashback was studied in detail and presented conclusion were also verified by the numerical simulation of a laminar premixed flame stabilized on a flat plate using ANSYS CFX [99]. As discussed in prior work [98], the critical velocity gradient model cannot capture the

characteristic of confined flames and temporally and spatially resolved measurements were used in this experiment to capture the onset of flashback. The setup used in this experiment was similar to configuration 1 in Figure 63. Both laminar and turbulent flames for hydrogen and methane were investigated in this study. The high spatial resolution required for capturing the structure of the near-wall region for the turbulent flame was achieved by spatial micron resolution PIV. Furthermore, Mie-scattering was also implemented to estimate the flame location in captured images. The detailed description of micro PIV setup and its alignments can be found in [99,191]. Since the channel length was not long enough, the flow did not reach a fully developed state.

Figure 75 presents the turbulent flame flashback behavior of hydrogen/air mixture by OH* chemiluminescence, Mie-scattering, and velocity contour images. The highly stochastic behavior of the turbulent flow is clear in this figure with presence of flame wrinkles. The concave wrinkles are formed into cusps, which can either travel upstream, move downstream, or stay at initial location. It was also observed that cusps in 40-50% of cases divided into new cusps, going through the same process of moving or remaining. The upstream movement comprising high fraction of formed cusps leads to the flame flashback. For first set of images at $t=0$, the flame tip is not located at the center of the channel (yellow line) and consequently the flame did not appear in Mie-scattering image. In the next three images, the flame location can be estimated by the low density fluid. The white line represents the zero axial velocity in the flow field and the formation of back flow region. The back flow region was observed for cases in which the cups propagated upstream; however, backflow was not formed as the cusps were washed downstream of the flame. The boundary layer separation was also observed for laminar flashback of methane/air flames.

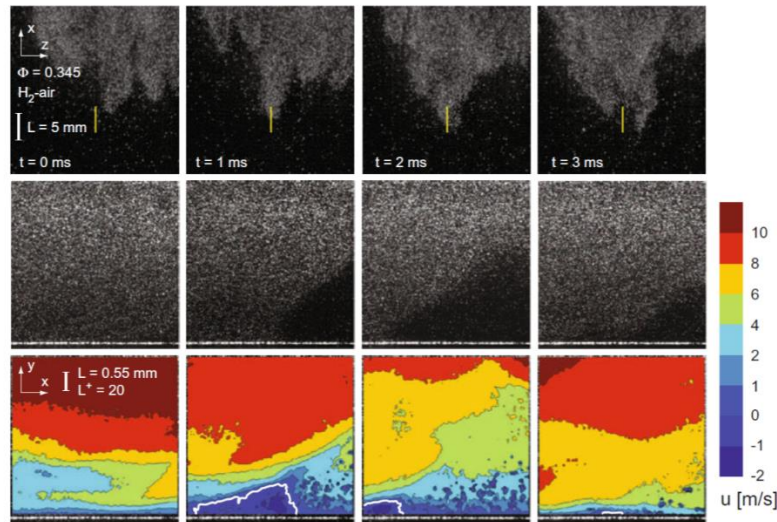


Figure 75: Micro-PIV and chemiluminescence measurements of turbulent boundary layer flashback for hydrogen/air [99].

This experiment shed some new light on the flame-flow interaction at the onset of flashback. An increase of pressure upstream of the flame front leads to the flow separation observed for both laminar and turbulent flames for studied cases. Hoferichter et al. recently developed a flashback prediction model for confined flames based on the separation of the turbulent boundary layer [111].

The boundary layer separation was further verified by numerical simulation of a laminar flame. In this model, the hydrogen/air mixture was stabilized on a flat plate with a hot wall segment used as a flame holder. The temperature of the mixture and the flat plate were kept constant at 293.15 K. The laminar premixed combustion was modeled by finite-rate chemistry with five species and three steps mechanisms based on Boivin et al. [192]. Full description of the implemented method can be found in [99].

An increase of pressure upstream of the flame front is a result of two dimension characteristics of the flame-flow interaction and was not predicted in simplified numerical models existing in the literature. The convex flame bulge towards reactants in the boundary layer results in the adverse pressure gradient and flow deceleration, which can potentially leads to the flow separation (see [138], chapter 7).

The contour of axial velocity of the premixed flame is indicated in Figure 76, normalized by the laminar flame speed (S_{L0}). The three first contours represent the stable condition, while the last one

shows the situation at flashback. The hot tile is indicated by a thick line at the right corner of the plate. The extension of the reaction zone can be seen from the black isoline of hydrogen molar fraction. As equivalence ratio increases the maximum height and length of back flow region increases. The ignition of the air/fuel mixture in the back flow region is a result of balance between the heat transfer from the flame front, heat loss to the wall and external region. The height of back flow region is considerably larger than the quenching distance. As the maximum height of the back flow region reaches a certain value; which is dictated by energy balance, the reaction can sustain itself inside the back flow region and flashback occurs. The back flow region at the onset of flashback is continuously moved upstream by the adverse pressure in front of the flame.

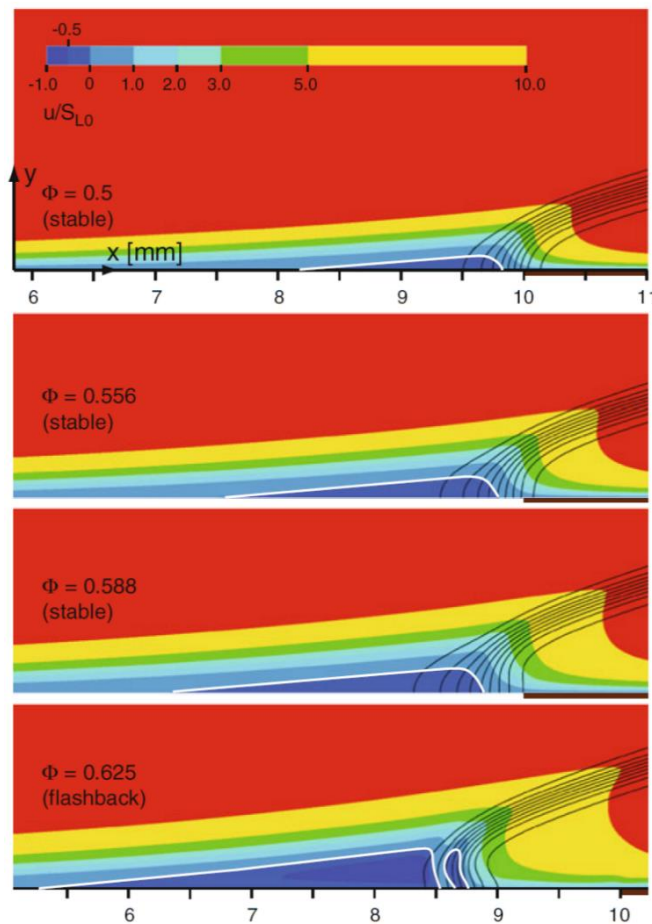


Figure 76: Axial velocity contour superimposed with isoline of hydrogen molar fraction for the laminar premixed flame at stable and flashback conditions [99].

Direct numerical simulation of boundary layer flashback in fully developed turbulent channel flow was carried out by Gruber et al [100]. The turbulent flame-wall interaction was studied for hydrogen/air with detailed chemical mechanisms developed by Li et al. [193]. The simulation was obtained for Reynolds number of 3200 (friction Reynolds number, $Re_\tau = 180$) based on the channel half width, preheat temperature of 750 K, and equivalence ratio of 1.5. The rich mixture was examined at pressures of 1 and 2 atm for channel half widths of 1 and 0.5 cm, respectively. The compressible Navier-Stokes equations were solved in three-dimensional domain to simulate the freely propagation of non-anchored flame through turbulent planar channel flow.

Figure 77 indicates the upstream flame propagation inside the channel at different time steps. The laminar flame is initially present in the flow field with alternating streaks in the near-wall region which are the coherent structures of the turbulent boundary layer [194]. As shown in Figure 77, no back flow region is observed in the flow field initially simulated for the non-reactive mixture. Then, the high speed sweeps causes the formation of flame cusps, the convex flame region towards the products, while the low speed ejections lead to the formation of bulges, the convex flame towards the reactants (upstream leading edge in Figure 77). Darrieus-Landau hydrodynamic instability [195] induced by this mechanism enhances the interaction between the turbulent boundary layer and premixed flame. Consequently, the backflow region is generated upstream of the bulges shown by blue surfaces in Figure 77. Indeed, these back flow pockets are results of the fluid expansion across the flame front occurring at the low-speed streaks locations.

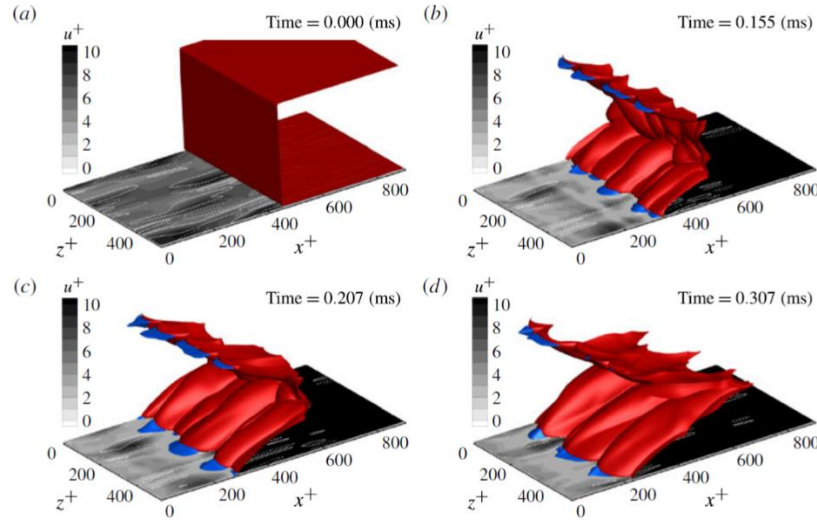


Figure 77: Temporal variation of the premixed flame (red color) in a channel, super imposed by the streamwise velocity contour (greyscale) on the $y^+ = 5$ plane, back flow region is shown by blue color [100].

These reverse flow regions are located near the wall with $y^+ < 20$ and extended up to 60-70 wall units in upstream direction. The height of backflow region reported in the experimental study by Eichler is $y^+ = 25$ to 30 [196]. The upstream propagation speed of the leading-edge flame is significantly enhanced by the presence of backflow pockets facilitating flashback. This is according to significantly higher critical velocity gradients associated with confined flames as shown in Figure 64 [98]. Figure 78 illustrates the streamwise velocity contour with location of the flame front (white line) relative to the back-flow region. The maximum negative velocity magnitude is $u^+ \sim -5$ or half of the laminar flame speed (Figure 78) and located at the wall distance of $y^+ = 9$, which matches well to the position of the upstream leading edge flame. It is noted that the flame is quenched at $y^+ = 2$, which is much smaller than the height of back flow region.

It was observed that the size of back flow region and the maximum strength of negative streamwise velocity are influenced by the mesh resolution of the boundary layer. Indeed, the course mesh suppresses the formation of back flow region; as a result, the LES may not be able to capture the structure of flame-flow interaction in the near-wall region.

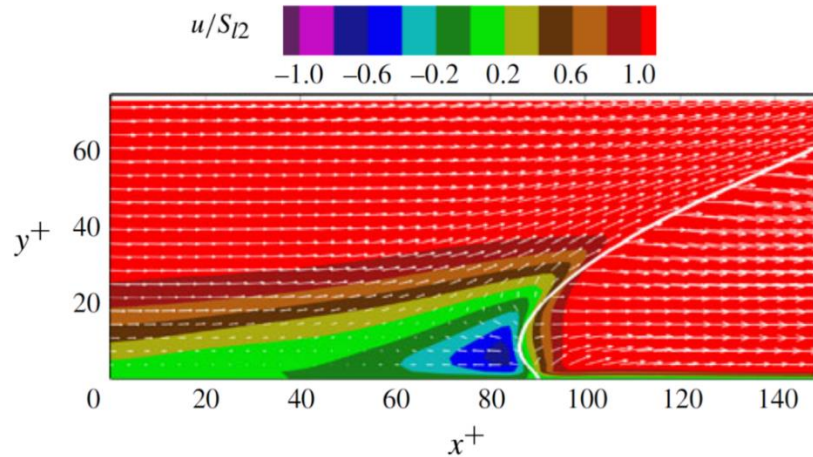


Figure 78: Streamwise velocity contours normalized by laminar flame speed. The flame is indicated by white line obtained from reaction progress variable of 0.7 [100].

5.4.2 Unconfined flames

In premixed combustors, the flame is generally stabilized downstream of the injector/premixer duct where the fuel/air mixture enters the combustion chamber with a larger cross sectional area. This sudden expansion is limited to the liner diameter, which is typically much larger than the premixer duct diameter. As this expansion ratio increases, the combustor resembles the case with free injection of the combustible mixture to the ambient air (Figure 8). Boundary layer flashback of jet flames has been extensively studied in the literature, but the detailed characteristics of the problem have not been well understood. The lack of numerical simulation addressing boundary layer flashback of jet flames is noted. Besides, all experimental studies in the literature are limited to investigating the flashback limits, in which the critical velocity gradient is used for comparison.

In order to the characterize the flame-flow interaction at the onset of flashback for unconfined jet flames, an experimental investigation at atmospheric pressure was conducted by Baumgartner et al. [106]. The high speed PIV, micro-PLIF along with chemiluminescence imaging were used to capture the spatial and temporal variations of the propagating hydrogen flame. The experimental setup includes a

high aspect ratio rectangular channel to resemble a two-dimensional flow through a planar channel. The flame is stabilized downstream of the burner exit to the ambient air.

Simultaneous OH* Chemiluminescence was used along the PIV or PLIF measurements. The temporal and spatial characteristics of the propagating flame were captured by the implemented high frequency lasers and long-distance microscopes. Further details of the laser diagnostics setup, measurement, and post processing of the data are found in [106].

Figure 79 presents the axial velocity contour with the propagation of the flame front at the onset of flashback for the bulk velocity of 7 m/s, equivalence ratio of 0.5, and temperature of the 293 K. The six images with the time interval of 0.33 ms show the flashback beginning and propagation of the flame front in which streamlines are indicated by white lines. The streamlines near the wall-region are highly deflected toward the expansion region, which is not observed for a jet flow without the reaction. The black dashed line in Figure 79 represent the approximate flame front location extracted from Mie scattering images. As the equivalence ratio is increased to the critical value, the leading flame tip is formed close to the edge of the lower wall and starts propagating upstream. With upstream propagation of leading flame, the flow decelerates as colored by red region in Figure 79. The upstream flame propagation through the duct enhances the reactants deceleration and can eventually lead to the boundary layer separation as previously observed from the experimental results for confined flames. This behavior is not observed in Figure 79 since the flame is partially confined by the lower wall.

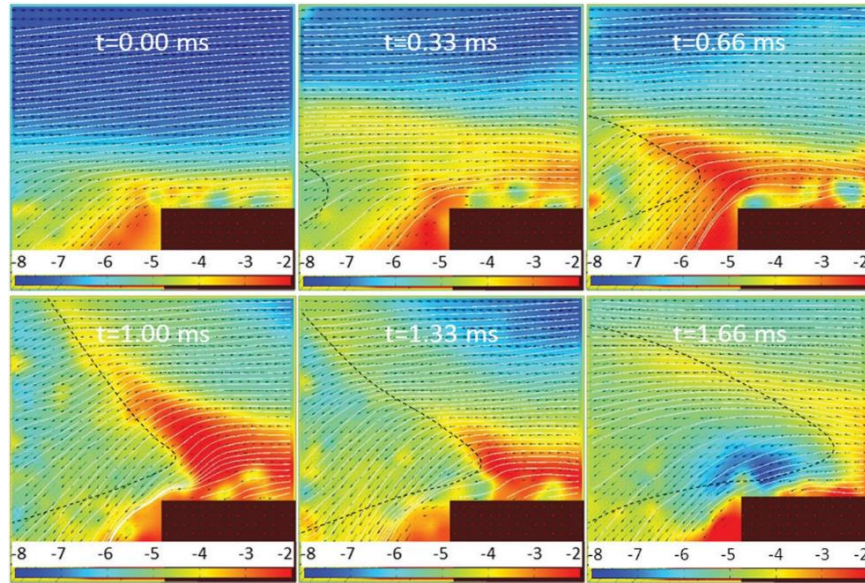


Figure 79: Flame propagation of boundary layer flashback for an unconfined flame (PIV measurement) [106].

The flame propagation in boundary layer flashback of an unconfined flame is not triggered by the immediate jump of the flame into the burner. Indeed, it initially starts with the formation of the leading flame front close to the wall appearing as a strongly curved flame tip towards the unburned mixture. This behavior is depicted with the schematic of flashback occurrence in Figure 80. The part of the flame anchored below the wall remains approximately undisturbed as the flame propagates upstream. The distance from the wall where the leading flame tip propagates is indicated by d_f and is approximately equal to 1 mm as reported in [106], which is in good agreement with the measured values (0.53-0.96 mm) for a confined flame in [196].

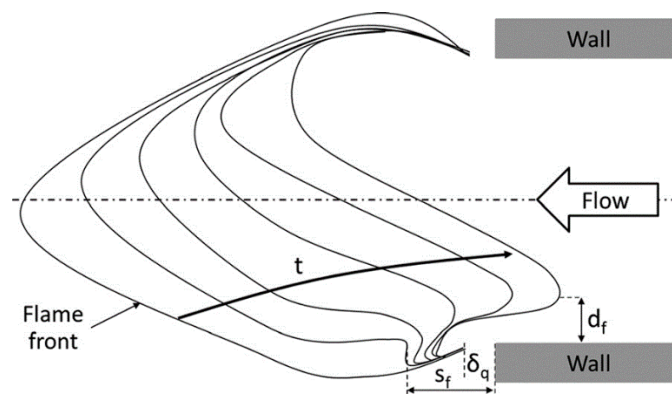


Figure 80: Schematic of flame front transition from the stable condition to flashback [106].

Based on this observation, a prediction model was proposed in [106]. Flashback starts at distance of s_f from the burner rim, which is larger than the head-on quenching distance (δ_q) as illustrated in Figure 80. At this location (s_f), the burning velocity is equal to the flow velocity providing the proper condition for the flame propagation. Based on experimental results, the streamline passing from the location (s_f) coincides with the distance (y_f) from the channel wall, which is slightly lower than the distance d_f in Figure 80. Assuming the axial velocity distribution inside the channel at the onset of flashback is not affected by the stable flame anchored downstream of the burner rim, the prediction model was proposed as following:

$$S_f(s_f) = u(s_f) \propto u(y_f) \quad \text{Eq. 54}$$

This equation states that the prediction of boundary layer flashback for unconfined flames is a function of the undisturbed velocity profile of the flow inside the channel. Experimental results showed that the distance (y_f) is between 0.6 and 0.7 mm and almost independent of equivalence ratio. This range of (y_f) corresponds to $y^+ = 13$ and $y^+ = 21$ and falls within the buffer layer region, while the critical velocity gradient model assumption is that the flow velocity at flashback can be approximated by the linear velocity profile in the viscous sublayer. Hoferichter et al. further developed this flashback prediction model by assuming that the location at which flashback is triggered corresponds to the maximum streamwise turbulence intensity in a fully developed duct flow. This model includes a balance between the local turbulent burning velocity and the local flow velocity at $y^+ = 16.4$ [109].

The flame-flow interaction in unconfined configurations at the onset of flashback does not significantly impact the upstream flow field. As a result, the velocity profile remains approximately undisturbed upstream of the anchored flame. This is the main reason why the concept of critical velocity gradient has been able to predict various data in the literature for the onset of flashback. The presence of backflow region has been experimentally observed and numerically verified for both laminar and

turbulent flame propagation through a channel. But, it is important to realize that the backflow region occurs only if the necessary conditions are met. Otherwise, the presence of backflow pockets cannot be generally extended to all fuel compositions and operating conditions. Indeed, the flame expansion ratio needs to be large enough compared to the fluid velocity close to the wall, which determines the flow retardation in that region.

5.5 Key Points and Status

- Flashback limit is dictated by the turbulent boundary layer structure and the critical velocity gradient concept is less able to capture flashback behavior of turbulent flames compared to laminar flames. This is due in part to the breakdown of assumptions underlying the theory such as linear velocity profiles and no consideration of turbulent flame characteristics.
- The role of heat transfer between the flame and the burner is similar to that in laminar flames. The net result is strong interactions between parameters that affect heat transfer such as flame temperature and proximity of the flame to the burner. The boundary layer heating can potentially lead to the flow separation in the boundary layer, inducing the upstream flame propagation. As a result, incorporation of key heat transfer effects is required to generalize any predictive model for capturing flashback tendencies.
- Work done on turbulent configurations underscores the importance of considering which regime the flame is in, whether it be unconfined or confined. It is evident that once the flame transitions from outside the premixer tube to within it, the propagation rate greatly increases. If the flame issues into an enclosure (e.g., a combustor), the impact on flow field and heat transfer again impacts the flashback behavior. Thus applying models based on open flames may not provide accurate results for flames issuing into an enclosure.
- More work has been carried on the pressure effect on turbulent flame flashback than for laminar flames. Increased flashback propensity at high pressure is a result of higher turbulent burning velocity and lower quenching distance. But, the contribution of each parameter to the overall trend (i.e. flashback as a function of pressure) is not well understood due to limited studies at high pressure.
- Some success in capturing these complex interactions has been found recently using Buckingham Pi theory as applied to data available over a range of fuel types, burner materials, operating pressure and reactant temperatures (e.g., [110,112]). A limitation of the approach is that the heat transfer effects are captured by the burner tip temperature which is really a culmination of many other factors associated with the overall heat transfer process.
- Recent insights into the behavior of confined turbulent flames has established a new framework to develop new theories and models for addressing flame propagation in these cases. These insights have been established with both detailed experiments (e.g., [99,104]) and high fidelity simulations (e.g., [100]). New approaches have been developed in light of these insights (e.g., [111]).

- The presence of backflow region for boundary layer flashback of confined flames has been observed. The backflow region occurs only if the flame expansion ratio is large enough compared to the fluid velocity in the near-wall region of a channel.

6. Closing Remarks and Recommendations

Boundary layer flashback has been studied since the 1940's. The effects of various parameters including fuel/oxidizer composition, preheat temperature, operating pressure, burner material, and geometrical burner configuration under laminar and turbulent conditions have been examined. The concept of a critical velocity gradient, neglecting the flame-flow interaction, has been used frequently in the literature to interpret experimental results. The transition of the stabilized flame at burner rim to the injector (unconfined configuration) has been addressed by numerous researchers, while the flame propagation through circular tubes or channels (confined configuration) has been the subject of both experimental and numerical simulations in the literature. These two regimes feature fundamentally different phenomenological behavior as discussed in Section 5.4. Recent experimental and numerical studies of confined flames have been particularly illuminating.

Laminar boundary layer flashback has been extensively studied in the literature for various fuel compositions (Sections 4). However, studies carried out for turbulent boundary layer flashback are limited (Section 5), especially those featuring numerical simulations. Numerical modeling of the flame-wall interaction for laminar boundary layer flashback has mainly focused on two-dimensional analysis using single-step chemical reaction mechanism. It is also noted that the most recent research have emphasized flashback propensity of high hydrogen content fuels and syngas fuels.

It is apparent that more studies are required at gas turbine premixer conditions to investigate the effect of pressure and high Reynolds numbers on flashback propensity. The effect of pressure can be isolated by controlling the burner rim temperature under conditions of interest. Such effects have not been studied at elevated pressure. In addition, laser diagnostic techniques such as PIV, PLIF, and flame

spectroscopy can be implemented to provide a better understanding of boundary layer flashback at high pressures.

The effect of turbulence intensity on flashback propensity, especially at elevated pressure, has not yet been studied in the literature. The strategy to alter the turbulence of the air/fuel mixture has proven difficult in narrow ducts as the fully developed flow needs to be attained, but with varying turbulence. It is noted that no numerical simulation has been conducted for the transition of the turbulent stabilized flame at burner rim to the injector as flashback occurs. Indeed, the numerical simulation addressing flame-wall interaction using detailed chemical kinetics for unconfined configurations is still required. These open questions warrant further work, but progress has been made in terms of design tool development as well as phenomenological understanding of the boundary layer flashback mechanism.

Acknowledgments

The contributions of Dr. Zhixuan Duan and Mr. Brendan Shaffar in discussions and review of this paper are acknowledged. Dr Elliot-Sullivan Lewis contributed to the overall interpretation of the results. Mr. Josh Holt assisted with extraction of data and plotting of results from various studies in the literature. Some of results shown in this paper were generated as a result of grants from the California Energy Commission and the U.S. Department of Energy. Dr. Peter Jahnson and Dr. Salvatore Danielle engaged in fruitful discussions and kindly provided data they obtained. Discussions with Prof. Thomas Sattelmayer, Dr. Christoph Hirsch and the engagement of Dr. Georg Baumgartner and Ms. Vera Hoferichter all from the Technical University of Munich in Garching are very much appreciated and added important insights to the interpretation of the flashback phenomena.

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